

(need to change)Project 2 Linear Mixed Model

** **

March 15, 2026

```
#download packages
library(tidyverse) #installed tidyverse
library(nlme)
library(ggplot2)
library(here)
library(kableExtra)
library(modelsummary)
library(dplyr)
library(performance)

file_path <- here("analysis_and_plots/heat_flow_ICE_EPFL.csv")

#Load the dataset
data <- read.csv(file_path)
```

1 Abstract

2 Introduction

3 Exploratory Data Analysis

An exploratory data analysis was conducted, aiming to understand the distributional properties of the heat flow measurements and identify preliminary trends regarding inter-individual and intra-individual variability, as well as the relationships between the different sensors.

3.1 Dataset Overview

The dataset consists of measurements from 24 participants (12 female, 12 male) monitored over two separate days. The only environmental difference between Day 1 and Day 2 is the temperature: Day 1 is 24 °C while Day 2 is 18 ° C. While the original experiment captured data every second, this analysis utilizes the mean heat flow per activity in each session monitored by 6 sensors to address the research goals.

98 out of 2736 observations have missing heat flow measurements, and were directly dropped.

3.1.1 Nikita note to add: Specifically, the activity walking only has data for sessions 1 & 3, and the activity standing does not have data for session 2 on day 2.

All variables are categorical: **day** has 2 levels ("1" and "2"); **session** has 4 levels ("1", "2", "3" and "4"); **activity** has 3 levels ("sitting", "standing" and "walking"); **sex** has 2 levels ("F" and "M"); **HF_type**(sensor ID) has 6 levels ("HF_2", "HF_3", "HF_5", "HF_7", "HF_8", "HF_16").

3.1.2 Nikita note to add: Note sensors correspond with XYZ body part.

3.2 Visual Evidence for Fixed Effects

To determine which variables influence heat flow, we examined the measurements across different experimental conditions. The exploratory plots indicate that heat flow is not constant and is influenced by every categorical variable in the study.

Figure 1 displays heat flow during the sitting activity, partitioned by Day, Session, Sensor, and Sex.

For Sensor Type, we can see there are distinct vertical offsets between panels (e.g., HF_16 vs. HF_1), showing that different body locations have different baseline heat flow. Comparing the top row (Day 1, 24°C) to the bottom row (Day 2, 18°C) reveals a clear shift in mean heat flow, confirming a strong environmental effect. The solid linear fit lines across the four sessions often show non-zero slopes, suggesting that heat flow changes over the course of the day even when activity is held constant. The red (Female) and blue (Male) lines are frequently separated, indicating that biological sex influences thermal regulation.

```
 #(Below is not needed anymore for checking linearity, we treat session as
 #factor)
 # make session to be continuous variable for now
 # spaghetti plot + faceting
data_sitting <- data_clean %>%
  filter(activity == "sitting")

ggplot(data_sitting, aes(x = session_num, y = HF_value, color = sex)) +
  geom_point(alpha = 0.1) +

   # Flexible LOESS curves (Dashed lines, one for M and one for F)
  geom_smooth(method = "loess", formula = y ~ x, se = FALSE, linetype = "dashed", size = 0.5) +

   # Straight Linear regression lines (Solid lines, one for M and one for F)
  geom_smooth(method = "lm", formula = y ~ x, se = FALSE, linetype = "solid", size = 0.5) +

   # Grid: Rows = Activity, Columns = Sensor
  facet_grid(day ~ HF_type) +

   # Manually set colors for clarity (e.g., Males = Blue, Females = Red)
  scale_color_manual(values = c("M" = "blue", "F" = "red")) +

  theme_minimal() +
  labs(
    title = "(Sitting) Heat Flow Over Time by Day, Session, Sensor and Sex",
    subtitle = "Solid lines = Linear Fit (lm) | Dashed lines = Flexible Fit (loess)",
    x = "Session (1 to 4)",
    y = "Heat Flow (HF_value)",
    color = "Sex"
  )
)
```

(Sitting) Heat Flow Over Time by Day, Session, Sensor and Sex

Solid lines = Linear Fit (lm) | Dashed lines = Flexible Fit (loess)

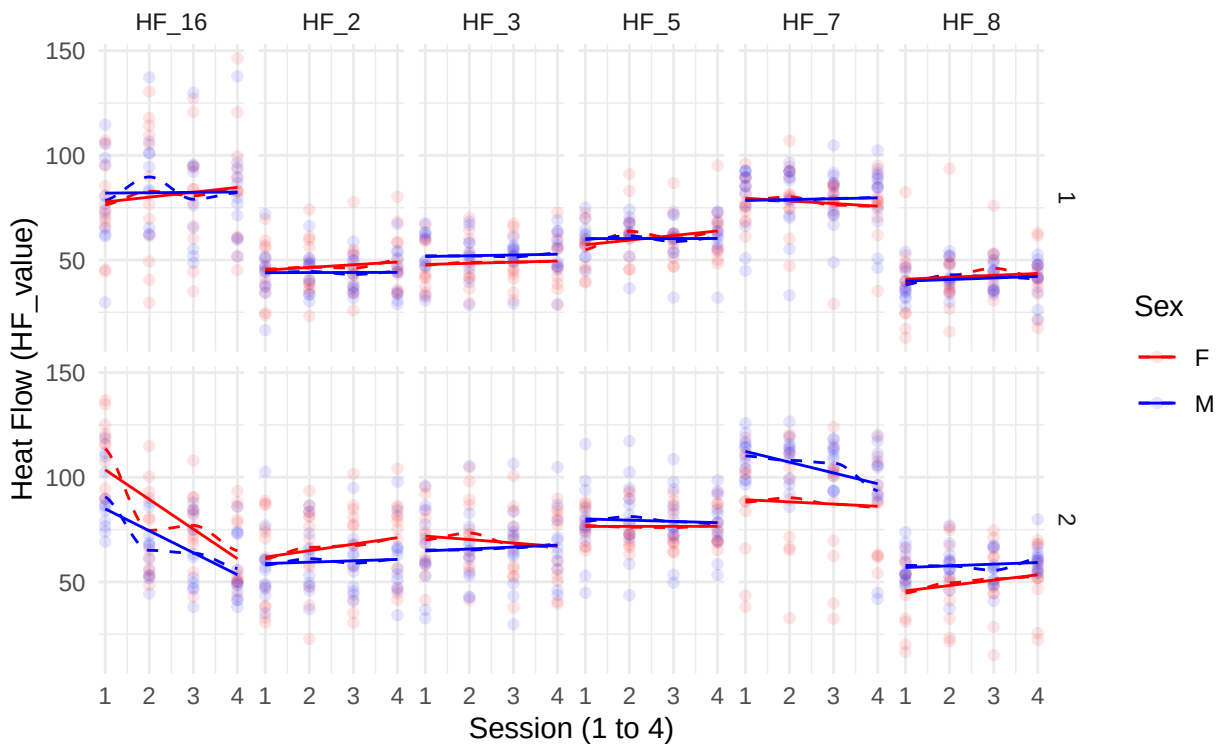


Figure 1: Sitting's Heat Flow Over Time by Day, Session, Sensor and Sex

Figure 2 focuses on Day 1 to highlight the impact of activity: Comparing the rows for sitting, standing, and walking reveals that heat flow varies with physical movement.

```
data_day1 <- data_clean %>% filter(day == "1")

ggplot(data_day1, aes(x = session_num, y = HF_value, color = sex)) +
  geom_point(alpha = 0.1) +

  geom_smooth(method = "loess", formula = y ~ x, se = FALSE, linetype = "dashed", size = 0.5) +

  # Straight Linear regression lines (Solid lines, one for M and one for F)
  geom_smooth(method = "lm", formula = y ~ x, se = FALSE, linetype = "solid", size = 0.5) +

  facet_grid(activity ~ HF_type) +

  scale_color_manual(values = c("M" = "blue", "F" = "red")) +

  theme_minimal() +
  labs(
    title = "(Day 1)Heat Flow Over Time by Activity, Sensor and Sex",
    subtitle = "Solid lines = Linear Fit | Dashed = LOESS | Grid = Activity+Day vs Sensor",
    x = "Session (1 to 4)",
    y = "Heat Flow (HF_value)",
    color = "Sex"
  )
```

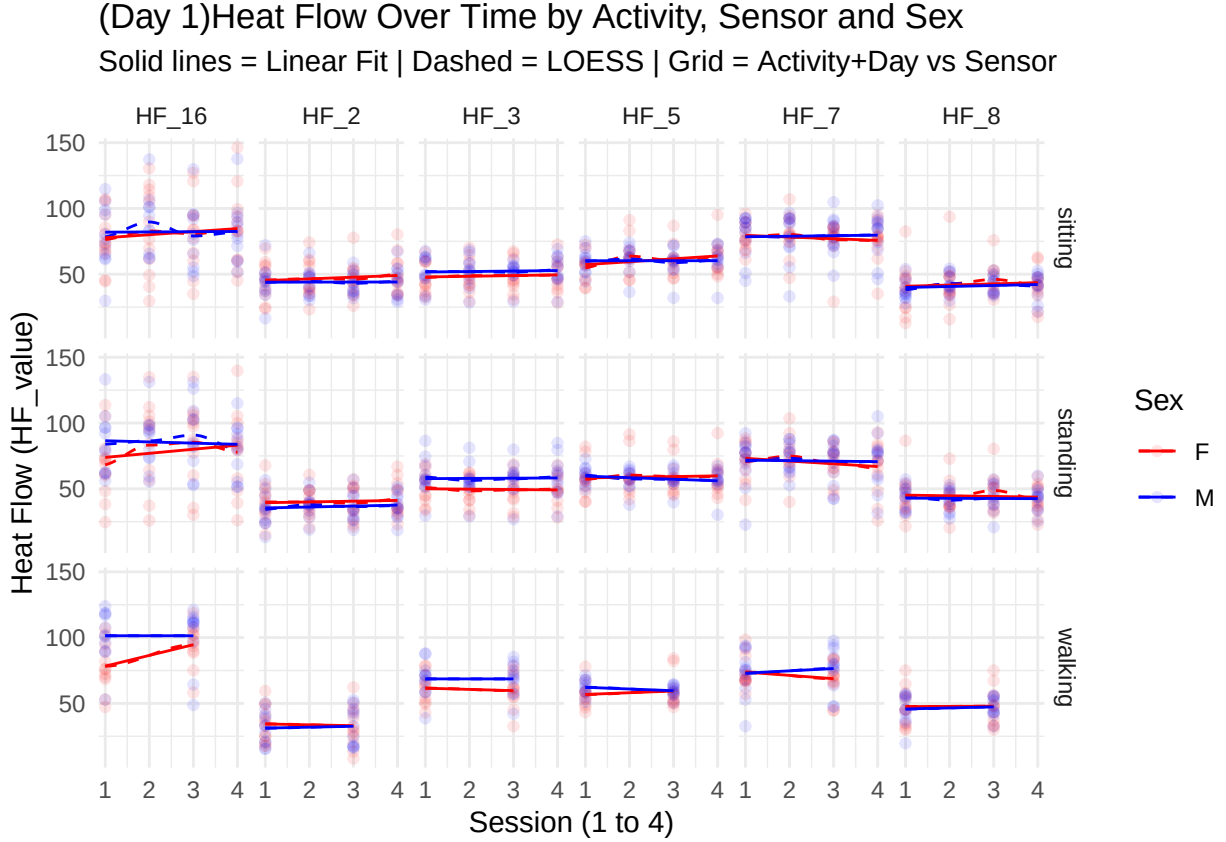


Figure 2: Day 1's Heat Flow Over Time by Activity, Sensor and Sex

Because the plots and physiological intuition suggest that every categorical variable exerts a fixed effect on the response, all predictors were carried forward into the modeling stage.

4 Statistical Description of the Methods

To analyze the factors influencing human heat flow while accounting for the complex hierarchical and temporal dependencies inherent to the experimental design, a Linear Mixed Model (LMM) was employed.

4.1 Fixed-Effects Optimization and Transformation

4.1.1 Nikita note: add note of what section the log stuff was shown, removed some description, discuss tests that zhuo ran

An initial naive ordinary least squares (OLS) regression was fitted to identify the baseline relationships between the predictors and the response variable and justify the inclusion of all 5 variables. Diagnostic plots of this initial model revealed significant heteroscedasticity and heavy-tailed residuals. To satisfy the assumptions of linear modeling and stabilize the variance, a natural logarithm transformation was applied to the heat flow measurements ($Y = \log(\text{HF_value})$), as further discussed in section XYZ.

4.1.2 Nikita note: discuss part F-tests. I think i used the likelihood ratio test, basically, from my anovas. I think that would be preferred - can we change this?

To determine the optimal fixed-effects structure, a systematic, multi-step model-building approach was employed. First, partial F-tests were utilized to evaluate the significance of predictors in a main-effects-only model.

For a given categorical predictor or interaction term with r degrees of freedom (corresponding to $k - 1$ coefficients for a variable with k levels), the partial F-test evaluates whether the inclusion of the term provides a statistically significant improvement to the model's goodness-of-fit. The null hypothesis posits that all r coefficients associated with the specific term are simultaneously zero:

$$H_0 : \beta_1 = \beta_2 = \dots = \beta_r = 0$$

To test this, the F-statistic is calculated by comparing the Residual Sum of Squares (RSS) of a reduced model (excluding the term in question) against a full model (including the term):

$$F = \frac{(RSS_{\text{reduced}} - RSS_{\text{full}})/r}{RSS_{\text{full}}/(n - p)}$$

(Where n represents the total number of observations, and p represents the total number of parameters estimated in the full model. Note that r is equivalent to $p_{\text{full}} - p_{\text{reduced}}$).

Under the null hypothesis, this test statistic follows an F-distribution with r and $n - p$ degrees of freedom ($F_{r, n-p}$). The mathematical validity of this test assumes that the model's residual errors are independent, identically distributed, and follow a normal distribution, $\epsilon \sim \mathcal{N}(0, \sigma^2)$.

4.1.3 nikita note: should we go through all of this? I think it is better to just explain what an interaction term is, and mention we used a similar methodology to the ANOVA done on the fixed effects... like, how is it diff?

Next, to identify relevant interaction terms, a preliminary marginal screening of ten theoretically plausible two-way interactions was conducted. To test whether the addition of each interaction term significantly improved the model (Null Hypothesis (H_0): The coefficient for the interaction term is exactly zero, $\beta_{\text{interaction}} = 0$), individual partial F-tests were performed.

4.1.4 Nikita note: I don't even look at the FWER.... we can mention later?

Because conducting multiple simultaneous hypothesis tests artificially inflates the probability of Type I errors (false positives), a Bonferroni correction was applied to control the family-wise error rate (FWER). Based on a standard family-wise alpha level of 0.05 distributed across 10 tests, the adjusted significance threshold was established at $p < 0.005$.

4.1.5 nikita note: again, i do not use partial F-test

To finalize the fixed-effects structure, a marginal backward elimination procedure (using partial F-test) was conducted for the resulting model. Furthermore, to adhere to the principle of marginality, any main effect involved in a significant two-way interaction was retained in the model, regardless of its independent main-effect significance.

4.2 Hierarchical Random-Effects Structure

The experimental design involved repeated measurements from the same 24 participants across multiple days, sessions, and physical activities. An assumption of standard linear regression is that all observations are independent. However, because multiple heat flow measurements were taken from the same individuals, unobserved subject-specific physiological traits (such as baseline metabolic rate or core temperature) were expected to cause intra-subject correlation.

To appropriately account for this unobserved heterogeneity and resolve the violation of independence, a Linear Mixed Model (LMM) was employed.

Mathematically, an LMM extends the general linear model by partitioning the total variance into fixed and random components. In matrix notation, the general model is defined as:

$$\mathbf{y} = \mathbf{X}\beta + \mathbf{Z}\mathbf{b} + \epsilon$$

Where:

$\mathbf{y}$ is the vector of the response variable ($\log(\text{HF_value})$).

$\mathbf{X}$ is the design matrix for the fixed effects (the controlled experimental conditions such as Day, Session, and Activity).

β is the vector of fixed-effect coefficients, representing the population-average responses.

$\mathbf{Z}$ is the design matrix for the random effects. This matrix can contain both indicator variables for grouping factors (yielding random intercepts) and continuous/categorical covariates (yielding random slopes).

$\mathbf{b}$ is the vector of random effects. In this study, it captures both a subject's unique baseline shift (a random intercept) and their specific physiological trajectory in response to a changing experimental condition (a random slope). It is assumed to be normally distributed with mean zero and covariance matrix $\mathbf{G}$, such that $\mathbf{b} \sim \mathcal{N}(\mathbf{0}, \mathbf{G})$.

ϵ is the vector of residual errors, assumed to be normally distributed with mean zero and covariance matrix $\mathbf{R}$, such that $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{R})$.

By introducing the random effects vector $\mathbf{b}$, the LMM defines the marginal covariance matrix of the response $\mathbf{y}$ as:

$$\text{Var}(\mathbf{y}) = \mathbf{Z}\mathbf{G}\mathbf{Z}^T + \mathbf{R}$$

Unlike standard Ordinary Least Squares (OLS) regression, which assumes a strictly diagonal covariance matrix where all observations are completely independent, this $\mathbf{Z}\mathbf{G}\mathbf{Z}^T$ structure mathematically permits non-zero covariances between observations belonging to the same participant. Furthermore, by allowing $\mathbf{G}$ to estimate both random intercepts and random slopes, the model accounts for the fact that participants not only start at different thermal baselines, but also adapt to the environmental temperature changes at entirely different physiological rates.

4.3 Residual Variance and Temporal Correlation

In standard linear mixed-effects models, the residual errors (ϵ_{ijk}) are conventionally assumed to be conditionally independent and identically distributed with a constant variance σ^2 . However, given the longitudinal nature of this experimental design, the assumption of residual independence may be overly restrictive. Physiologically, heat flow is a continuous, dynamic process; therefore, it is hypothesized that short-term, unexplained biological variations exhibit temporal autocorrelation. Specifically, a participant's unobserved physiological state during one session is likely to influence their state in the immediately subsequent session, with this residual correlation decaying as the time gap increases.

To mathematically account for this temporal decay across consecutive measurements, a first-order Autoregressive AR(1) correlation structure was implemented. This temporal memory was explicitly nested within the experimental day (person/HF_type/day). This boundary specification ensures that residual autocorrelation decays continuously across consecutive sessions within a single day, but is forced to assume complete independence across the overnight gap between Day 1 and Day 2.

Let ρ be the autoregressive parameter, and $|k - k'|$ be the chronological time lag between two measurement instances. Under AR(1) assumption, the covariance between any two residuals from the same sensor on the same person is defined as:

$$\text{Cov}(\epsilon_{ijk}, \epsilon_{ijk'}) = \sigma^2 \rho^{|k-k'|} I(\text{Day}_{ijk} = \text{Day}_{ijk'})$$

Where the indicator function $I(\cdot)$ returns 1 if the measurements occurred on the same day, and 0 if they occurred on different days, breaking the temporal correlation overnight.

While the AR(1) specification successfully captures temporal memory, it still imposes a strict assumption of homoscedasticity, forcing a single, universal residual variance (σ^2) across all observations. Physiologically, different sensor placements (e.g., wrist versus chest) experience varying degrees of ambient noise interference and physical contact stability.

To resolve this limitation, the model's residual structure was upgraded to an Autoregressive Heteroscedastic ARH(1) formulation. This structure retains the autoregressive temporal decay (ρ) but relaxes the constant variance assumption by calculating a unique, independent residual variance (σ_j^2) for each specific sensor type j .

The final residual errors are distributed as a vector $\epsilon_{ij} \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_{ij})$, where the elements of the block-diagonal residual covariance matrix $\mathbf{R}_{ij}$ are formalized as:

$$\text{Cov}(\epsilon_{ijk}, \epsilon_{ijk'}) = \sigma_j^2 \rho^{|k-k'|} I(\text{Day}_{ijk} = \text{Day}_{ijk'})$$

This final $\mathbf{R}$ matrix allows the baseline magnitude of the physiological noise to scale up or down depending on the specific physical sensor, while preserving the chronological decay of that heat flow noise as the experimental day progresses.

5 Nikita's Notes (outside of results)

- How to justify use of LMM - discuss more...
- Add coefficients for each equation, and discuss whether used ML or REML (and impacts?)
- Explain ICC
- When discussing ICCs, just create tables to show it?? Each row is a model, and each column is a variable?
- Add final summary table to show coefficients, etc of final model; add discussion!
- discuss adding activity to ar1 - better correlation?
- Explain AIC, BIC? likelihood ratio test stat? Log likelihood?

```
#are NAs by sex?
sex_na_summary <- data %>%
  group_by(sex) %>%
  summarise(
    Total_Rows = n(),
    Missing_HF = sum(is.na(HF_value)),
    Percent_Missing = round((Missing_HF / Total_Rows) * 100, 2)
  )
print(sex_na_summary)
```

```
# A tibble: 2 x 4
  sex    Total_Rows Missing_HF Percent_Missing
<chr>    <int>      <int>         <dbl>
1 F         1368         32          2.34
2 M         1368         66          4.82
```

```
#are NAs by activity and session and day?
activity_session_day_na_summary <- data %>%
  group_by(activity, session, day) %>%
  summarise(
```

```

    Total_Rows = n(),
    Missing_HF = sum(is.na(HF_value)),
    Percent_Missing = round((Missing_HF / Total_Rows) * 100, 2)
  )
print(activity_session_day_na_summary)

```

A tibble: 19 x 6

Groups: activity, session [10]

activity	session	day	Total_Rows	Missing_HF	Percent_Missing
<chr>	<int>	<int>	<int>	<int>	<dbl>
1 sitting	1	1	144	8	5.56
2 sitting	1	2	144	1	0.69
3 sitting	2	1	144	9	6.25
4 sitting	2	2	144	2	1.39
5 sitting	3	1	144	12	8.33
6 sitting	3	2	144	1	0.69
7 sitting	4	1	144	8	5.56
8 sitting	4	2	144	2	1.39
9 standing	1	1	144	8	5.56
10 standing	1	2	144	1	0.69
11 standing	2	1	144	10	6.94
12 standing	3	1	144	10	6.94
13 standing	3	2	144	1	0.69
14 standing	4	1	144	8	5.56
15 standing	4	2	144	1	0.69
16 walking	1	1	144	8	5.56
17 walking	1	2	144	0	0
18 walking	3	1	144	8	5.56
19 walking	3	2	144	0	0

```

person_na_summary <- data %>%
  group_by(person) %>%
  summarise(
    Total_Rows = n(),
    Missing_HF = sum(is.na(HF_value)),
    Percent_Missing = round((Missing_HF / Total_Rows) * 100, 2)
  )
print(person_na_summary)

```

A tibble: 24 x 4

person	Total_Rows	Missing_HF	Percent_Missing
<chr>	<int>	<int>	<dbl>
1 F1	114	0	0
2 F10	114	0	0
3 F11	114	14	12.3
4 F12	114	1	0.88
5 F2	114	0	0
6 F3	114	0	0
7 F4	114	10	8.77
8 F5	114	3	2.63
9 F6	114	2	1.75
10 F7	114	0	0

i 14 more rows


```

activity_session_day_sensor_na_summary <- data %>%
  group_by(activity, session, day, HF_type) %>%
  summarise(
    Total_Rows = n(),
    Missing_HF = sum(is.na(HF_value)),
    Percent_Missing = round((Missing_HF / Total_Rows) * 100, 2)
  )
print(activity_session_day_sensor_na_summary, n =120)

```

A tibble: 114 x 7

Groups: activity, session, day [19]

	activity	session	day	HF_type	Total_Rows	Missing_HF	Percent_Missing
	<chr>	<int>	<int>	<chr>	<int>	<int>	<dbl>
1	sitting	1	1	HF_16	24	1	4.17
2	sitting	1	1	HF_2	24	1	4.17
3	sitting	1	1	HF_3	24	2	8.33
4	sitting	1	1	HF_5	24	3	12.5
5	sitting	1	1	HF_7	24	0	0
6	sitting	1	1	HF_8	24	1	4.17
7	sitting	1	2	HF_16	24	1	4.17
8	sitting	1	2	HF_2	24	0	0
9	sitting	1	2	HF_3	24	0	0
10	sitting	1	2	HF_5	24	0	0
11	sitting	1	2	HF_7	24	0	0
12	sitting	1	2	HF_8	24	0	0
13	sitting	2	1	HF_16	24	2	8.33
14	sitting	2	1	HF_2	24	1	4.17
15	sitting	2	1	HF_3	24	2	8.33
16	sitting	2	1	HF_5	24	3	12.5
17	sitting	2	1	HF_7	24	0	0
18	sitting	2	1	HF_8	24	1	4.17
19	sitting	2	2	HF_16	24	1	4.17
20	sitting	2	2	HF_2	24	0	0
21	sitting	2	2	HF_3	24	0	0
22	sitting	2	2	HF_5	24	0	0
23	sitting	2	2	HF_7	24	1	4.17
24	sitting	2	2	HF_8	24	0	0
25	sitting	3	1	HF_16	24	3	12.5
26	sitting	3	1	HF_2	24	1	4.17
27	sitting	3	1	HF_3	24	2	8.33
28	sitting	3	1	HF_5	24	3	12.5
29	sitting	3	1	HF_7	24	0	0
30	sitting	3	1	HF_8	24	3	12.5
31	sitting	3	2	HF_16	24	0	0
32	sitting	3	2	HF_2	24	0	0
33	sitting	3	2	HF_3	24	0	0
34	sitting	3	2	HF_5	24	0	0
35	sitting	3	2	HF_7	24	0	0
36	sitting	3	2	HF_8	24	1	4.17
37	sitting	4	1	HF_16	24	1	4.17
38	sitting	4	1	HF_2	24	1	4.17
39	sitting	4	1	HF_3	24	2	8.33
40	sitting	4	1	HF_5	24	3	12.5
41	sitting	4	1	HF_7	24	0	0

42 sitting	4	1 HF_8	24	1	4.17
43 sitting	4	2 HF_16	24	0	0
44 sitting	4	2 HF_2	24	0	0
45 sitting	4	2 HF_3	24	1	4.17
46 sitting	4	2 HF_5	24	0	0
47 sitting	4	2 HF_7	24	1	4.17
48 sitting	4	2 HF_8	24	0	0
49 standing	1	1 HF_16	24	1	4.17
50 standing	1	1 HF_2	24	1	4.17
51 standing	1	1 HF_3	24	2	8.33
52 standing	1	1 HF_5	24	3	12.5
53 standing	1	1 HF_7	24	0	0
54 standing	1	1 HF_8	24	1	4.17
55 standing	1	2 HF_16	24	1	4.17
56 standing	1	2 HF_2	24	0	0
57 standing	1	2 HF_3	24	0	0
58 standing	1	2 HF_5	24	0	0
59 standing	1	2 HF_7	24	0	0
60 standing	1	2 HF_8	24	0	0
61 standing	2	1 HF_16	24	2	8.33
62 standing	2	1 HF_2	24	1	4.17
63 standing	2	1 HF_3	24	2	8.33
64 standing	2	1 HF_5	24	3	12.5
65 standing	2	1 HF_7	24	1	4.17
66 standing	2	1 HF_8	24	1	4.17
67 standing	3	1 HF_16	24	1	4.17
68 standing	3	1 HF_2	24	1	4.17
69 standing	3	1 HF_3	24	2	8.33
70 standing	3	1 HF_5	24	3	12.5
71 standing	3	1 HF_7	24	0	0
72 standing	3	1 HF_8	24	3	12.5
73 standing	3	2 HF_16	24	1	4.17
74 standing	3	2 HF_2	24	0	0
75 standing	3	2 HF_3	24	0	0
76 standing	3	2 HF_5	24	0	0
77 standing	3	2 HF_7	24	0	0
78 standing	3	2 HF_8	24	0	0
79 standing	4	1 HF_16	24	1	4.17
80 standing	4	1 HF_2	24	1	4.17
81 standing	4	1 HF_3	24	2	8.33
82 standing	4	1 HF_5	24	3	12.5
83 standing	4	1 HF_7	24	0	0
84 standing	4	1 HF_8	24	1	4.17
85 standing	4	2 HF_16	24	0	0
86 standing	4	2 HF_2	24	0	0
87 standing	4	2 HF_3	24	1	4.17
88 standing	4	2 HF_5	24	0	0
89 standing	4	2 HF_7	24	0	0
90 standing	4	2 HF_8	24	0	0
91 walking	1	1 HF_16	24	1	4.17
92 walking	1	1 HF_2	24	1	4.17
93 walking	1	1 HF_3	24	2	8.33
94 walking	1	1 HF_5	24	3	12.5
95 walking	1	1 HF_7	24	0	0

96 walking	1	1 HF_8	24	1	4.17
97 walking	1	2 HF_16	24	0	0
98 walking	1	2 HF_2	24	0	0
99 walking	1	2 HF_3	24	0	0
100 walking	1	2 HF_5	24	0	0
101 walking	1	2 HF_7	24	0	0
102 walking	1	2 HF_8	24	0	0
103 walking	3	1 HF_16	24	1	4.17
104 walking	3	1 HF_2	24	1	4.17
105 walking	3	1 HF_3	24	2	8.33
106 walking	3	1 HF_5	24	3	12.5
107 walking	3	1 HF_7	24	0	0
108 walking	3	1 HF_8	24	1	4.17
109 walking	3	2 HF_16	24	0	0
110 walking	3	2 HF_2	24	0	0
111 walking	3	2 HF_3	24	0	0
112 walking	3	2 HF_5	24	0	0
113 walking	3	2 HF_7	24	0	0
114 walking	3	2 HF_8	24	0	0

6 Results

6.1 Initial Model

```
naive_lm <- lm(HF_value ~ session + activity + day + HF_type + sex, data = data_clean)

summary(naive_lm)
```

Call:

```
lm(formula = HF_value ~ session + activity + day + HF_type +
    sex, data = data_clean)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-79.530	-10.704	-0.129	11.157	77.121

Coefficients:

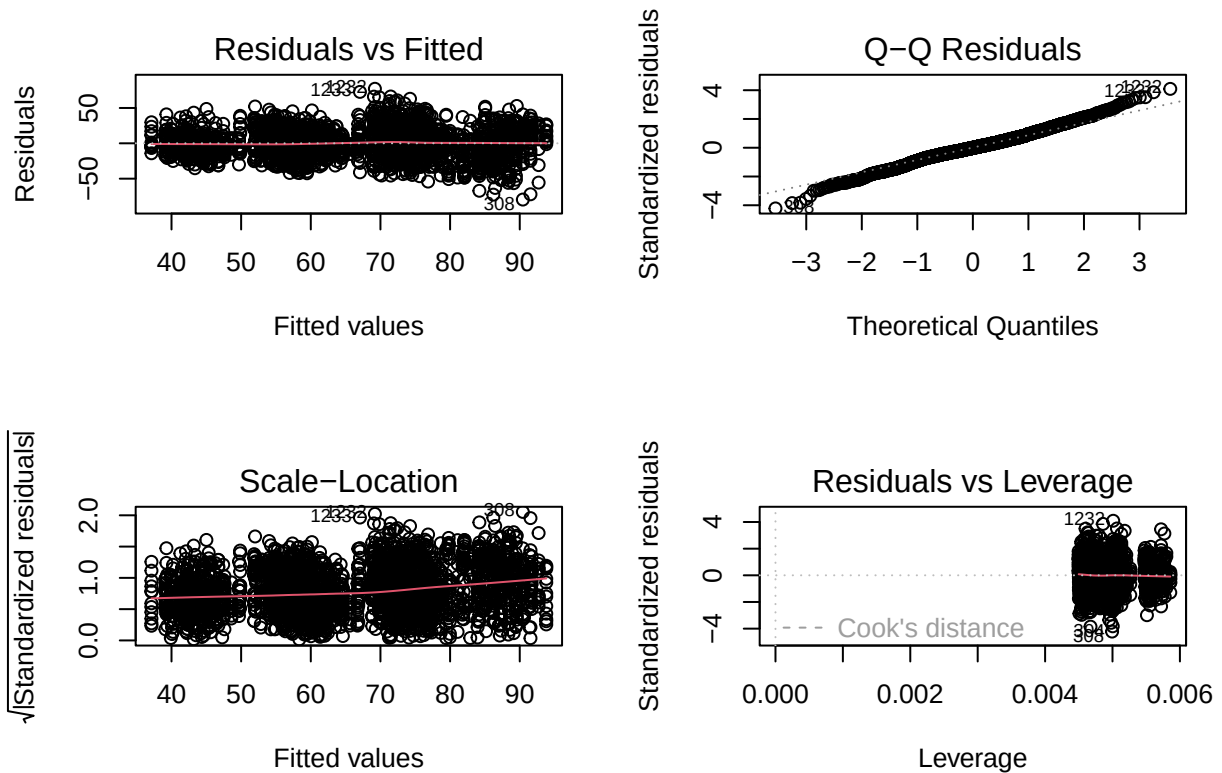
	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	72.3069	1.2999	55.624	< 2e-16 ***
session2	-0.3232	1.2009	-0.269	0.787869
session3	-1.0986	0.9248	-1.188	0.235003
session4	-3.1023	1.0837	-2.863	0.004233 **
activitystanding	-2.1447	0.8419	-2.547	0.010914 *
activitywalking	2.0898	1.0664	1.960	0.050146 .
day2	12.7357	0.7443	17.111	< 2e-16 ***
HF_typeHF_2	-29.9354	1.2712	-23.549	< 2e-16 ***
HF_typeHF_3	-12.9806	1.2799	-10.142	< 2e-16 ***
HF_typeHF_5	-9.9628	1.2860	-7.747	1.33e-14 ***
HF_typeHF_7	4.4112	1.2664	3.483	0.000503 ***
HF_typeHF_8	-26.9716	1.2748	-21.158	< 2e-16 ***
sexM	2.2649	0.7360	3.077	0.002111 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 18.9 on 2625 degrees of freedom
Multiple R-squared: 0.3705, Adjusted R-squared: 0.3677
F-statistic: 128.8 on 12 and 2625 DF, p-value: < 2.2e-16

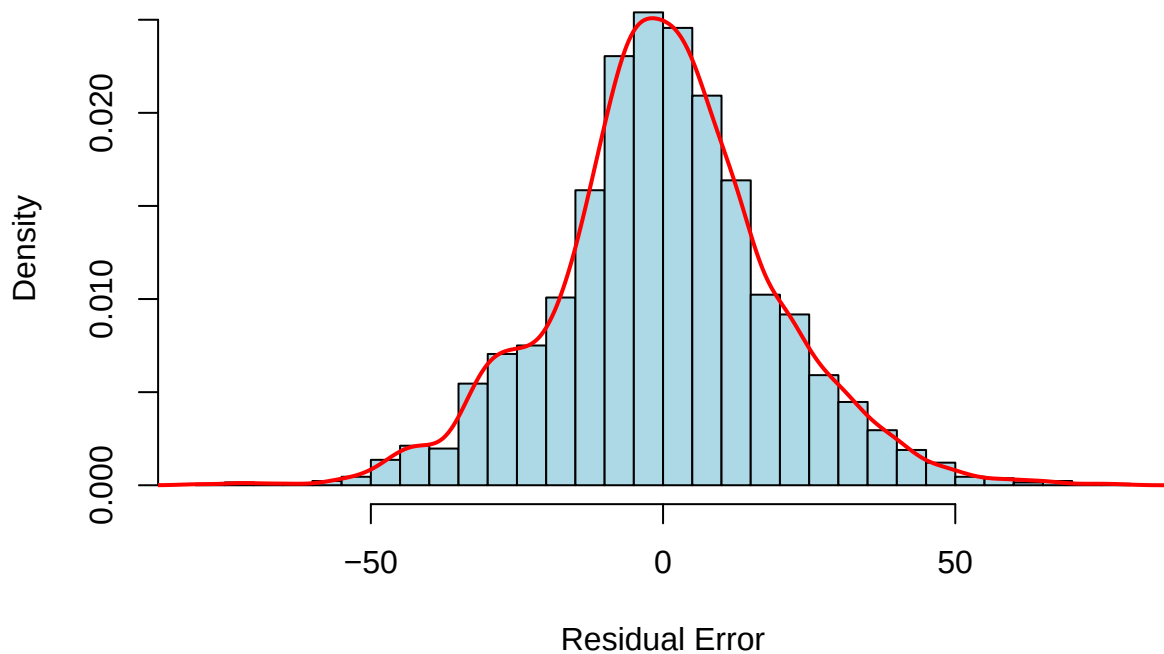
#Also based on the significance for each coefficients, we should include all of the predictors.

```
par(mfrow = c(2, 2))
plot(naive_lm)
```



```
hist(resid(naive_lm), breaks = 30, probability = TRUE, col = "lightblue", main = "Histogram of Residuals")
lines(density(resid(naive_lm)), col = "red", lwd = 2)
```

Histogram of Residuals



```
log_naive_lm <- lm(log(HF_value) ~ session + activity + day + HF_type + sex, data = data_clean)
summary(log_naive_lm)
```

Call:

```
lm(formula = log(HF_value) ~ session + activity + day + HF_type +
    sex, data = data_clean)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.09094	-0.14471	0.03708	0.20098	0.81742

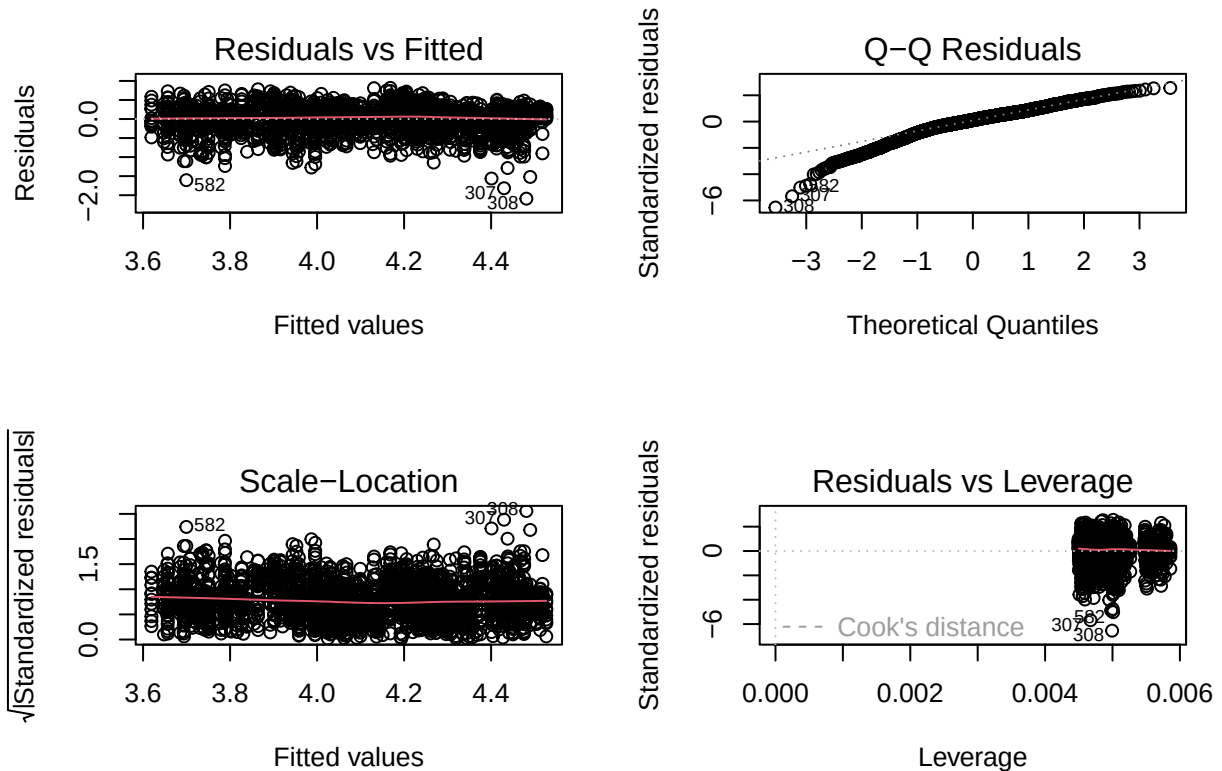
Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.2059986	0.0220531	190.721	< 2e-16 ***
session2	-0.0002083	0.0203732	-0.010	0.99184
session3	-0.0087793	0.0156898	-0.560	0.57583
session4	-0.0375880	0.0183842	-2.045	0.04100 *
activitystanding	-0.0374859	0.0142836	-2.624	0.00873 **
activitywalking	0.0138636	0.0180920	0.766	0.44358
day2	0.2073803	0.0126274	16.423	< 2e-16 ***
HF_typeHF_2	-0.5116352	0.0215655	-23.725	< 2e-16 ***
HF_typeHF_3	-0.1751289	0.0217134	-8.065	1.10e-15 ***
HF_typeHF_5	-0.1060988	0.0218168	-4.863	1.22e-06 ***
HF_typeHF_7	0.0626621	0.0214838	2.917	0.00357 **
HF_typeHF_8	-0.4176090	0.0216262	-19.310	< 2e-16 ***
sexM	0.0369273	0.0124864	2.957	0.00313 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3206 on 2625 degrees of freedom
Multiple R-squared: 0.3549, Adjusted R-squared: 0.352
F-statistic: 120.4 on 12 and 2625 DF, p-value: < 2.2e-16

```
par(mfrow = c(2, 2))
plot(log_naive_lm)
```



```
par(mfrow = c(2, 2), mar = c(4, 4, 3, 2), oma = c(5, 0, 3, 0))
#2 rows 2 columns
#oma = outer margins to have space for caption

#NAIVE (row 1)
# Plot 2: Normal Q-Q
plot(naive_lm, which = 2, main = "Model 0 (Raw): Normality (QQ)", sub = "", caption = "") #sub means do
# Plot 3: Scale-Location (Spread-Location)
plot(naive_lm, which = 3, main = "Model 0 (Raw): Scale-Location", sub = "", caption = "") #caption mean

#LOG (row 2)
# Plot 2: Normal Q-Q
plot(log_naive_lm, which = 2, main = "Model 1 (Log): Normality (QQ)", sub = "", caption = "")
# Plot 3: Scale-Location (Spread-Location)
plot(log_naive_lm, which = 3, main = "Model 1 (Log): Scale-Location", sub = "", caption = "")

# caption_text<-

# wrapped_caption <- paste(strwrap(caption_text, width = 150), collapse = "\n")

mtext("Diagnostic Comparison of Raw & Log Linear Models",
      side = 3, line = 0, outer = TRUE, cex = 1.2, font = 2)
```

Diagnostic Comparison of Raw & Log Linear Models

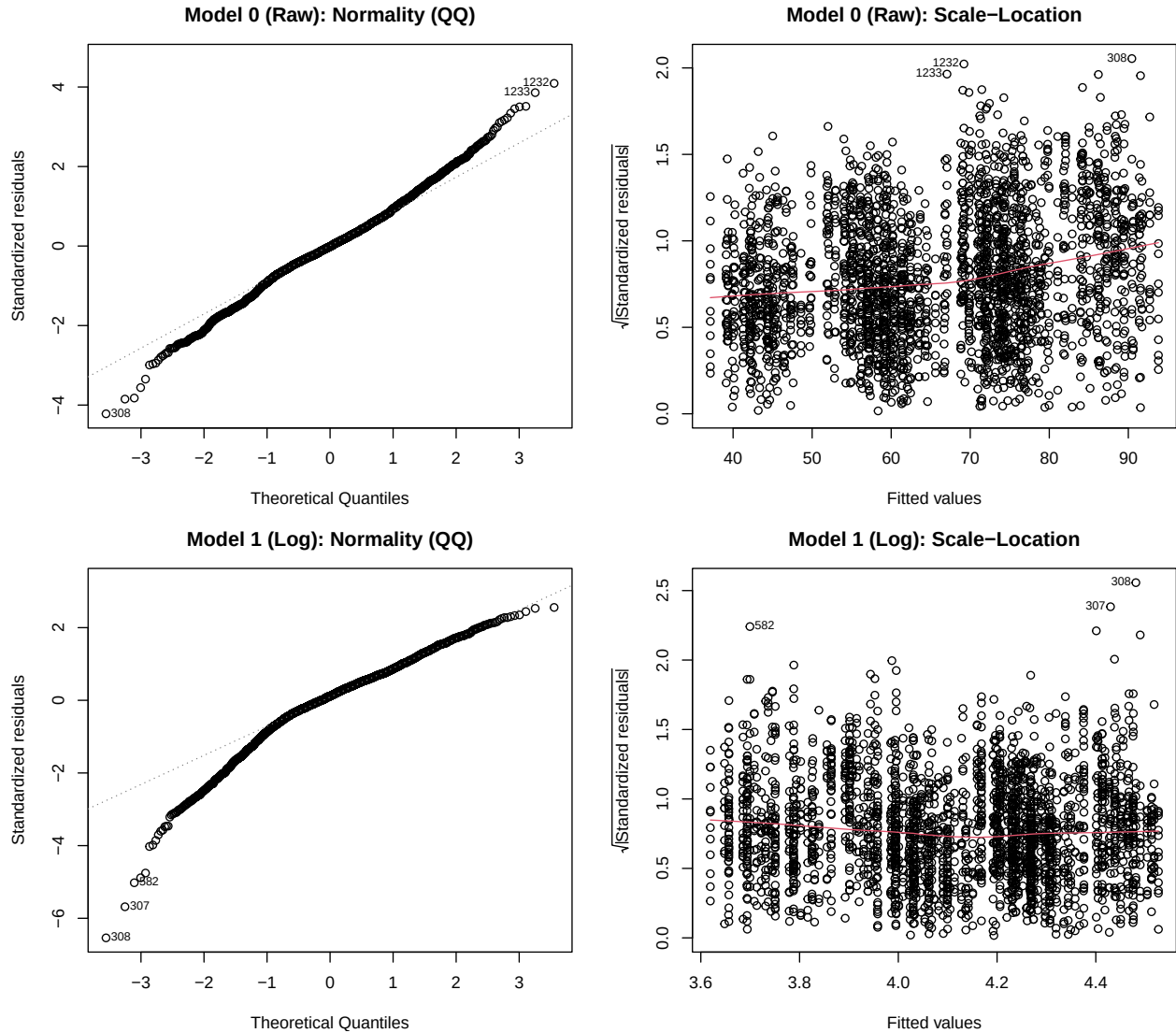


Figure 3: The QQ-plot of the Model 0 (the raw (untransformed) model) shows an approximately normal model with slightly heavy tails, but the Scale-Location plot shows heteroscedastic residuals, with variance increasing as fitted values increase. Model 1 (the log-transformed model) corrects the right heavy tail but the left tail deviates more, and the Scale-Location plot shows fairly homoscedastic residuals.

```
# mtext(wrapped_caption,
#       side = 1, line = 2, outer = TRUE, cex = 0.8)
```

As seen in Figure 3 and Table 1, applying a simple linear model (Model 0: $\text{HF_value} \sim \text{session} + \text{activity} + \text{day} + \text{HF_type} + \text{sex}$) shows that the residuals are approximately normal (with heavy tails), but heteroscedastic. Transforming the response variable (Model 1: $\log(\text{HF_value})$) resulted in homoscedasticity and corrected the heavy right tail, but introduced more left-skew in the residuals. We prioritize homoscedasticity and use Model 1 (the log model) going forward, to ensure reliable parameter estimation and significance testing.

```
fit_naive <- fitted(naive_lm)
res_naive <- residuals(naive_lm)

fit_log <- fitted(log_naive_lm)
```

```

res_log <- residuals(log_naive_lm)

# Calculate Correlation for Heteroscedasticity - closer to 0 means heteroscedastic
het_corr_naive <- cor(fit_naive, abs(res_naive))
het_corr_log <- cor(fit_log, abs(res_log))

#Calculate Correlation for Normality (Q-Q Correlation) - closer to 1 means more normal
qq_naive <- cor(sort(res_naive), qnorm(ppoints(length(res_naive))))
qq_log <- cor(sort(res_log), qnorm(ppoints(length(res_log))))

log_naive_comparison <- data.frame(
  Metric = c("Normality", "Heteroscedasticity"),
  Naive_Model = c(round(qq_naive, 4), round(het_corr_naive, 4)),
  Log_Model = c(round(qq_log, 4), round(het_corr_log, 4))
)

kable(log_naive_comparison,
      col.names = c("Metric", "Model 0 (Raw)", "Model 1 (Log)"),
      align = "lrr",
      booktabs = TRUE, #right aligns numbers
      caption = "\\footnotesize{Mathematical diagnostics comparison between Model 0 (raw) and Model 1 (Log)}",
      kable_styling(latex_options = "hold_position",
                    full_width = FALSE,
                    table.envir = "table") %>%
      add_header_above(c("Mathematical Diagnostic Comparison of Model 0 & Model 1" = 3),
                      bold = TRUE, align = "l") %>%
      row_spec(0, bold = TRUE)

```

Table 1: Mathematical diagnostics comparison between Model 0 (raw) and Model 1 (log-transformed model), comparing normality (correlation between actual and theoretically normal residuals, with values closer to 1 indicating normality) and heteroscedasticity (correlation between fitted values and residuals, with values closer to 0 indicating heteroscedasticity). While Model 1 shows slightly worse normality, it is significantly closer to homoscedasticity.

Mathematical Diagnostic Comparison of Model 0 & Model 1		
Metric	Model 0 (Raw)	Model 1 (Log)
Normality	0.9966	0.9768
Heteroscedasticity	0.2274	-0.0524

6.2 Fixed Effects

```

mdl1_log <- gls(
  log(HF_value) ~ day + session + activity + HF_type + sex,
  data = data_clean,
  method = "ML"
)
mdl1_log_no_day <- gls(
  log(HF_value) ~ session + (activity) + (HF_type) + (sex),
  data = data_clean,
  method = "ML"
)
mdl1_log_no_session <- gls(
  log(HF_value) ~ (day) + (activity) + (HF_type) + (sex),

```



```

data = data_clean,
method = "ML"
)
mdl1_log_no_activity <- gls(
  log(HF_value) ~ (day) + (session) + (HF_type) + (sex),
  data = data_clean,
  method = "ML"
)
mdl1_log_no_sensor <- gls(
  log(HF_value) ~ (day) + (session) + (activity) + (sex),
  data = data_clean,
  method = "ML"
)
mdl1_log_no_sex <- gls(
  log(HF_value) ~ (day) + (session) + (activity) + (HF_type),
  data = data_clean,
  method = "ML"
)

#helper function to get anova outputs
get_baseline_metrics <- function(orig, name) {
  data.frame(
    Model = name,
    Df = attr(logLik(orig), "df"),
    AIC = round(AIC(orig)),
    BIC = round(BIC(orig)),
    LogLik = round(as.numeric(logLik(orig))),
    stringsAsFactors = FALSE
  )
}

#note that type=marginal means it is an F test!!
get_anova_outputs <- function(orig, new, name) {
  res <- anova(orig, new) #, type = "marginal" doesnt matter if pass 2 models; uses chi-squared
  data.frame(
    Model = name,
    Df = attr(logLik(new), "df"),
    AIC = round(res$AIC[2]),
    BIC = round(res$BIC[2]),
    LogLik = round(res$logLik[2]),
    LRT_Stat = round(res$L.Ratio[2], 2),
    p_value = res$p-value[2]
  )
}

anova_summary_manual_mdl1 <- bind_rows(
  get_baseline_metrics(mdl1_log, "Model 1 (Full Model)"),
  get_anova_outputs(mdl1_log, mdl1_log_no_day, "Drop Day"),
  get_anova_outputs(mdl1_log, mdl1_log_no_session, "Drop Session"),
  get_anova_outputs(mdl1_log, mdl1_log_no_activity, "Drop Activity"),
  get_anova_outputs(mdl1_log, mdl1_log_no_sensor, "Drop Sensor"),
  get_anova_outputs(mdl1_log, mdl1_log_no_sex, "Drop Sex")
)

```

```
print(anova_summary_manual_md11)
```

	Model	Df	AIC	BIC	LogLik	LRT_Stat	p_value
1	Model 1 (Full Model)	14	1499	1581	-736	NA	NA
2	Drop Day	13	1755	1832	-865	258.01	4.648055e-58
3	Drop Session	11	1498	1563	-738	5.04	1.691164e-01
4	Drop Activity	12	1506	1576	-741	10.78	4.566764e-03
5	Drop Sensor	9	2438	2491	-1210	948.97	6.687274e-203
6	Drop Sex	13	1506	1582	-740	8.77	3.053999e-03

```
anova_summary_manual_md11_cleaned <- anova_summary_manual_md11 %>%
  mutate(
    LRT_Stat = ifelse(is.na(LRT_Stat), "--", as.character(LRT_Stat)),
    p_value = ifelse(is.na(p_value), "--",
      # eps defines the threshold for "<"
      # digits defines the rounding for values above that threshold
      format.pval(p_value, eps = 0.0001, digits = 4))
  )
```

```
col_labels <- c("Model Version",
  "\\shortstack{Degrees of\\\\Freedom}",
  "AIC",
  "BIC",
  "Log-Likelihood",
  "\\shortstack{Likelihood Ratio\\\\Test Stat}",
  "P-Value")
```

```
kable(anova_summary_manual_md11_cleaned,
  format="latex",
  col.names = col_labels,
  align = "lrrrrrr",
  booktabs = TRUE, #, #right aligns numbers
  escape = FALSE,
  caption = "\\footnotesize{Model selection of fixed effects via backwards elimination. Model 1 (th
  kable_styling(latex_options = c("scale_down", "hold_position"),
    full_width = FALSE,
    table.envir = "table") %>%
  add_header_above(c("Backwards Selection of Fixed Effects: Likelihood Ratio Test Summary" = 7),
    bold = TRUE, align = "l") %>%
  row_spec(0, bold = TRUE) %>% #highlight main row
  row_spec(1, italic = TRUE)
```

As seen in Table 2, although Model 1 includes all fixed effects ($\log(\text{HF_value}) \sim \text{session} + \text{activity} + \text{day} + \text{sensor} + \text{sex}$), a backwards selection process was conducted to ensure parsimony. We evaluated each variable by comparing the full model to nested models where one term was removed, using the Likelihood Ratio Test (LRT) and changes in information criteria (AIC and BIC).

At first glance, all variables are statistically significant except for the model dropping the session variable ($\log(\text{HF_value}) \sim \text{activity} + \text{day} + \text{sensor} + \text{sex}$), as the p-value from the likelihood ratio test of this model versus the full model is not below 0.05 ($p = 0.1691$). Additionally, although the models that drop activity and drop sex are technically statistically significant (p -values < 0.01), these models show a unique characteristic of having slightly higher AICs than the full model, but lower or similar BICs. This discrepancy implies that although these variables improve prediction, the improvement may not be large enough to justify including

Table 2: Model selection of fixed effects via backwards elimination. Model 1 (the 'Full Model') includes all predictors. Each subsequent row represents a model where only one term was dropped from Model 1. The LRT Statistic and P-Value are derived from Likelihood Ratio Tests comparing the reduced model to Model 1. Note all models were fit using Maximum Likelihood (ML), to compare across models.

Backwards Selection of Fixed Effects: Likelihood Ratio Test Summary						
Model Version	Degrees of Freedom	AIC	BIC	Log-Likelihood	Likelihood Ratio Test Stat	P-Value
<i>Model 1 (Full Model)</i>	14	1499	1581	-736	–	–
Drop Day	13	1755	1832	-865	258.01	< 1e-04
Drop Session	11	1498	1563	-738	5.04	0.169116
Drop Activity	12	1506	1576	-741	10.78	0.004567
Drop Sensor	9	2438	2491	-1210	948.97	< 1e-04
Drop Sex	13	1506	1582	-740	8.77	0.003054

the variables. In this case (and similar cases further in the analysis), the variables are included but will be re-examined in final model pruning.

As such, the final fixed effect model was Model 2 ($\log(\text{HF_value}) \sim \text{activity} + \text{day} + \text{sensor} + \text{sex}$).

6.3 Interaction Terms

We further examined fixed effects by examining the value of adding all possible interaction terms to the model, as shown in Table 3. Testing adding each interaction term against against Model 2 ($\text{HF_value} \sim \text{activity} + \text{day} + \text{sensor} + \text{sex}$), we saw that adding the following interaction terms results in models with worse (or similar) AIC & BIC than the current model without interaction terms, and thus did not add them to the model: Activity & Day, Activity & Sex, Session & Sex, Day & Sex, Day & Session.

Technically, adding the interaction of Day & Session results in a p-value of 0.0118, below 0.05. However, this interaction increased the BIC and barely decreases the AIC, so we chose not to include it in the model. Note that the interaction of Activity & Session was not tested - because there is no data for certain interactions (ex. walking & session 2, walking & session 4), the model cannot run.

The interaction of Sensor with the other possible variables (Day, Activity, Sex, and Session) all have low p-values (<0.001) and result in decreased AIC, although two interaction terms (Sensor and Sex, and Sensor and Session) result in increased BIC. Table 2 showed a similar story - both Session and Sex decreased AIC slightly, but increased BIC. Similarly, we include these interaction terms for now and will prune the model at the end.

```
#sex*session
mdl2_log_no_session_interact_sex_session<- gls(
  log(HF_value) ~ (day) + (activity) + (HF_type) + (sex) + (sex * session) ,
  data = data_clean,
  method = "ML"
)

#day*sex
dl2_log_no_session_interact_sex_day<- gls(
  log(HF_value) ~ (day) + (activity) + (HF_type) + (sex) + (sex * day) ,
  data = data_clean,
  method = "ML"
)

#day*activity
```

```

dl2_log_no_session_interact_day_activity<- gls(
  log(HF_value) ~ (day) + (activity) + (HF_type) + (sex) + (activity * day) ,
  data = data_clean,
  method = "ML"
)

#HF_type*day
mdl2_log_no_session_interact_sensor_day <- gls(
  log(HF_value) ~ (day) + (activity) + (HF_type) + (sex) + (HF_type * day) ,
  data = data_clean,
  method = "ML"
)

#HF_type*activity
mdl2_log_no_session_interact_sensor_activity <- gls(
  log(HF_value) ~ (day) + (activity) + (HF_type) + (sex) + (HF_type * activity) ,
  data = data_clean,
  method = "ML"
)

#HF_type*sex
mdl2_log_no_session_interact_sensor_sex <- gls(
  log(HF_value) ~ (day) + (activity) + (HF_type) + (sex) + (HF_type * sex) ,
  data = data_clean,
  method = "ML"
)

#HF_type*session
mdl2_log_no_session_interact_sensor_session <- gls(
  log(HF_value) ~ (day) + (activity) + (HF_type) + (sex) + (HF_type * session) ,
  data = data_clean,
  method = "ML"
)

#activity*sex
mdl2_log_no_session_interact_activity_sex <- gls(
  log(HF_value) ~ (day) + (activity) + (HF_type) + (sex) + (activity * sex) ,
  data = data_clean,
  method = "ML"
)

#sex*session
mdl2_log_no_session_interact_session_sex <- gls(
  log(HF_value) ~ (day) + (activity) + (HF_type) + (sex) + (session * sex) ,
  data = data_clean,
  method = "ML"
)

#day*session
mdl2_log_no_session_interact_session_day <- gls(

```

```

log(HF_value) ~ (day) + (activity) + (HF_type) + (sex) + (session * day) ,
data = data_clean,
method = "ML"
)

#session*activity
#error: Error in `glsEstimate()`! computed "gls" fit is singular, rank 18
#we are missing some data (NAs) - ex walking for session 2 & 4, likely the reason?
# mdl2_log_no_session_interact_session_activity <- gls(
#   log(HF_value) ~ (day) + (activity) + (HF_type) + (sex) + (session * activity) ,
#   data = data_clean,
#   method = "ML"
# )

anova_summary_manual_md11_nosession_interact1 <- bind_rows(
  get_baseline_metrics(md11_log_no_session, "Model 2 (No Interaction Terms)"),
  get_anova_outputs(md11_log_no_session, mdl2_log_no_session_interact_sensor_day, "Interact Sensor and Day"),
  get_anova_outputs(md11_log_no_session, mdl2_log_no_session_interact_sensor_activity, "Interact Sensor and Activity"),
  get_anova_outputs(md11_log_no_session, mdl2_log_no_session_interact_sensor_sex, "Interact Sensor and Sex"),
  get_anova_outputs(md11_log_no_session, mdl2_log_no_session_interact_sensor_session, "Interact Sensor and Session"),
  get_anova_outputs(md11_log_no_session, mdl2_log_no_session_interact_day_activity, "Interact Activity and Day"),
  get_anova_outputs(md11_log_no_session, mdl2_log_no_session_interact_activity_sex, "Interact Activity and Sex"),
  get_anova_outputs(md11_log_no_session, mdl2_log_no_session_interact_sex_session, "Interact Session and Sex"),
  get_anova_outputs(md11_log_no_session, mdl2_log_no_session_interact_sex_day, "Interact Day and Sex"),
  get_anova_outputs(md11_log_no_session, mdl2_log_no_session_interact_session_day, "Interact Day and Session")
)

print(anova_summary_manual_md11_nosession_interact1)

```

	Model	Df	AIC	BIC	LogLik	LRT_Stat	p_value
1	Model 2 (No Interaction Terms)	11	1498	1563	-738	NA	NA
2	Interact Sensor and Day	16	1311	1405	-640	196.89	1.312356e-40
3	Interact Sensor and Activity	21	1340	1463	-649	178.67	4.434286e-33
4	Interact Sensor and Sex	16	1482	1576	-725	25.78	9.824603e-05
5	Interact Sensor and Session	29	1455	1625	-698	79.58	1.013875e-09
6	Interact Activity and Day	13	1502	1578	-738	0.38	8.285852e-01
7	Interact Activity and Sex	13	1499	1575	-736	3.43	1.799382e-01
8	Interact Session and Sex	17	1504	1604	-735	6.53	3.665424e-01
9	Interact Day and Sex	12	1500	1570	-738	0.63	4.266220e-01
10	Interact Day and Session	17	1494	1594	-730	16.40	1.177444e-02

```

anova_summary_manual_md11_nosession_interact1_cleaned <- anova_summary_manual_md11_nosession_interact1 %>%
  mutate(
    LRT_Stat = ifelse(is.na(LRT_Stat), "--", as.character(LRT_Stat)),
    p_value = ifelse(is.na(p_value), "--",
      # eps defines the threshold for "<"
      # digits defines the rounding for values above that threshold
      format.pval(p_value, eps = 0.0001, digits = 4))
  )

print("hi")

```

```
[1] "hi"
```

```
print(ncol(anova_summary_manual_md11_nosession_interact1_cleaned))

[1] 7

# Clean the Data Frame

col_labels <- c("Model Version",
                "\\shortstack{Degrees of\\\\Freedom}",
                "AIC",
                "BIC",
                "Log-Likelihood",
                "\\shortstack{Likelihood Ratio\\\\Test Stat}",
                "P-Value")

kable(anova_summary_manual_md11_nosession_interact1_cleaned,
      format="latex",
      col.names = col_labels,
      align = "lrrrrrr",
      booktabs = TRUE, #, #right aligns numbers
      row.names = FALSE,
      escape = FALSE,
      caption = "\\footnotesize{The top row (Model 2) is the row without any interaction terms. Each subsequent row represents a model where only one interaction term was added to Model 2. The LRT Statistic and P-Value are derived from Likelihood Ratio Tests comparing the new model to Model 2 via ANOVA. Note all models were fit using Maximum Likelihood (ML), to compare across models.}",
      kable_styling(latex_options = c("scale_down", "hold_position"),
                    full_width = FALSE,
                    table.envir = "table") %>%
      add_header_above(c("Interaction Terms: Likelihood Ratio Test Summary" = 7),
                       bold = TRUE, align = "l") %>%
      row_spec(0, bold = TRUE) %>% #highlight main row
      row_spec(1, italic = TRUE)
```

Table 3: The top row (Model 2) is the row without any interaction terms. Each subsequent row represents a model where only one interaction term was added to Model 2. The LRT Statistic and P-Value are derived from Likelihood Ratio Tests comparing the new model to Model 2 via ANOVA. Note all models were fit using Maximum Likelihood (ML), to compare across models.

Interaction Terms: Likelihood Ratio Test Summary						
Model Version	Degrees of Freedom	AIC	BIC	Log-Likelihood	Likelihood Ratio Test Stat	P-Value
<i>Model 2 (No Interaction Terms)</i>	11	1498	1563	-738	–	–
Interact Sensor and Day	16	1311	1405	-640	196.89	< 1e-04
Interact Sensor and Activity	21	1340	1463	-649	178.67	< 1e-04
Interact Sensor and Sex	16	1482	1576	-725	25.78	< 1e-04
Interact Sensor and Session	29	1455	1625	-698	79.58	< 1e-04
Interact Activity and Day	13	1502	1578	-738	0.38	0.82859
Interact Activity and Sex	13	1499	1575	-736	3.43	0.17994
Interact Session and Sex	17	1504	1604	-735	6.53	0.36654
Interact Day and Sex	12	1500	1570	-738	0.63	0.42662
Interact Day and Session	17	1494	1594	-730	16.4	0.01177

This results in Model 3: $\log(\text{HF_value}) \sim \text{activity} + \text{day} + \text{sensor} + \text{sex} + \text{sensor} * \text{activity} + \text{sensor} * \text{day} + \text{sensor} * \text{sex} + \text{sensor} * \text{session} + \text{session}$. Note that session is added back into the model automatically, as the interaction of sensor and session is included, due to the principle of marginality. Table 4 shows the results of checking if removing any of the four interaction terms improves Model 3. This showed a similar story to the individual tests of adding interaction terms—the inclusion of the interactions of Sensor and Sex, and Sensor and Session, slightly decrease AIC but increase BIC.

```

#HF_type*day, HF_type * activity, HF_type * sex, HF_type * session, day * session
mdl3 <- gls(
  log(HF_value) ~ (day) + (activity) + (HF_type) + (sex) + (session) + (HF_type * day) + (HF_type *
  data = data_clean,
  method = "ML"
)

#remove hf_type * day
mdl3a <- gls(
  log(HF_value) ~ (day) + (activity) + (HF_type) + (sex) +(session) +(HF_type * activity) + (HF_type
  data = data_clean,
  method = "ML"
)

#remove hf_type * activity
mdl3b <- gls(
  log(HF_value) ~ (day) + (activity) + (HF_type) + (sex) +(session) + (HF_type * day) + (HF_type *
  data = data_clean,
  method = "ML"
)

#remove hf_type * sex
mdl3c <- gls(
  log(HF_value) ~ (day) + (activity) + (HF_type) + (sex) +(session) + (HF_type * day) + (HF_type *
  data = data_clean,
  method = "ML"
)

#remove hf_type * session, and session
mdl3d <- gls(
  log(HF_value) ~ (day) + (activity) + (HF_type) + (sex) + (HF_type * day) + (HF_type * activity) +
  data = data_clean,
  method = "ML"
)

anova_summary_manual_md11_nosession_interact2 <- bind_rows(
  get_baseline_metrics(mdl3, "Model 3 (Four Interaction Terms)"),
  get_anova_outputs(mdl3, mdl3a, "Drop Sensor and Day"),
  get_anova_outputs(mdl3, mdl3b, "Drop Sensor and Activity"),
  get_anova_outputs(mdl3, mdl3c, "Drop Sensor and Sex"),
  get_anova_outputs(mdl3, mdl3d, "Drop Sensor and Session")
)
print(anova_summary_manual_md11_nosession_interact2)

```

	Model	Df	AIC	BIC	LogLik	LRT_Stat	p_value
1	Model 3 (Four Interaction Terms)	49	1107	1395	-505	NA	NA
2	Drop Sensor and Day	44	1314	1573	-613	216.74	7.420141e-45
3	Drop Sensor and Activity	39	1248	1477	-585	160.62	2.409731e-29
4	Drop Sensor and Sex	44	1127	1386	-520	30.02	1.460823e-05
5	Drop Sensor and Session	31	1113	1295	-525	41.49	1.303934e-03

```

anova_summary_manual_md11_nosession_interact2_cleaned <- anova_summary_manual_md11_nosession_interact2
mutate(
  LRT_Stat = ifelse(is.na(LRT_Stat), "--", as.character(LRT_Stat)),

```

```

p_value = ifelse(is.na(p_value), "--",
                 # eps defines the threshold for "<"
                 # digits defines the rounding for values above that threshold
                 format.pval(p_value, eps = 0.0001, digits = 4))
)

col_labels <- c("Model Version",
               "\\shortstack{Degrees of\\\\Freedom}",
               "AIC",
               "BIC",
               "Log-Likelihood",
               "\\shortstack{Likelihood Ratio\\\\Test Stat}",
               "P-Value")

kable(anova_summary_manual_md11_nosession_interact2_cleaned,
      format="latex",
      col.names = col_labels,
      align = "lrrrrrr",
      booktabs = TRUE, #, #right aligns numbers
      escape = FALSE,
      caption = "\\footnotesize{Model selection of interaction terms via backwards elimination. Model 3
kable_styling(latex_options = c("scale_down", "hold_position"),
              full_width = FALSE,
              table.envir = "table") %>%
add_header_above(c("Backwards Selection of Interaction Terms: Likelihood Ratio Test Summary" = 7),
                 bold = TRUE, align = "l") %>%
row_spec(0, bold = TRUE) %>% #highlight main row
row_spec(1, italic = TRUE)

```

Table 4: Model selection of interaction terms via backwards elimination. Model 3 includes the four interaction terms previously noted. Each subsequent row represents a model where only one interaction term was dropped from Model 3. Note that for the last row, Drop Sensor and Session, the fixed effect Session is dropped as well (Session was only included in Model 3 because the interaction of Sensor and Session was added). The LRT Statistic and P-Value are derived from Likelihood Ratio Tests comparing the reduced model to Model 3. Note all models were fit using Maximum Likelihood (ML), to compare across models.

Backwards Selection of Interaction Terms: Likelihood Ratio Test Summary						
Model Version	Degrees of Freedom	AIC	BIC	Log-Likelihood	Likelihood Ratio Test Stat	P-Value
<i>Model 3 (Four Interaction Terms)</i>	49	1107	1395	-505	–	–
Drop Sensor and Day	44	1314	1573	-613	216.74	< 1e-04
Drop Sensor and Activity	39	1248	1477	-585	160.62	< 1e-04
Drop Sensor and Sex	44	1127	1386	-520	30.02	< 1e-04
Drop Sensor and Session	31	1113	1295	-525	41.49	0.001304

6.4 Random Intercepts

6.4.1 TO DO: ADD JUSTIFICATION FOR LMM

```

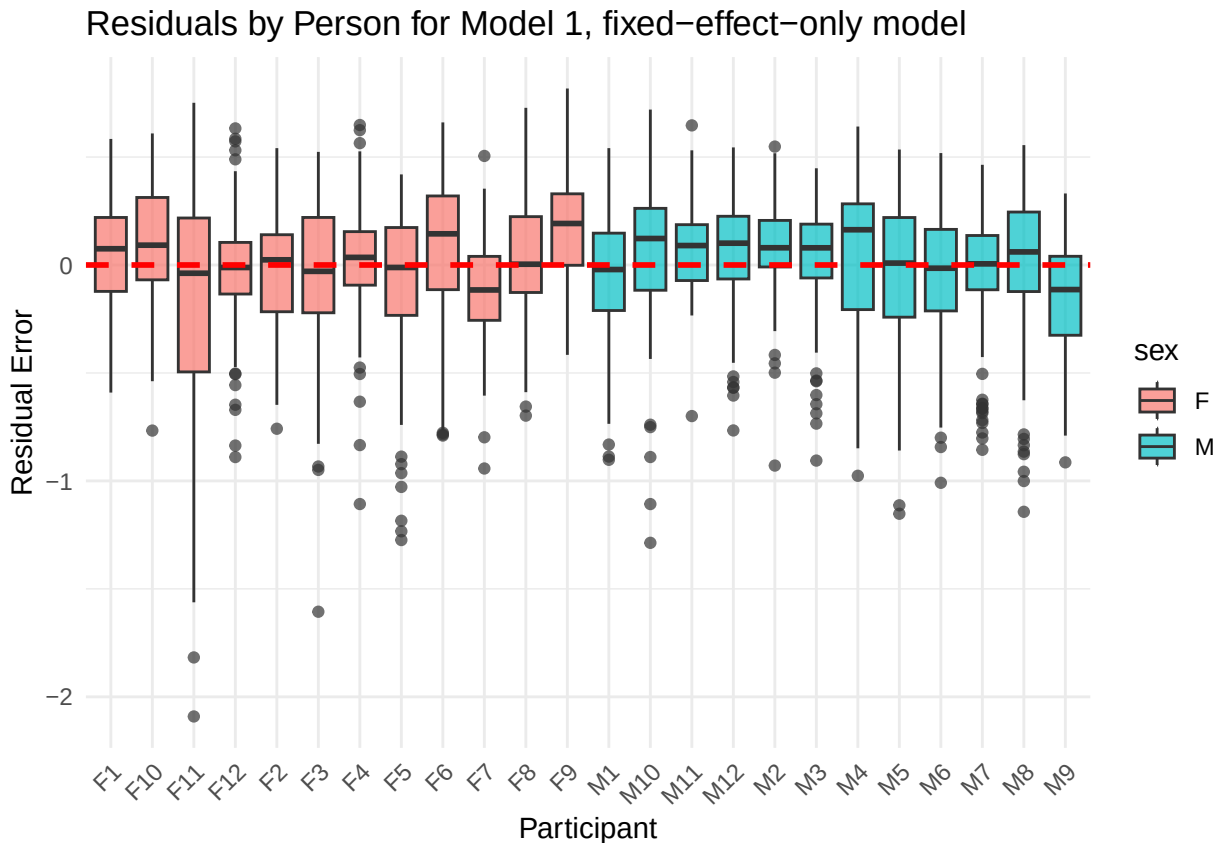
# residuals added to dataframe
data_clean$residuals_model1 <- resid(log_naive_lm)

ggplot(data_clean, aes(x = person, y = residuals_model1, fill = sex)) +

```



```
geom_boxplot(alpha = 0.7) +
# red line: perfect independent residuals should be centered at 0
geom_hline(yintercept = 0, color = "red", linetype = "dashed", size = 1) +
theme_minimal() +
theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
labs(
  title = "Residuals by Person for Model 1, fixed-effect-only model",
  x = "Participant",
  y = "Residual Error"
)
```

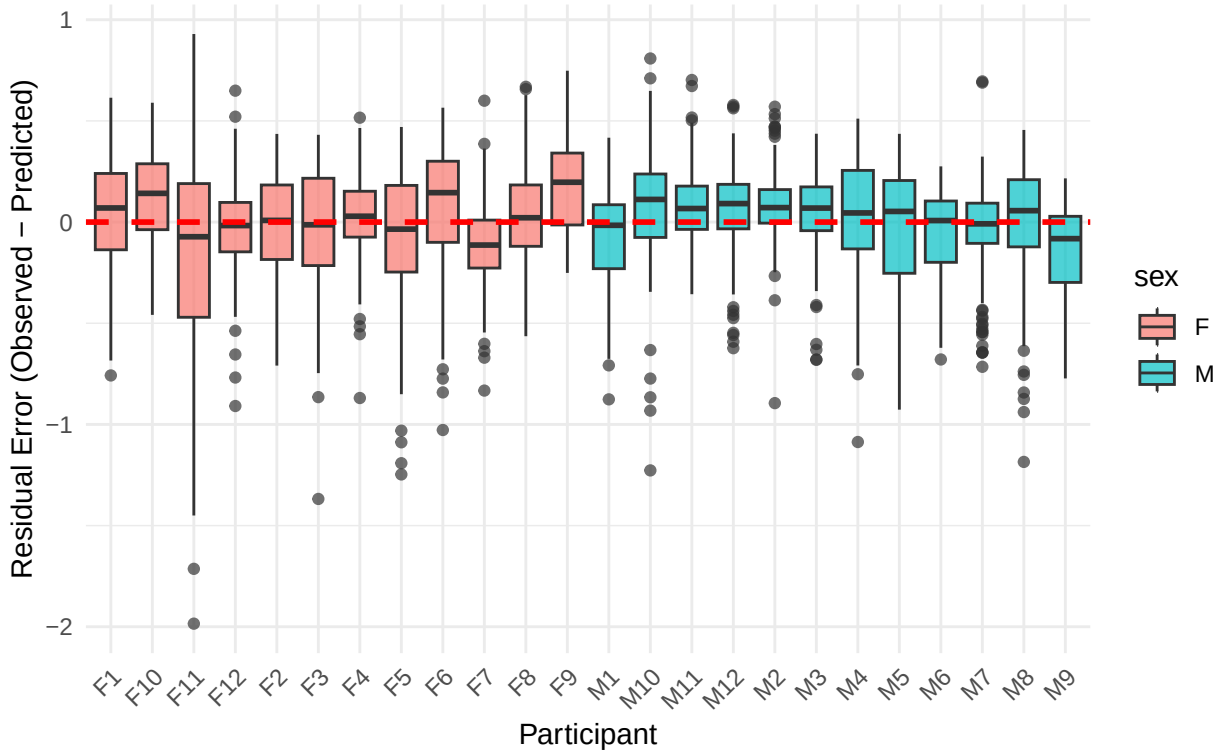


```
data_clean$residuals_model3 <- resid mdl3)

ggplot(data_clean, aes(x = person, y = residuals_model3, fill = sex)) +
  geom_boxplot(alpha = 0.7) +
  # red line: perfect independent residuals should be centered at 0
  geom_hline(yintercept = 0, color = "red", linetype = "dashed", size = 1) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(
    title = "Motivation for LMM: Residuals by Person (Model 3)",
    subtitle = "Systematic shifts away from zero prove that errors within an individual are correlated.",
    x = "Participant",
    y = "Residual Error (Observed - Predicted)"
  )
```

Motivation for LMM: Residuals by Person (Model 3)

Systematic shifts away from zero prove that errors within an individual are correlated.



To determine if there is inter-individual variability, we added a random intercept for each person, creating Model 4: $\log(\text{HF_value}) \sim \text{activity} + \text{day} + \text{sensor} + \text{sex} + \text{sensor} * \text{activity} + \text{sensor} * \text{day} + \text{sensor} * \text{sex} + \text{sensor} * \text{session} + \text{session}$. This assumes that all individuals have unique baselines for heat flow, regardless of if they are performing the same activity and/or are at the same temperature. Note the models in this section were run using REML to ensure estimates of variance components are not biased, unless explicitly stated otherwise.

```
mdl3_reml <- gls(log(HF_value) ~ (day + session + activity + sex) * HF_type,
  data = data_clean,
  method = "REML")

mdl4a_reml <- lme(fixed = log(HF_value) ~ (day + session + activity + sex) * HF_type,
  random = ~ 1 | person, #unique intercept for each person
  data = data_clean,
  method = "REML")

summary(mdl4a_reml)
```

Linear mixed-effects model fit by REML

```
Data: data_clean
      AIC      BIC    logLik
1207.812 1500.783 -553.9061
```

Random effects:

```
Formula: ~1 | person
      (Intercept) Residual
StdDev:  0.08333832 0.2844007
```

```

Fixed effects:  log(HF_value) ~ (day + session + activity + sex) * HF_type
               Value Std.Error DF   t-value p-value
(Intercept)      4.479602 0.04418779 2568 101.37648 0.0000
day2             -0.165397 0.02753267 2568  -6.00732 0.0000
session2         -0.068984 0.04462140 2568  -1.54599 0.1222
session3         -0.067929 0.03417981 2568  -1.98740 0.0470
session4         -0.201175 0.03989567 2568  -5.04252 0.0000
activitystanding -0.087103 0.03114911 2568  -2.79631 0.0052
activitywalking   0.038501 0.03934063 2568   0.97866 0.3278
sexM              0.004110 0.04357422   22   0.09433 0.9257
HF_typeHF_2      -0.711877 0.05217546 2568 -13.64390 0.0000
HF_typeHF_3      -0.641528 0.05277767 2568 -12.15529 0.0000
HF_typeHF_5      -0.410268 0.05298564 2568  -7.74300 0.0000
HF_typeHF_7      -0.156055 0.05196167 2568  -3.00328 0.0027
HF_typeHF_8      -0.892809 0.05235319 2568 -17.05358 0.0000
day2:HF_typeHF_2   0.492416 0.03875616 2568  12.70549 0.0000
day2:HF_typeHF_3   0.501060 0.03903910 2568  12.83484 0.0000
day2:HF_typeHF_5   0.427314 0.03924014 2568  10.88972 0.0000
day2:HF_typeHF_7   0.342664 0.03859369 2568   8.87875 0.0000
day2:HF_typeHF_8   0.479292 0.03886928 2568  12.33086 0.0000
session2:HF_typeHF_2 0.126035 0.06257996 2568   2.01399 0.0441
session3:HF_typeHF_2 0.101480 0.04812013 2568   2.10888 0.0351
session4:HF_typeHF_2 0.283330 0.05629551 2568   5.03290 0.0000
session2:HF_typeHF_3 0.046317 0.06300399 2568   0.73514 0.4623
session3:HF_typeHF_3 0.042632 0.04837859 2568   0.88121 0.3783
session4:HF_typeHF_3 0.184793 0.05677569 2568   3.25479 0.0011
session2:HF_typeHF_5 0.086032 0.06344480 2568   1.35602 0.1752
session3:HF_typeHF_5 0.061820 0.04864707 2568   1.27079 0.2039
session4:HF_typeHF_5 0.198626 0.05691482 2568   3.48988 0.0005
session2:HF_typeHF_7 0.067088 0.06246244 2568   1.07406 0.2829
session3:HF_typeHF_7 0.023360 0.04787113 2568   0.48799 0.6256
session4:HF_typeHF_7 0.090332 0.05608570 2568   1.61061 0.1074
session2:HF_typeHF_8 0.093123 0.06260596 2568   1.48745 0.1370
session3:HF_typeHF_8 0.128704 0.04834497 2568   2.66219 0.0078
session4:HF_typeHF_8 0.232480 0.05630772 2568   4.12874 0.0000
activitystanding:HF_typeHF_2 -0.091764 0.04379096 2568  -2.09549 0.0362
activitywalking:HF_typeHF_2 -0.377798 0.05545474 2568  -6.81272 0.0000
activitystanding:HF_typeHF_3 0.189686 0.04412383 2568   4.29895 0.0000
activitywalking:HF_typeHF_3 0.232009 0.05577278 2568   4.15989 0.0000
activitystanding:HF_typeHF_5 0.062717 0.04433521 2568   1.41462 0.1573
activitywalking:HF_typeHF_5 -0.038941 0.05607659 2568  -0.69442 0.4875
activitystanding:HF_typeHF_7 -0.043330 0.04362560 2568  -0.99323 0.3207
activitywalking:HF_typeHF_7 -0.120250 0.05517846 2568  -2.17930 0.0294
activitystanding:HF_typeHF_8 0.182402 0.04394940 2568   4.15027 0.0000
activitywalking:HF_typeHF_8 0.166013 0.05562698 2568   2.98440 0.0029
sexM:HF_typeHF_2   -0.074248 0.03833085 2568  -1.93702 0.0529
sexM:HF_typeHF_3    0.083189 0.03859034 2568   2.15569 0.0312
sexM:HF_typeHF_5    0.013351 0.03881501 2568   0.34396 0.7309
sexM:HF_typeHF_7    0.113521 0.03815055 2568   2.97562 0.0030
sexM:HF_typeHF_8    0.074246 0.03843794 2568   1.93159 0.0535
Correlation:
(Intr) day2   sessn2 sessn3 sessn4 actvtys actvtyw
day2          -0.336
session2      -0.453  0.108

```

session3	-0.381	-0.009	0.379				
session4	-0.415	-0.006	0.410	0.426			
activitystanding	-0.377	0.088	0.129	-0.004	0.000		
activitywalking	-0.405	0.033	0.302	-0.004	0.281	0.398	
sexM	-0.485	-0.018	0.006	0.002	0.004	-0.006	0.001
HF_typeHF_2	-0.596	0.284	0.384	0.323	0.352	0.320	0.343
HF_typeHF_3	-0.589	0.282	0.379	0.319	0.347	0.316	0.339
HF_typeHF_5	-0.587	0.280	0.378	0.318	0.346	0.315	0.337
HF_typeHF_7	-0.598	0.286	0.385	0.324	0.353	0.321	0.344
HF_typeHF_8	-0.594	0.283	0.382	0.322	0.350	0.318	0.342
day2:HF_typeHF_2	0.238	-0.710	-0.077	0.006	0.004	-0.062	-0.023
day2:HF_typeHF_3	0.236	-0.706	-0.076	0.006	0.004	-0.062	-0.023
day2:HF_typeHF_5	0.236	-0.702	-0.076	0.006	0.004	-0.061	-0.023
day2:HF_typeHF_7	0.240	-0.713	-0.077	0.007	0.004	-0.062	-0.023
day2:HF_typeHF_8	0.238	-0.708	-0.077	0.006	0.004	-0.062	-0.023
session2:HF_typeHF_2	0.323	-0.077	-0.713	-0.271	-0.293	-0.092	-0.215
session3:HF_typeHF_2	0.271	0.007	-0.270	-0.710	-0.303	0.003	0.003
session4:HF_typeHF_2	0.294	0.004	-0.291	-0.302	-0.709	0.000	-0.199
session2:HF_typeHF_3	0.321	-0.077	-0.708	-0.269	-0.291	-0.091	-0.214
session3:HF_typeHF_3	0.269	0.007	-0.268	-0.707	-0.301	0.003	0.003
session4:HF_typeHF_3	0.292	0.004	-0.288	-0.299	-0.703	0.000	-0.197
session2:HF_typeHF_5	0.319	-0.076	-0.703	-0.267	-0.289	-0.091	-0.213
session3:HF_typeHF_5	0.268	0.007	-0.267	-0.703	-0.299	0.003	0.003
session4:HF_typeHF_5	0.291	0.004	-0.288	-0.299	-0.701	0.000	-0.197
session2:HF_typeHF_7	0.324	-0.077	-0.714	-0.271	-0.293	-0.092	-0.216
session3:HF_typeHF_7	0.272	0.007	-0.271	-0.714	-0.304	0.003	0.003
session4:HF_typeHF_7	0.295	0.004	-0.292	-0.303	-0.711	0.000	-0.200
session2:HF_typeHF_8	0.323	-0.077	-0.713	-0.270	-0.292	-0.092	-0.215
session3:HF_typeHF_8	0.270	0.007	-0.268	-0.707	-0.301	0.003	0.003
session4:HF_typeHF_8	0.294	0.004	-0.291	-0.302	-0.709	0.000	-0.199
activitystanding:HF_typeHF_2	0.268	-0.062	-0.092	0.003	0.000	-0.711	-0.283
activitywalking:HF_typeHF_2	0.287	-0.023	-0.214	0.003	-0.199	-0.282	-0.709
activitystanding:HF_typeHF_3	0.266	-0.062	-0.091	0.003	0.000	-0.706	-0.281
activitywalking:HF_typeHF_3	0.285	-0.023	-0.213	0.003	-0.198	-0.281	-0.705
activitystanding:HF_typeHF_5	0.265	-0.062	-0.091	0.003	0.000	-0.703	-0.280
activitywalking:HF_typeHF_5	0.284	-0.023	-0.212	0.003	-0.197	-0.279	-0.702
activitystanding:HF_typeHF_7	0.269	-0.063	-0.092	0.003	0.000	-0.714	-0.284
activitywalking:HF_typeHF_7	0.289	-0.023	-0.215	0.003	-0.200	-0.284	-0.713
activitystanding:HF_typeHF_8	0.267	-0.062	-0.091	0.003	0.000	-0.709	-0.282
activitywalking:HF_typeHF_8	0.286	-0.023	-0.214	0.003	-0.199	-0.281	-0.707
sexM:HF_typeHF_2	0.209	0.020	-0.007	-0.002	-0.005	0.007	-0.001
sexM:HF_typeHF_3	0.208	0.020	-0.006	-0.002	-0.005	0.007	-0.001
sexM:HF_typeHF_5	0.207	0.020	-0.007	-0.002	-0.005	0.007	-0.001
sexM:HF_typeHF_7	0.210	0.020	-0.007	-0.002	-0.005	0.007	-0.001
sexM:HF_typeHF_8	0.209	0.020	-0.007	-0.002	-0.005	0.007	-0.001
	sexM	HF_HF_2	HF_HF_3	HF_HF_5	HF_HF_7	HF_HF_8	
day2							
session2							
session3							
session4							
activitystanding							
activitywalking							
sexM							
HF_typeHF_2	0.156						

HF_typeHF_3	0.154	0.499				
HF_typeHF_5	0.154	0.497	0.491			
HF_typeHF_7	0.157	0.507	0.501	0.499		
HF_typeHF_8	0.155	0.503	0.497	0.495	0.505	
day2:HF_typeHF_2	0.013	-0.403	-0.200	-0.199	-0.203	-0.202
day2:HF_typeHF_3	0.013	-0.201	-0.411	-0.198	-0.201	-0.200
day2:HF_typeHF_5	0.013	-0.200	-0.198	-0.411	-0.200	-0.199
day2:HF_typeHF_7	0.013	-0.203	-0.201	-0.200	-0.401	-0.202
day2:HF_typeHF_8	0.013	-0.202	-0.200	-0.199	-0.202	-0.404
session2:HF_typeHF_2	-0.004	-0.541	-0.270	-0.269	-0.275	-0.273
session3:HF_typeHF_2	-0.001	-0.458	-0.227	-0.226	-0.230	-0.229
session4:HF_typeHF_2	-0.003	-0.493	-0.246	-0.245	-0.250	-0.248
session2:HF_typeHF_3	-0.004	-0.272	-0.539	-0.268	-0.273	-0.271
session3:HF_typeHF_3	-0.001	-0.228	-0.455	-0.225	-0.229	-0.227
session4:HF_typeHF_3	-0.003	-0.247	-0.491	-0.243	-0.248	-0.246
session2:HF_typeHF_5	-0.004	-0.270	-0.267	-0.540	-0.271	-0.269
session3:HF_typeHF_5	-0.001	-0.227	-0.224	-0.456	-0.228	-0.226
session4:HF_typeHF_5	-0.003	-0.246	-0.243	-0.491	-0.247	-0.246
session2:HF_typeHF_7	-0.004	-0.274	-0.271	-0.270	-0.537	-0.273
session3:HF_typeHF_7	-0.001	-0.231	-0.228	-0.227	-0.457	-0.230
session4:HF_typeHF_7	-0.003	-0.250	-0.247	-0.246	-0.491	-0.249
session2:HF_typeHF_8	-0.004	-0.273	-0.270	-0.269	-0.275	-0.541
session3:HF_typeHF_8	-0.001	-0.228	-0.226	-0.225	-0.229	-0.448
session4:HF_typeHF_8	-0.003	-0.249	-0.246	-0.245	-0.250	-0.493
activitystanding:HF_typeHF_2	0.005	-0.450	-0.225	-0.224	-0.228	-0.227
activitywalking:HF_typeHF_2	-0.001	-0.480	-0.240	-0.239	-0.244	-0.242
activitystanding:HF_typeHF_3	0.004	-0.226	-0.449	-0.222	-0.227	-0.225
activitywalking:HF_typeHF_3	-0.001	-0.242	-0.477	-0.238	-0.243	-0.241
activitystanding:HF_typeHF_5	0.004	-0.224	-0.222	-0.449	-0.225	-0.224
activitywalking:HF_typeHF_5	-0.001	-0.240	-0.237	-0.478	-0.241	-0.240
activitystanding:HF_typeHF_7	0.005	-0.228	-0.226	-0.225	-0.448	-0.227
activitywalking:HF_typeHF_7	-0.001	-0.244	-0.241	-0.241	-0.479	-0.244
activitystanding:HF_typeHF_8	0.004	-0.226	-0.224	-0.223	-0.227	-0.452
activitywalking:HF_typeHF_8	-0.001	-0.242	-0.239	-0.239	-0.243	-0.483
sexM:HF_typeHF_2	-0.444	-0.350	-0.174	-0.174	-0.178	-0.176
sexM:HF_typeHF_3	-0.441	-0.175	-0.358	-0.173	-0.177	-0.175
sexM:HF_typeHF_5	-0.438	-0.175	-0.172	-0.349	-0.176	-0.174
sexM:HF_typeHF_7	-0.446	-0.178	-0.176	-0.175	-0.358	-0.177
sexM:HF_typeHF_8	-0.443	-0.176	-0.174	-0.174	-0.178	-0.353
	d2:HF_HF_2	d2:HF_HF_3	d2:HF_HF_5	d2:HF_HF_7		
day2						
session2						
session3						
session4						
activitystanding						
activitywalking						
sexM						
HF_typeHF_2						
HF_typeHF_3						
HF_typeHF_5						
HF_typeHF_7						
HF_typeHF_8						
day2:HF_typeHF_2						
day2:HF_typeHF_3	0.502					

day2:HF_typeHF_5	0.500	0.496		
day2:HF_typeHF_7	0.507	0.503	0.501	
day2:HF_typeHF_8	0.504	0.501	0.498	0.505
session2:HF_typeHF_2	0.111	0.054	0.054	0.055
session3:HF_typeHF_2	-0.005	-0.005	-0.004	-0.005
session4:HF_typeHF_2	-0.003	-0.003	-0.003	-0.003
session2:HF_typeHF_3	0.054	0.111	0.054	0.055
session3:HF_typeHF_3	-0.004	-0.005	-0.004	-0.005
session4:HF_typeHF_3	-0.003	0.001	-0.003	-0.003
session2:HF_typeHF_5	0.054	0.053	0.112	0.054
session3:HF_typeHF_5	-0.004	-0.005	-0.004	-0.005
session4:HF_typeHF_5	-0.003	-0.003	-0.003	-0.003
session2:HF_typeHF_7	0.055	0.054	0.054	0.112
session3:HF_typeHF_7	-0.005	-0.005	-0.004	-0.005
session4:HF_typeHF_7	-0.003	-0.003	-0.003	-0.001
session2:HF_typeHF_8	0.055	0.054	0.054	0.055
session3:HF_typeHF_8	-0.004	-0.005	-0.004	-0.005
session4:HF_typeHF_8	-0.003	-0.003	-0.003	-0.003
activitystanding:HF_typeHF_2	0.084	0.044	0.044	0.044
activitywalking:HF_typeHF_2	0.032	0.016	0.016	0.017
activitystanding:HF_typeHF_3	0.044	0.085	0.043	0.044
activitywalking:HF_typeHF_3	0.016	0.032	0.016	0.017
activitystanding:HF_typeHF_5	0.044	0.043	0.085	0.044
activitywalking:HF_typeHF_5	0.016	0.016	0.033	0.016
activitystanding:HF_typeHF_7	0.044	0.044	0.044	0.081
activitywalking:HF_typeHF_7	0.017	0.016	0.016	0.031
activitystanding:HF_typeHF_8	0.044	0.044	0.043	0.044
activitywalking:HF_typeHF_8	0.016	0.016	0.016	0.017
sexM:HF_typeHF_2	-0.026	-0.016	-0.015	-0.014
sexM:HF_typeHF_3	-0.015	-0.012	-0.015	-0.014
sexM:HF_typeHF_5	-0.015	-0.016	-0.027	-0.014
sexM:HF_typeHF_7	-0.014	-0.014	-0.014	-0.016
sexM:HF_typeHF_8	-0.016	-0.016	-0.015	-0.014
d2:HF_HF_8 s2:HF_HF_2 s3:HF_HF_2 s4:HF_HF_2				
day2				
session2				
session3				
session4				
activitystanding				
activitywalking				
sexM				
HF_typeHF_2				
HF_typeHF_3				
HF_typeHF_5				
HF_typeHF_7				
HF_typeHF_8				
day2:HF_typeHF_2				
day2:HF_typeHF_3				
day2:HF_typeHF_5				
day2:HF_typeHF_7				
day2:HF_typeHF_8				
session2:HF_typeHF_2	0.055			
session3:HF_typeHF_2	-0.005	0.383		
session4:HF_typeHF_2	-0.003	0.411	0.426	

session2:HF_typeHF_3	0.054	0.505	0.191	0.206
session3:HF_typeHF_3	-0.004	0.191	0.502	0.213
session4:HF_typeHF_3	-0.003	0.205	0.213	0.498
session2:HF_typeHF_5	0.054	0.501	0.190	0.204
session3:HF_typeHF_5	-0.004	0.190	0.499	0.212
session4:HF_typeHF_5	-0.003	0.205	0.212	0.497
session2:HF_typeHF_7	0.055	0.509	0.193	0.208
session3:HF_typeHF_7	-0.005	0.193	0.507	0.216
session4:HF_typeHF_7	-0.003	0.208	0.215	0.504
session2:HF_typeHF_8	0.112	0.508	0.192	0.207
session3:HF_typeHF_8	-0.009	0.191	0.502	0.213
session4:HF_typeHF_8	-0.002	0.207	0.214	0.502
activitystanding:HF_typeHF_2	0.044	0.129	-0.002	0.000
activitywalking:HF_typeHF_2	0.016	0.302	-0.002	0.279
activitystanding:HF_typeHF_3	0.044	0.065	-0.002	0.000
activitywalking:HF_typeHF_3	0.016	0.152	-0.002	0.140
activitystanding:HF_typeHF_5	0.043	0.065	-0.002	0.000
activitywalking:HF_typeHF_5	0.016	0.151	-0.002	0.140
activitystanding:HF_typeHF_7	0.044	0.066	-0.002	0.000
activitywalking:HF_typeHF_7	0.016	0.154	-0.002	0.142
activitystanding:HF_typeHF_8	0.083	0.065	-0.002	0.000
activitywalking:HF_typeHF_8	0.035	0.152	-0.002	0.141
sexM:HF_typeHF_2	-0.016	0.005	0.001	0.004
sexM:HF_typeHF_3	-0.015	0.005	0.001	0.003
sexM:HF_typeHF_5	-0.015	0.005	0.001	0.004
sexM:HF_typeHF_7	-0.014	0.005	0.002	0.003
sexM:HF_typeHF_8	-0.025	0.005	0.001	0.004
s2:HF_HF_3 s3:HF_HF_3 s4:HF_HF_3 s2:HF_HF_5				
day2				
session2				
session3				
session4				
activitystanding				
activitywalking				
sexM				
HF_typeHF_2				
HF_typeHF_3				
HF_typeHF_5				
HF_typeHF_7				
HF_typeHF_8				
day2:HF_typeHF_2				
day2:HF_typeHF_3				
day2:HF_typeHF_5				
day2:HF_typeHF_7				
day2:HF_typeHF_8				
session2:HF_typeHF_2				
session3:HF_typeHF_2				
session4:HF_typeHF_2				
session2:HF_typeHF_3				
session3:HF_typeHF_3	0.382			
session4:HF_typeHF_3	0.409	0.425		
session2:HF_typeHF_5	0.498	0.189	0.203	
session3:HF_typeHF_5	0.189	0.496	0.210	0.382
session4:HF_typeHF_5	0.204	0.211	0.493	0.410

session2:HF_typeHF_7	0.506	0.192	0.206	0.502
session3:HF_typeHF_7	0.192	0.504	0.214	0.191
session4:HF_typeHF_7	0.207	0.214	0.500	0.205
session2:HF_typeHF_8	0.505	0.191	0.205	0.501
session3:HF_typeHF_8	0.190	0.499	0.212	0.189
session4:HF_typeHF_8	0.206	0.213	0.498	0.204
activitystanding:HF_typeHF_2	0.065	-0.002	0.000	0.064
activitywalking:HF_typeHF_2	0.152	-0.002	0.140	0.151
activitystanding:HF_typeHF_3	0.131	-0.002	0.000	0.064
activitywalking:HF_typeHF_3	0.302	-0.002	0.278	0.150
activitystanding:HF_typeHF_5	0.064	-0.002	0.000	0.132
activitywalking:HF_typeHF_5	0.150	-0.002	0.138	0.303
activitystanding:HF_typeHF_7	0.065	-0.002	0.000	0.065
activitywalking:HF_typeHF_7	0.153	-0.002	0.141	0.152
activitystanding:HF_typeHF_8	0.065	-0.002	0.000	0.064
activitywalking:HF_typeHF_8	0.151	-0.002	0.140	0.150
sexM:HF_typeHF_2	0.005	0.001	0.003	0.005
sexM:HF_typeHF_3	0.005	0.001	0.007	0.005
sexM:HF_typeHF_5	0.005	0.001	0.003	0.005
sexM:HF_typeHF_7	0.005	0.002	0.003	0.005
sexM:HF_typeHF_8	0.005	0.001	0.003	0.005
s3:HF_HF_5 s4:HF_HF_5 s2:HF_HF_7 s3:HF_HF_7				
day2				
session2				
session3				
session4				
activitystanding				
activitywalking				
sexM				
HF_typeHF_2				
HF_typeHF_3				
HF_typeHF_5				
HF_typeHF_7				
HF_typeHF_8				
day2:HF_typeHF_2				
day2:HF_typeHF_3				
day2:HF_typeHF_5				
day2:HF_typeHF_7				
day2:HF_typeHF_8				
session2:HF_typeHF_2				
session3:HF_typeHF_2				
session4:HF_typeHF_2				
session2:HF_typeHF_3				
session3:HF_typeHF_3				
session4:HF_typeHF_3				
session2:HF_typeHF_5				
session3:HF_typeHF_5				
session4:HF_typeHF_5	0.426			
session2:HF_typeHF_7	0.190	0.205		
session3:HF_typeHF_7	0.502	0.213	0.381	
session4:HF_typeHF_7	0.213	0.499	0.409	0.426
session2:HF_typeHF_8	0.190	0.205	0.509	0.193
session3:HF_typeHF_8	0.497	0.211	0.192	0.505
session4:HF_typeHF_8	0.212	0.497	0.208	0.216

activitystanding:HF_typeHF_2	-0.002	0.000	0.066	-0.002
activitywalking:HF_typeHF_2	-0.002	0.140	0.153	-0.002
activitystanding:HF_typeHF_3	-0.002	0.000	0.065	-0.002
activitywalking:HF_typeHF_3	-0.002	0.139	0.152	-0.002
activitystanding:HF_typeHF_5	-0.002	0.000	0.065	-0.002
activitywalking:HF_typeHF_5	-0.002	0.279	0.151	-0.002
activitystanding:HF_typeHF_7	-0.002	0.000	0.128	-0.002
activitywalking:HF_typeHF_7	-0.002	0.140	0.301	-0.002
activitystanding:HF_typeHF_8	-0.002	0.000	0.065	-0.002
activitywalking:HF_typeHF_8	-0.002	0.139	0.153	-0.002
sexM:HF_typeHF_2	0.001	0.003	0.005	0.001
sexM:HF_typeHF_3	0.001	0.003	0.005	0.001
sexM:HF_typeHF_5	0.001	0.003	0.005	0.001
sexM:HF_typeHF_7	0.002	0.003	0.000	0.002
sexM:HF_typeHF_8	0.001	0.003	0.005	0.001
s4:HF_HF_7 s2:HF_HF_8 s3:HF_HF_8 s4:HF_HF_8				
day2				
session2				
session3				
session4				
activitystanding				
activitywalking				
sexM				
HF_typeHF_2				
HF_typeHF_3				
HF_typeHF_5				
HF_typeHF_7				
HF_typeHF_8				
day2:HF_typeHF_2				
day2:HF_typeHF_3				
day2:HF_typeHF_5				
day2:HF_typeHF_7				
day2:HF_typeHF_8				
session2:HF_typeHF_2				
session3:HF_typeHF_2				
session4:HF_typeHF_2				
session2:HF_typeHF_3				
session3:HF_typeHF_3				
session4:HF_typeHF_3				
session2:HF_typeHF_5				
session3:HF_typeHF_5				
session4:HF_typeHF_5				
session2:HF_typeHF_7				
session3:HF_typeHF_7				
session4:HF_typeHF_7				
session2:HF_typeHF_8	0.208			
session3:HF_typeHF_8	0.214	0.378		
session4:HF_typeHF_8	0.504	0.411	0.422	
activitystanding:HF_typeHF_2	0.000	0.065	-0.002	0.000
activitywalking:HF_typeHF_2	0.142	0.153	-0.002	0.141
activitystanding:HF_typeHF_3	0.000	0.065	-0.002	0.000
activitywalking:HF_typeHF_3	0.141	0.152	-0.002	0.140
activitystanding:HF_typeHF_5	0.000	0.065	-0.002	0.000
activitywalking:HF_typeHF_5	0.140	0.151	-0.002	0.140

activitystanding:HF_typeHF_7	-0.002	0.066	-0.002	0.000
activitywalking:HF_typeHF_7	0.278	0.154	-0.002	0.142
activitystanding:HF_typeHF_8	0.000	0.130	-0.004	0.001
activitywalking:HF_typeHF_8	0.141	0.303	-0.009	0.280
sexM:HF_typeHF_2	0.004	0.005	0.001	0.004
sexM:HF_typeHF_3	0.003	0.005	0.001	0.003
sexM:HF_typeHF_5	0.004	0.005	0.002	0.003
sexM:HF_typeHF_7	0.002	0.005	0.002	0.003
sexM:HF_typeHF_8	0.004	0.006	-0.003	0.004
		actvtys:HF_HF_2	actvtyw:HF_HF_2	actvtys:HF_HF_3
day2				
session2				
session3				
session4				
activitystanding				
activitywalking				
sexM				
HF_typeHF_2				
HF_typeHF_3				
HF_typeHF_5				
HF_typeHF_7				
HF_typeHF_8				
day2:HF_typeHF_2				
day2:HF_typeHF_3				
day2:HF_typeHF_5				
day2:HF_typeHF_7				
day2:HF_typeHF_8				
session2:HF_typeHF_2				
session3:HF_typeHF_2				
session4:HF_typeHF_2				
session2:HF_typeHF_3				
session3:HF_typeHF_3				
session4:HF_typeHF_3				
session2:HF_typeHF_5				
session3:HF_typeHF_5				
session4:HF_typeHF_5				
session2:HF_typeHF_7				
session3:HF_typeHF_7				
session4:HF_typeHF_7				
session2:HF_typeHF_8				
session3:HF_typeHF_8				
session4:HF_typeHF_8				
activitystanding:HF_typeHF_2				
activitywalking:HF_typeHF_2	0.396			
activitystanding:HF_typeHF_3	0.502	0.199		
activitywalking:HF_typeHF_3	0.200	0.500	0.397	
activitystanding:HF_typeHF_5	0.500	0.198	0.496	
activitywalking:HF_typeHF_5	0.199	0.498	0.197	
activitystanding:HF_typeHF_7	0.508	0.201	0.504	
activitywalking:HF_typeHF_7	0.202	0.506	0.200	
activitystanding:HF_typeHF_8	0.504	0.200	0.500	
activitywalking:HF_typeHF_8	0.200	0.502	0.199	
sexM:HF_typeHF_2	-0.005	0.001	-0.005	
sexM:HF_typeHF_3	-0.005	0.001	-0.005	

sexM:HF_typeHF_5	-0.005	0.001	-0.005
sexM:HF_typeHF_7	-0.005	0.001	-0.005
sexM:HF_typeHF_8	-0.005	0.001	-0.005
	actvttyw:HF_HF_3	actvtys:HF_HF_5	actvttyw:HF_HF_5
day2			
session2			
session3			
session4			
activitystanding			
activitywalking			
sexM			
HF_typeHF_2			
HF_typeHF_3			
HF_typeHF_5			
HF_typeHF_7			
HF_typeHF_8			
day2:HF_typeHF_2			
day2:HF_typeHF_3			
day2:HF_typeHF_5			
day2:HF_typeHF_7			
day2:HF_typeHF_8			
session2:HF_typeHF_2			
session3:HF_typeHF_2			
session4:HF_typeHF_2			
session2:HF_typeHF_3			
session3:HF_typeHF_3			
session4:HF_typeHF_3			
session2:HF_typeHF_5			
session3:HF_typeHF_5			
session4:HF_typeHF_5			
session2:HF_typeHF_7			
session3:HF_typeHF_7			
session4:HF_typeHF_7			
session2:HF_typeHF_8			
session3:HF_typeHF_8			
session4:HF_typeHF_8			
activitystanding:HF_typeHF_2			
activitywalking:HF_typeHF_2			
activitystanding:HF_typeHF_3			
activitywalking:HF_typeHF_3			
activitystanding:HF_typeHF_5	0.197		
activitywalking:HF_typeHF_5	0.495	0.396	
activitystanding:HF_typeHF_7	0.200	0.502	0.199
activitywalking:HF_typeHF_7	0.503	0.199	0.500
activitystanding:HF_typeHF_8	0.199	0.498	0.198
activitywalking:HF_typeHF_8	0.499	0.198	0.496
sexM:HF_typeHF_2	0.001	-0.005	0.001
sexM:HF_typeHF_3	0.001	-0.005	0.001
sexM:HF_typeHF_5	0.001	-0.005	0.001
sexM:HF_typeHF_7	0.001	-0.005	0.001
sexM:HF_typeHF_8	0.001	-0.005	0.001
	actvtys:HF_HF_7	actvttyw:HF_HF_7	actvtys:HF_HF_8
day2			
session2			

```

session3
session4
activitystanding
activitywalking
sexM
HF_typeHF_2
HF_typeHF_3
HF_typeHF_5
HF_typeHF_7
HF_typeHF_8
day2:HF_typeHF_2
day2:HF_typeHF_3
day2:HF_typeHF_5
day2:HF_typeHF_7
day2:HF_typeHF_8
session2:HF_typeHF_2
session3:HF_typeHF_2
session4:HF_typeHF_2
session2:HF_typeHF_3
session3:HF_typeHF_3
session4:HF_typeHF_3
session2:HF_typeHF_5
session3:HF_typeHF_5
session4:HF_typeHF_5
session2:HF_typeHF_7
session3:HF_typeHF_7
session4:HF_typeHF_7
session2:HF_typeHF_8
session3:HF_typeHF_8
session4:HF_typeHF_8
activitystanding:HF_typeHF_2
activitywalking:HF_typeHF_2
activitystanding:HF_typeHF_3
activitywalking:HF_typeHF_3
activitystanding:HF_typeHF_5
activitywalking:HF_typeHF_5
activitystanding:HF_typeHF_7
activitywalking:HF_typeHF_7    0.396
activitystanding:HF_typeHF_8    0.506            0.201
activitywalking:HF_typeHF_8    0.201            0.504            0.397
sexM:HF_typeHF_2                -0.005            0.001            -0.005
sexM:HF_typeHF_3                -0.005            0.001            -0.005
sexM:HF_typeHF_5                -0.005            0.001            -0.005
sexM:HF_typeHF_7                -0.005            0.001            -0.005
sexM:HF_typeHF_8                -0.005            0.001            -0.001
                                actvtyw:HF_HF_8 sM:HF_HF_2 sM:HF_HF_3 sM:HF_HF_5

day2
session2
session3
session4
activitystanding
activitywalking
sexM
HF_typeHF_2

```

```

HF_typeHF_3
HF_typeHF_5
HF_typeHF_7
HF_typeHF_8
day2:HF_typeHF_2
day2:HF_typeHF_3
day2:HF_typeHF_5
day2:HF_typeHF_7
day2:HF_typeHF_8
session2:HF_typeHF_2
session3:HF_typeHF_2
session4:HF_typeHF_2
session2:HF_typeHF_3
session3:HF_typeHF_3
session4:HF_typeHF_3
session2:HF_typeHF_5
session3:HF_typeHF_5
session4:HF_typeHF_5
session2:HF_typeHF_7
session3:HF_typeHF_7
session4:HF_typeHF_7
session2:HF_typeHF_8
session3:HF_typeHF_8
session4:HF_typeHF_8
activitystanding:HF_typeHF_2
activitywalking:HF_typeHF_2
activitystanding:HF_typeHF_3
activitywalking:HF_typeHF_3
activitystanding:HF_typeHF_5
activitywalking:HF_typeHF_5
activitystanding:HF_typeHF_7
activitywalking:HF_typeHF_7
activitystanding:HF_typeHF_8
activitywalking:HF_typeHF_8
sexM:HF_typeHF_2      0.001
sexM:HF_typeHF_3      0.001      0.502
sexM:HF_typeHF_5      0.001      0.500      0.496
sexM:HF_typeHF_7      0.001      0.507      0.503      0.501
sexM:HF_typeHF_8      0.005      0.504      0.501      0.498
                        sM:HF_HF_7
day2
session2
session3
session4
activitystanding
activitywalking
sexM
HF_typeHF_2
HF_typeHF_3
HF_typeHF_5
HF_typeHF_7
HF_typeHF_8
day2:HF_typeHF_2
day2:HF_typeHF_3

```

```

day2:HF_typeHF_5
day2:HF_typeHF_7
day2:HF_typeHF_8
session2:HF_typeHF_2
session3:HF_typeHF_2
session4:HF_typeHF_2
session2:HF_typeHF_3
session3:HF_typeHF_3
session4:HF_typeHF_3
session2:HF_typeHF_5
session3:HF_typeHF_5
session4:HF_typeHF_5
session2:HF_typeHF_7
session3:HF_typeHF_7
session4:HF_typeHF_7
session2:HF_typeHF_8
session3:HF_typeHF_8
session4:HF_typeHF_8
activitystanding:HF_typeHF_2
activitywalking:HF_typeHF_2
activitystanding:HF_typeHF_3
activitywalking:HF_typeHF_3
activitystanding:HF_typeHF_5
activitywalking:HF_typeHF_5
activitystanding:HF_typeHF_7
activitywalking:HF_typeHF_7
activitystanding:HF_typeHF_8
activitywalking:HF_typeHF_8
sexM:HF_typeHF_2
sexM:HF_typeHF_3
sexM:HF_typeHF_5
sexM:HF_typeHF_7
sexM:HF_typeHF_8          0.505

```

Standardized Within-Group Residuals:

	Min	Q1	Med	Q3	Max
	-6.4785480	-0.4907863	0.1117616	0.6319402	3.7636443

Number of Observations: 2638

Number of Groups: 24

```
anova mdl3_reml, mdl4a_reml)
```

	Model	df	AIC	BIC	logLik	Test	L.Ratio	p-value
mdl3_reml	1	49	1355.773	1642.885	-628.8867			
mdl4a_reml	2	50	1207.812	1500.783	-553.9061	1 vs 2	149.9612	<.0001

```
print(VarCorr(mdl4a_reml))
```

```
person = pdLogChol(1)
```

	Variance	StdDev
(Intercept)	0.006945276	0.08333832
Residual	0.080883753	0.28440069

```
mdl4a_var <- as.numeric(VarCorr(mdl4a_reml))
```

```
print(mdl4a_var )
```

```

[1] 0.006945276 0.080883753 0.083338320 0.284400690
#these are the variances
print(md14a_var[1])

[1] 0.006945276
print(md14a_var[2])

[1] 0.08088375
md14a_var_person <- md14a_var[1]
md14a_var_residual <- md14a_var[2]

md14a_icc <- md14a_var_person/(md14a_var_person+md14a_var_residual)
md14a_icc_res <- md14a_var_residual/(md14a_var_person+md14a_var_residual)
print(md14a_icc)

[1] 0.07907723
print(md14a_icc_res)

[1] 0.9209228
anova_md13_md14 <- bind_rows(
  get_baseline_metrics(md13_reml, "Model 3"),
  get_anova_outputs(md13_reml, md14a_reml, "Model 4")
)

anova_md13_md14_cleaned <- anova_md13_md14 %>%
  mutate(
    LRT_Stat = ifelse(is.na(LRT_Stat), "--", as.character(LRT_Stat)),
    p_value = ifelse(is.na(p_value), "--",
      # eps defines the threshold for "<"
      # digits defines the rounding for values above that threshold
      format.pval(p_value, eps = 0.0001, digits = 4))
  )

col_labels <- c("Model Version",
  "\\shortstack{Degrees of\\Freedom}",
  "AIC",
  "BIC",
  "Log-Likelihood",
  "\\shortstack{Likelihood Ratio\\Test Stat}",
  "P-Value")

kable(anova_md13_md14_cleaned,
  format="latex",
  col.names = col_labels,
  align = "lrrrrrr",
  booktabs = TRUE, #, #right aligns numbers
  escape = FALSE,
  caption = "\\footnotesize{Impact of adding a random intercept for person (Model 4) to the fixed-e",
  kable_styling(latex_options = c("scale_down","hold_position"),
    full_width = FALSE,
    table.envir = "table") %>%
  add_header_above(c("Likelihood Ratio Test for Random Intercept for Person" = 7),

```

```
bold = TRUE, align = "l") %>%
row_spec(0, bold = TRUE)
```

Table 5: Impact of adding a random intercept for person (Model 4) to the fixed-effect only model (Model 3). Note this uses REML to compare models, as fixed effects are not changing. As such, the final p-values should be divided by 2. Given the p-value is already extremely low, this has no impact on the takeaway that we should include the Random Intercept for Person. Note models were fit using Restricted Maximum Likelihood (REML), as fixed effects are the same between models.

Likelihood Ratio Test for Random Intercept for Person						
Model Version	Degrees of Freedom	AIC	BIC	Log-Likelihood	Likelihood Ratio Test Stat	P-Value
Model 3	49	1356	1643	-629	—	—
Model 4	50	1208	1501	-554	149.96	< 1e-04

The Intraclass Correlation Coefficient (ICC) (also known to as intra-individual coefficient here) for Model 4 was 0.0791, implying that ~8% of the variance is due to differences between people (and 92% is unexplained). This suggests there is some inter-individual variability. Table 5 shows that this also reduces AIC and BIC significantly,

```
mdl4b_reml <- lme(fixed = log(HF_value) ~ (day + session + activity + sex) * HF_type,
  random = ~ 1 | person/activity, #unique intercept for each person
  data = data_clean,
  method = "REML")
```

```
print(VarCorr(mdl4b_reml))
```

```
      Variance      StdDev
person = pdLogChol(1)
(Intercept) 6.945276e-03 8.333832e-02
activity = pdLogChol(1)
(Intercept) 2.265352e-10 1.505109e-05
Residual    8.088375e-02 2.844007e-01
```

```
mdl4b_var <- as.numeric(VarCorr(mdl4b_reml))
print(mdl4b_var )
```

```
[1]      NA 6.945276e-03      NA 2.265352e-10 8.088375e-02
[6]      NA 8.333832e-02      NA 1.505109e-05 2.844007e-01
```

```
#these are the variances
```

```
print(mdl4b_var[1])
```

```
[1] NA
```

```
print(mdl4b_var[2])
```

```
[1] 0.006945276
```

```
print(mdl4b_var[3])
```

```
[1] NA
```

```
print(mdl4b_var[4])
```

```
[1] 2.265352e-10
```



```
print(md14b_var[5])
```

```
[1] 0.08088375
```

```
print(md14b_var[6])
```

```
[1] NA
```

```
md14b_var_person <- md14b_var[2]
```

```
md14b_var_activity <- md14b_var[4]
```

```
md14b_var_residual <- md14b_var[5]
```

```
md14b_var_total <- md14b_var_person + md14b_var_activity + md14b_var_residual
```

```
md14b_icc_person <- round(md14b_var_person / md14b_var_total, 4)
```

```
md14b_icc_activity <- round(md14b_var_activity / md14b_var_total, 4)
```

```
md14b_icc_res <- round(md14b_var_residual / md14b_var_total, 4)
```

```
print("icc person")
```

```
[1] "icc person"
```

```
print(md14b_icc_person)
```

```
[1] 0.0791
```

```
print("icc activity")
```

```
[1] "icc activity"
```

```
print(md14b_icc_activity)
```

```
[1] 0
```

```
print("icc residual")
```

```
[1] "icc residual"
```

```
print(md14b_icc_res)
```

```
[1] 0.9209
```

We then examined if a nested activity-level random effect was necessary. The ICC for person remains the same as in model 4 (0.0791), and the ICC of activity (within person) is 0, implying adding activity as a random intercept adds no value - the effect of activity does not vary randomly between people.

```
md14c_reml <- lme(fixed = log(HF_value) ~ (day + session + activity + sex) * HF_type,  
  random = ~ 1 | person/HF_type, #unique intercept for each person  
  data = data_clean,  
  method = "REML")
```

```
print(VarCorr(md14c_reml))
```

	Variance	StdDev
person =	pdLogChol(1)	
(Intercept)	0.001667335	0.04083301
HF_type =	pdLogChol(1)	
(Intercept)	0.034177320	0.18487109
Residual	0.054767267	0.23402407

```

mdl4c_var <- as.numeric(VarCorr(mdl4c_reml))
print(mdl4c_var )

[1] NA 0.001667335 NA 0.034177320 0.054767267 NA
[7] 0.040833010 NA 0.184871090 0.234024070

#these are the variances
print(mdl4c_var[1])

[1] NA
print(mdl4c_var[2])

[1] 0.001667335
print(mdl4c_var[3])

[1] NA
print(mdl4c_var[4])

[1] 0.03417732
print(mdl4c_var[5])

[1] 0.05476727
print(mdl4c_var[6])

[1] NA
mdl4c_var_person <- mdl4c_var[2]
mdl4c_var_sensor <- mdl4c_var[4]
mdl4c_var_residual <- mdl4c_var[5]
mdl4c_var_total <- mdl4c_var_person + mdl4c_var_sensor + mdl4c_var_residual

mdl4c_icc_person <- round(mdl4c_var_person / mdl4c_var_total, 4)
mdl4c_icc_sensor <- round(mdl4c_var_sensor / mdl4c_var_total, 4)
mdl4c_icc_res <- round(mdl4c_var_residual / mdl4c_var_total, 4)

print("icc person")

[1] "icc person"
print(mdl4c_icc_person)

[1] 0.0184
print("icc sensor")

[1] "icc sensor"
print(mdl4c_icc_sensor)

[1] 0.3772
print("icc residual")

[1] "icc residual"
print(mdl4c_icc_res)

[1] 0.6044

```

```
mdl4d_reml <- lme(fixed = log(HF_value) ~ (day + session + activity + sex) * HF_type,
  random = ~ 1 | HF_type, #unique intercept for each person
  data = data_clean,
  method = "REML")
```

```
print(VarCorr(mdl4d_reml))
```

```
HF_type = pdLogChol(1)
      Variance StdDev
(Intercept) 0.001414016 0.03760341
Residual    0.087425966 0.29567882
```

```
mdl4d_var <- as.numeric(VarCorr(mdl4d_reml))
print(mdl4d_var )
```

```
[1] 0.001414016 0.087425966 0.037603410 0.295678820
```

```
#these are the variances
```

```
print(mdl4d_var[1])
```

```
[1] 0.001414016
```

```
print(mdl4d_var[2])
```

```
[1] 0.08742597
```

```
print(mdl4d_var[3])
```

```
[1] 0.03760341
```

```
print(mdl4d_var[4])
```

```
[1] 0.2956788
```

```
mdl4d_var_sensor <- mdl4d_var[1]
mdl4d_var_residual <- mdl4d_var[2]
mdl4d_var_total <- mdl4d_var_sensor + mdl4d_var_residual
```

```
mdl4d_icc_sensor <- round(mdl4d_var_sensor / mdl4d_var_total, 4)
mdl4d_icc_res <- round(mdl4d_var_residual / mdl4d_var_total, 4)
```

```
print("icc sensor")
```

```
[1] "icc sensor"
```

```
print(mdl4d_icc_sensor)
```

```
[1] 0.0159
```

```
print("icc residaul")
```

```
[1] "icc residaul"
```

```
print(mdl4d_icc_res)
```

```
[1] 0.9841
```

```
print(anova(mdl3_reml, mdl4a_reml))
```

	Model	df	AIC	BIC	logLik	Test	L.Ratio	p-value
mdl3_reml		1 49	1355.773	1642.885	-628.8867			

```
mdl4a_reml      2 50 1207.812 1500.783 -553.9061 1 vs 2 149.9612 <.0001
```

```
print(anova(mdl4a_reml, mdl4c_reml))
```

	Model	df	AIC	BIC	logLik	Test	L.Ratio	p-value
mdl4a_reml	1	50	1207.812	1500.783	-553.9061			
mdl4c_reml	2	51	484.768	783.598	-191.3840	1 vs 2	725.0442	<.0001

```
print(anova(mdl4c_reml, mdl4d_reml))
```

	Model	df	AIC	BIC	logLik	Test	L.Ratio	p-value
mdl4c_reml	1	51	484.768	783.598	-191.3840			
mdl4d_reml	2	50	1357.773	1650.744	-628.8867	1 vs 2	875.0054	<.0001

We then examined if a nested sensor-level random intercept was necessary (Model 5). The ICC for person is now 0.0184, lower than in model 4 (0.0791), and the ICC of sensor (within person) is 0.3772, with 60% of the variance unexplained. This implies that the major source of clustering is at the sensor level, and not individual level.

Removing the person random intercept (resulting in a sensor-only random intercept model) results in the significantly lower ICC for sensor (0.0159), with 98% of the variance left unexplained. This implies that the sensor by itself is not the primary driver of variance; instead, there is a shared baseline amongst sensors belonging to the same individual. Thus, we kept the nested sensor-level random intercept model (model 5).

```
anova_md15 <- anova(mdl4c_reml, mdl4d_reml)
anova_md15_clean <- data.frame(
  Model = c("Model 5: Random Intercept: Person and Nested Sensor", "Random Intercept: Sensor only"),
  Df = as.numeric(anova_md15$df),
  AIC = round(as.numeric(anova_md15$AIC)),
  BIC = round(as.numeric(anova_md15$BIC)),
  LogLik = round(as.numeric(anova_md15$logLik)),
  LRT_Stat = c("--", round(anova_md15$L.Ratio[2], 2)),
  p_value = anova_md15$p-value[2])
```

6.5 Random Slopes

The current model has random intercepts for person and sensor (nested), implying that measurements from the same sensor are correlated, and sensors belonging to the same person are correlated. However, it assumes that the impact of temperature (the day variable) is identical for all individuals. We tested day as a random slope (given person) to determine if the effect of day varies by person, meaning some people adapt to the temperature environment differently than others.

```
#mdl 4c is model 5
```

```
mdl6_reml <- lme(
  fixed = log(HF_value) ~ (day + session + activity + sex) * HF_type,
  random = list(
    person = ~ 1 + day,
    HF_type = ~ 1
  ),
  data = data_clean,
  method = "REML"
)
```

```
print(anova(mdl4c_reml, mdl6_reml))
```

	Model	df	AIC	BIC	logLik	Test	L.Ratio	p-value
mdl4c_reml	1	51	484.768	783.5980	-191.384			

```
mdl6_reml      2 53 311.948 622.4968 -102.974 1 vs 2 176.82 <.0001
```

```
print(VarCorr(mdl6_reml))
```

```
      Variance      StdDev      Corr
person = pdLogChol(1 + day)
(Intercept) 0.004649289    0.0681857 (Intr)
day2         0.020644014    0.1436803 -0.884
HF_type = pdLogChol(1)
(Intercept) 0.035971708    0.1896621
Residual     0.049731894    0.2230065
```

```
mdl6_var <- as.numeric(VarCorr(mdl6_reml))
```

```
print(mdl6_var)
```

```
[1]      NA 0.004649289 0.020644014      NA 0.035971708
[6] 0.049731894      NA 0.068185700 0.143680300      NA
[11] 0.189662100 0.223006500      NA      NA -0.884000000
[16]      NA      NA      NA
```

```
#these are the variances
```

```
print(mdl6_var[1])
```

```
[1] NA
```

```
print(mdl6_var[2])
```

```
[1] 0.004649289
```

```
print(mdl6_var[3])
```

```
[1] 0.02064401
```

```
print(mdl6_var[4])
```

```
[1] NA
```

```
print(mdl6_var[5])
```

```
[1] 0.03597171
```

```
print(mdl6_var[6])
```

```
[1] 0.04973189
```

```
print(mdl6_var[15]) #correlation
```

```
[1] -0.884
```

```
mdl6_var_person <- mdl6_var[2]
```

```
mdl6_var_day <- mdl6_var[3]
```

```
mdl6_var_sensor <- mdl6_var[5]
```

```
mdl6_var_residual <- mdl6_var[6]
```

```
mdl6_var_total <- mdl6_var_person + mdl6_var_sensor + mdl6_var_day + mdl6_var_residual
```

```
mdl6_icc_person <- round(mdl6_var_person / mdl6_var_total, 4)
```

```
mdl6_icc_sensor <- round(mdl6_var_sensor / mdl6_var_total, 4)
```

```
mdl6_icc_day <- round(mdl6_var_day / mdl6_var_total, 4)
```

```
mdl6_icc_res <- round(mdl6_var_residual / mdl6_var_total, 4)
```

```
mdl6_corr_day_person <- mdl6_var[15]
```

```
print("icc person mdl6")
```

```
[1] "icc person mdl6"
```

```
print mdl6_icc_person)
```

```
[1] 0.0419
```

```
print("icc day mdl6")
```

```
[1] "icc day mdl6"
```

```
print mdl6_icc_day)
```

```
[1] 0.186
```

```
print("icc sensor mdl6")
```

```
[1] "icc sensor mdl6"
```

```
print mdl6_icc_sensor)
```

```
[1] 0.3241
```

```
print("icc residual mdl6")
```

```
[1] "icc residual mdl6"
```

```
print mdl6_icc_res)
```

```
[1] 0.448
```

```
print("icc person mdl4c")
```

```
[1] "icc person mdl4c"
```

```
print mdl4c_icc_person)
```

```
[1] 0.0184
```

```
print("icc sensor mdl4c")
```

```
[1] "icc sensor mdl4c"
```

```
print mdl4c_icc_sensor)
```

```
[1] 0.3772
```

```
print("icc residual mdl4c")
```

```
[1] "icc residual mdl4c"
```

```
print mdl4c_icc_res)
```

```
[1] 0.6044
```

```
print("corr day person")
```

```
[1] "corr day person"
```

```
print mdl6_corr_day_person)
```

```
[1] -0.884
```

```
anova_md16 <- anova(md14c_reml, md16_reml)
anova_md16_clean <- data.frame(
  Model = c("Model 5: Random Intercept: Person and Nested Sensor",
            "Model 6: Add Random Slope for Day (Given Person)"),
  Df = as.numeric(anova_md16$df),
  AIC = round(as.numeric(anova_md16$AIC)),
  BIC = round(as.numeric(anova_md16$BIC)),
  LogLik = round(as.numeric(anova_md16$logLik)),
  LRT_Stat = c(NA, round(anova_md16$L.Ratio[2], 2)),
  # FIX: Pass a vector of TWO values instead of just one
  p_value = c(NA, anova_md16$p-value[2])
)

print(anova_md16)
```

	Model	df	AIC	BIC	logLik	Test L.Ratio	p-value
md14c_reml	1	51	484.768	783.5980	-191.384		
md16_reml	2	53	311.948	622.4968	-102.974	1 vs 2	176.82 <.0001

```
anova_md16_clean <- anova_md16_clean %>%
  mutate(
    LRT_Stat = ifelse(is.na(LRT_Stat), "--", as.character(LRT_Stat)),
    p_value = ifelse(is.na(p_value), "--",
                     # eps defines the threshold for "<"
                     # digits defines the rounding for values above that threshold
                     format.pval(p_value, eps = 0.0001, digits = 4))
  )

col_labels <- c("Model Version",
                "\\shortstack{Degrees of\\Freedom}",
                "AIC",
                "BIC",
                "Log-Likelihood",
                "\\shortstack{Likelihood Ratio\\Test Stat}",
                "P-Value")
```

```
kable(anova_md16_clean,
      format="latex",
      col.names = col_labels,
      align = "lrrrrrr",
      booktabs = TRUE, #, #right aligns numbers
      escape = FALSE,
      caption = "\\footnotesize{Likelihood test between Model 5 and Model 6, which adds day as a random",
      kable_styling(latex_options = c("scale_down", "hold_position"),
                    full_width = FALSE,
                    table.envir = "table") %>%
      add_header_above(c("Likelihood Ratio Test for Random Slope for day" = 7),
                       bold = TRUE, align = "l") %>%
      row_spec(0, bold = TRUE)
```

Adding day as a random slope (Model 6) results in 45% unexplained variance, compared to 60% from Model 5. The ICC for day is 0.186, which is fairly large; ~19% of the variation in heat flow measurements is explained by the fact that people have different heat flow reactions depending on the external temperature. Additionally, the correlation between the person-level intercept and the day-level random slope is -0.884. This implies that the higher a person's heat flow was on day 1 (hot day), the steeper their drop was on day

Table 6: Likelihood test between Model 5 and Model 6, which adds day as a random slope. Model 6 has significantly lower AIC and BIC, and a very small p-value, so it is statistically significant. Note that models are run using REML, as they have the same fixed effects.

Likelihood Ratio Test for Random Slope for day						
Model Version	Degrees of Freedom	AIC	BIC	Log-Likelihood	Likelihood Ratio Test Stat	P-Value
Model 5: Random Intercept: Person and Nested Sensor	51	485	784	-191	–	–
Model 6: Add Random Slope for Day (Given Person)	53	312	622	-103	176.82	< 1e-04

2 (cold day), suggesting convergence.

The ICC for person also increased to 0.0419 (compared to 0.0184 from model 5), suggesting that the noise from people reacting differently to temperature masked their baseline differences. The ICC for sensor decreased to 0.3241 (from 0.3772 in model 5), suggesting that temperature explained some of the heat flow variance that was previously attributed to sensor.

Additionally as shown in Table 6, Model 6 has better AIC and BIC than Model 5, so we included day as a random slope.

6.6 Covariance Structures

```
mdl7_reml <- lme(
  fixed = log(HF_value) ~ (day + session + activity + sex) * HF_type,
  random = list(
    person = ~ 1 + day,
    HF_type = ~ 1
  ),
  correlation = corAR1(form = ~ 1 | person/HF_type/day),
  data = data_clean,
  method = "REML"
)
print(summary(mdl7_reml))
```

Linear mixed-effects model fit by REML

```
Data: data_clean
      AIC      BIC   logLik
-401.9296 -85.52129 254.9648
```

Random effects:

```
Formula: ~1 + day | person
Structure: General positive-definite, Log-Cholesky parametrization
      StdDev      Corr
(Intercept) 0.06198345 (Intr)
day2         0.11897929 -0.756
```

```
Formula: ~1 | HF_type %in% person
      (Intercept) Residual
StdDev:   0.1680532 0.2422861
```

Correlation Structure: AR(1)

```
Formula: ~1 | person/HF_type/day
Parameter estimate(s):
      Phi
```


0.5745104

Fixed effects: log(HF_value) ~ (day + session + activity + sex) * HF_type

	Value	Std.Error	DF	t-value	p-value
(Intercept)	4.488957	0.06592302	2458	68.09392	0.0000
day2	-0.151431	0.04648553	2458	-3.25760	0.0011
session2	-0.060633	0.03322583	2458	-1.82487	0.0681
session3	-0.090015	0.03515867	2458	-2.56026	0.0105
session4	-0.222219	0.04355403	2458	-5.10214	0.0000
activitystanding	-0.090135	0.01631579	2458	-5.52438	0.0000
activitywalking	0.039045	0.02366080	2458	1.65018	0.0990
sexM	0.000090	0.08097786	22	0.00112	0.9991
HF_typeHF_2	-0.717902	0.09061296	110	-7.92273	0.0000
HF_typeHF_3	-0.626148	0.09150243	110	-6.84296	0.0000
HF_typeHF_5	-0.422609	0.09161587	110	-4.61283	0.0000
HF_typeHF_7	-0.163327	0.09049073	110	-1.80491	0.0738
HF_typeHF_8	-0.910688	0.09068759	110	-10.04204	0.0000
day2:HF_typeHF_2	0.479848	0.05586706	2458	8.58910	0.0000
day2:HF_typeHF_3	0.468231	0.05645889	2458	8.29330	0.0000
day2:HF_typeHF_5	0.416277	0.05703019	2458	7.29924	0.0000
day2:HF_typeHF_7	0.317024	0.05540063	2458	5.72238	0.0000
day2:HF_typeHF_8	0.467158	0.05599344	2458	8.34309	0.0000
session2:HF_typeHF_2	0.101537	0.04677783	2458	2.17062	0.0301
session3:HF_typeHF_2	0.111083	0.04974142	2458	2.23320	0.0256
session4:HF_typeHF_2	0.306178	0.06150624	2458	4.97801	0.0000
session2:HF_typeHF_3	0.034506	0.04704642	2458	0.73344	0.4634
session3:HF_typeHF_3	0.072422	0.05001746	2458	1.44794	0.1478
session4:HF_typeHF_3	0.206123	0.06198508	2458	3.32536	0.0009
session2:HF_typeHF_5	0.069958	0.04731438	2458	1.47857	0.1394
session3:HF_typeHF_5	0.084023	0.05028350	2458	1.67099	0.0949
session4:HF_typeHF_5	0.222845	0.06218066	2458	3.58383	0.0003
session2:HF_typeHF_7	0.035422	0.04666370	2458	0.75910	0.4479
session3:HF_typeHF_7	0.028809	0.04940146	2458	0.58315	0.5598
session4:HF_typeHF_7	0.099322	0.06119266	2458	1.62310	0.1047
session2:HF_typeHF_8	0.070063	0.04682996	2458	1.49610	0.1348
session3:HF_typeHF_8	0.152757	0.04978100	2458	3.06858	0.0022
session4:HF_typeHF_8	0.263035	0.06146595	2458	4.27937	0.0000
activitystanding:HF_typeHF_2	-0.088902	0.02286430	2458	-3.88827	0.0001
activitywalking:HF_typeHF_2	-0.378341	0.03335347	2458	-11.34337	0.0000
activitystanding:HF_typeHF_3	0.189720	0.02304431	2458	8.23285	0.0000
activitywalking:HF_typeHF_3	0.219512	0.03356350	2458	6.54020	0.0000
activitystanding:HF_typeHF_5	0.063982	0.02316086	2458	2.76251	0.0058
activitywalking:HF_typeHF_5	-0.050777	0.03374416	2458	-1.50477	0.1325
activitystanding:HF_typeHF_7	-0.031730	0.02278715	2458	-1.39247	0.1639
activitywalking:HF_typeHF_7	-0.115347	0.03318837	2458	-3.47554	0.0005
activitystanding:HF_typeHF_8	0.176346	0.02297667	2458	7.67500	0.0000
activitywalking:HF_typeHF_8	0.144645	0.03342563	2458	4.32736	0.0000
sexM:HF_typeHF_2	-0.082158	0.11194091	110	-0.73395	0.4645
sexM:HF_typeHF_3	0.063028	0.11223169	110	0.56159	0.5755
sexM:HF_typeHF_5	0.017006	0.11248752	110	0.15118	0.8801
sexM:HF_typeHF_7	0.124206	0.11168658	110	1.11210	0.2685
sexM:HF_typeHF_8	0.085935	0.11200041	110	0.76727	0.4446

Correlation:

	(Intr)	day2	sessn2	sessn3	sessn4	actvtys	actvtyw
day2		-0.321					

session2	-0.275	0.025					
session3	-0.303	-0.004	0.580				
session4	-0.339	-0.010	0.525	0.661			
activitystanding	-0.166	0.015	0.088	0.075	0.126		
activitywalking	-0.230	0.004	0.394	0.159	0.449	0.402	
sexM	-0.604	-0.018	0.003	0.001	0.002	-0.001	0.002
HF_typeHF_2	-0.689	0.179	0.200	0.221	0.247	0.121	0.167
HF_typeHF_3	-0.683	0.177	0.198	0.219	0.245	0.119	0.166
HF_typeHF_5	-0.682	0.177	0.198	0.218	0.244	0.119	0.166
HF_typeHF_7	-0.690	0.179	0.200	0.221	0.247	0.121	0.168
HF_typeHF_8	-0.689	0.178	0.200	0.220	0.247	0.121	0.167
day2:HF_typeHF_2	0.204	-0.605	-0.021	0.003	0.008	-0.013	-0.004
day2:HF_typeHF_3	0.202	-0.598	-0.021	0.003	0.008	-0.013	-0.004
day2:HF_typeHF_5	0.200	-0.593	-0.021	0.003	0.008	-0.012	-0.003
day2:HF_typeHF_7	0.206	-0.610	-0.021	0.003	0.008	-0.013	-0.004
day2:HF_typeHF_8	0.204	-0.604	-0.021	0.003	0.008	-0.013	-0.004
session2:HF_typeHF_2	0.195	-0.018	-0.710	-0.412	-0.373	-0.062	-0.280
session3:HF_typeHF_2	0.214	0.003	-0.410	-0.707	-0.467	-0.053	-0.112
session4:HF_typeHF_2	0.240	0.007	-0.372	-0.468	-0.708	-0.089	-0.318
session2:HF_typeHF_3	0.194	-0.018	-0.706	-0.409	-0.371	-0.062	-0.278
session3:HF_typeHF_3	0.213	0.003	-0.407	-0.703	-0.464	-0.053	-0.112
session4:HF_typeHF_3	0.238	0.007	-0.369	-0.464	-0.703	-0.088	-0.315
session2:HF_typeHF_5	0.193	-0.018	-0.702	-0.407	-0.369	-0.062	-0.276
session3:HF_typeHF_5	0.212	0.003	-0.405	-0.699	-0.462	-0.052	-0.111
session4:HF_typeHF_5	0.238	0.007	-0.368	-0.463	-0.700	-0.088	-0.314
session2:HF_typeHF_7	0.196	-0.018	-0.712	-0.413	-0.374	-0.063	-0.280
session3:HF_typeHF_7	0.216	0.003	-0.412	-0.712	-0.470	-0.053	-0.113
session4:HF_typeHF_7	0.241	0.007	-0.374	-0.470	-0.712	-0.090	-0.320
session2:HF_typeHF_8	0.195	-0.018	-0.709	-0.411	-0.373	-0.062	-0.279
session3:HF_typeHF_8	0.214	0.003	-0.409	-0.706	-0.467	-0.053	-0.112
session4:HF_typeHF_8	0.240	0.007	-0.372	-0.468	-0.709	-0.089	-0.318
activitystanding:HF_typeHF_2	0.118	-0.011	-0.063	-0.054	-0.090	-0.714	-0.287
activitywalking:HF_typeHF_2	0.163	-0.003	-0.279	-0.113	-0.318	-0.286	-0.709
activitystanding:HF_typeHF_3	0.117	-0.011	-0.062	-0.053	-0.089	-0.708	-0.285
activitywalking:HF_typeHF_3	0.162	-0.003	-0.277	-0.112	-0.316	-0.284	-0.705
activitystanding:HF_typeHF_5	0.117	-0.011	-0.062	-0.053	-0.089	-0.704	-0.284
activitywalking:HF_typeHF_5	0.161	-0.003	-0.276	-0.111	-0.315	-0.282	-0.701
activitystanding:HF_typeHF_7	0.119	-0.011	-0.063	-0.054	-0.090	-0.716	-0.288
activitywalking:HF_typeHF_7	0.164	-0.003	-0.281	-0.113	-0.320	-0.287	-0.713
activitystanding:HF_typeHF_8	0.118	-0.011	-0.062	-0.053	-0.089	-0.710	-0.286
activitywalking:HF_typeHF_8	0.163	-0.003	-0.279	-0.112	-0.318	-0.285	-0.708
sexM:HF_typeHF_2	0.418	0.013	-0.002	-0.001	-0.002	0.000	-0.002
sexM:HF_typeHF_3	0.417	0.013	-0.002	-0.001	-0.002	0.000	-0.002
sexM:HF_typeHF_5	0.415	0.013	-0.002	-0.001	-0.002	0.000	-0.002
sexM:HF_typeHF_7	0.419	0.013	-0.002	-0.001	-0.002	0.000	-0.002
sexM:HF_typeHF_8	0.417	0.013	-0.002	-0.001	-0.002	0.000	-0.002
	sexM	HF_HF_2	HF_HF_3	HF_HF_5	HF_HF_7	HF_HF_8	
day2							
session2							
session3							
session4							
activitystanding							
activitywalking							
sexM							

HF_typeHF_2	0.420					
HF_typeHF_3	0.416	0.497				
HF_typeHF_5	0.415	0.496	0.491			
HF_typeHF_7	0.420	0.502	0.497	0.497		
HF_typeHF_8	0.420	0.501	0.496	0.496	0.502	
day2:HF_typeHF_2	0.015	-0.297	-0.148	-0.148	-0.149	-0.149
day2:HF_typeHF_3	0.015	-0.148	-0.310	-0.146	-0.147	-0.148
day2:HF_typeHF_5	0.014	-0.146	-0.145	-0.312	-0.146	-0.146
day2:HF_typeHF_7	0.015	-0.150	-0.148	-0.148	-0.295	-0.150
day2:HF_typeHF_8	0.015	-0.149	-0.147	-0.147	-0.148	-0.298
session2:HF_typeHF_2	-0.002	-0.283	-0.141	-0.140	-0.142	-0.142
session3:HF_typeHF_2	-0.001	-0.313	-0.154	-0.154	-0.156	-0.156
session4:HF_typeHF_2	-0.002	-0.348	-0.173	-0.173	-0.175	-0.175
session2:HF_typeHF_3	-0.002	-0.141	-0.282	-0.140	-0.141	-0.141
session3:HF_typeHF_3	-0.001	-0.155	-0.312	-0.153	-0.155	-0.155
session4:HF_typeHF_3	-0.002	-0.173	-0.348	-0.172	-0.174	-0.173
session2:HF_typeHF_5	-0.002	-0.140	-0.139	-0.283	-0.141	-0.140
session3:HF_typeHF_5	-0.001	-0.154	-0.153	-0.312	-0.154	-0.154
session4:HF_typeHF_5	-0.002	-0.173	-0.171	-0.348	-0.173	-0.173
session2:HF_typeHF_7	-0.002	-0.142	-0.141	-0.141	-0.280	-0.142
session3:HF_typeHF_7	-0.001	-0.157	-0.156	-0.155	-0.311	-0.157
session4:HF_typeHF_7	-0.002	-0.176	-0.174	-0.174	-0.346	-0.176
session2:HF_typeHF_8	-0.002	-0.142	-0.141	-0.140	-0.142	-0.283
session3:HF_typeHF_8	-0.001	-0.156	-0.154	-0.154	-0.156	-0.309
session4:HF_typeHF_8	-0.002	-0.175	-0.173	-0.173	-0.175	-0.347
activitystanding:HF_typeHF_2	0.000	-0.170	-0.085	-0.085	-0.086	-0.086
activitywalking:HF_typeHF_2	-0.002	-0.234	-0.118	-0.117	-0.119	-0.119
activitystanding:HF_typeHF_3	0.000	-0.085	-0.170	-0.084	-0.086	-0.085
activitywalking:HF_typeHF_3	-0.002	-0.118	-0.234	-0.117	-0.118	-0.118
activitystanding:HF_typeHF_5	0.000	-0.085	-0.084	-0.170	-0.085	-0.085
activitywalking:HF_typeHF_5	-0.002	-0.117	-0.117	-0.234	-0.118	-0.117
activitystanding:HF_typeHF_7	0.000	-0.086	-0.086	-0.085	-0.170	-0.086
activitywalking:HF_typeHF_7	-0.002	-0.119	-0.119	-0.118	-0.234	-0.119
activitystanding:HF_typeHF_8	0.000	-0.086	-0.085	-0.085	-0.086	-0.170
activitywalking:HF_typeHF_8	-0.002	-0.119	-0.118	-0.117	-0.119	-0.234
sexM:HF_typeHF_2	-0.692	-0.606	-0.301	-0.300	-0.304	-0.303
sexM:HF_typeHF_3	-0.690	-0.303	-0.608	-0.299	-0.304	-0.303
sexM:HF_typeHF_5	-0.689	-0.302	-0.299	-0.602	-0.303	-0.302
sexM:HF_typeHF_7	-0.694	-0.305	-0.302	-0.301	-0.612	-0.304
sexM:HF_typeHF_8	-0.692	-0.303	-0.300	-0.300	-0.304	-0.606
	d2:HF_HF_2	d2:HF_HF_3	d2:HF_HF_5	d2:HF_HF_7		
day2						
session2						
session3						
session4						
activitystanding						
activitywalking						
sexM						
HF_typeHF_2						
HF_typeHF_3						
HF_typeHF_5						
HF_typeHF_7						
HF_typeHF_8						
day2:HF_typeHF_2						

day2:HF_typeHF_3	0.500			
day2:HF_typeHF_5	0.496	0.490		
day2:HF_typeHF_7	0.508	0.502	0.497	
day2:HF_typeHF_8	0.505	0.499	0.495	0.506
session2:HF_typeHF_2	0.032	0.015	0.015	0.015
session3:HF_typeHF_2	-0.002	-0.002	-0.002	-0.002
session4:HF_typeHF_2	-0.009	-0.006	-0.006	-0.006
session2:HF_typeHF_3	0.015	0.033	0.015	0.015
session3:HF_typeHF_3	-0.002	-0.001	-0.002	-0.002
session4:HF_typeHF_3	-0.006	-0.005	-0.006	-0.006
session2:HF_typeHF_5	0.015	0.015	0.032	0.015
session3:HF_typeHF_5	-0.002	-0.002	-0.002	-0.002
session4:HF_typeHF_5	-0.006	-0.006	-0.009	-0.006
session2:HF_typeHF_7	0.015	0.015	0.015	0.033
session3:HF_typeHF_7	-0.002	-0.002	-0.002	-0.001
session4:HF_typeHF_7	-0.006	-0.006	-0.006	-0.008
session2:HF_typeHF_8	0.015	0.015	0.015	0.015
session3:HF_typeHF_8	-0.002	-0.002	-0.002	-0.002
session4:HF_typeHF_8	-0.006	-0.006	-0.006	-0.006
activitystanding:HF_typeHF_2	0.018	0.009	0.009	0.009
activitywalking:HF_typeHF_2	0.007	0.003	0.002	0.003
activitystanding:HF_typeHF_3	0.009	0.019	0.009	0.009
activitywalking:HF_typeHF_3	0.003	0.008	0.002	0.003
activitystanding:HF_typeHF_5	0.009	0.009	0.018	0.009
activitywalking:HF_typeHF_5	0.003	0.003	0.007	0.003
activitystanding:HF_typeHF_7	0.009	0.009	0.009	0.017
activitywalking:HF_typeHF_7	0.003	0.003	0.002	0.007
activitystanding:HF_typeHF_8	0.009	0.009	0.009	0.009
activitywalking:HF_typeHF_8	0.003	0.003	0.002	0.003
sexM:HF_typeHF_2	-0.021	-0.012	-0.012	-0.011
sexM:HF_typeHF_3	-0.012	-0.010	-0.012	-0.011
sexM:HF_typeHF_5	-0.012	-0.012	-0.021	-0.011
sexM:HF_typeHF_7	-0.011	-0.011	-0.011	-0.011
sexM:HF_typeHF_8	-0.012	-0.012	-0.012	-0.011
d2:HF_HF_8 s2:HF_HF_2 s3:HF_HF_2 s4:HF_HF_2				
day2				
session2				
session3				
session4				
activitystanding				
activitywalking				
sexM				
HF_typeHF_2				
HF_typeHF_3				
HF_typeHF_5				
HF_typeHF_7				
HF_typeHF_8				
day2:HF_typeHF_2				
day2:HF_typeHF_3				
day2:HF_typeHF_5				
day2:HF_typeHF_7				
day2:HF_typeHF_8				
session2:HF_typeHF_2	0.015			
session3:HF_typeHF_2	-0.002	0.581		

session4:HF_typeHF_2	-0.006	0.526	0.661	
session2:HF_typeHF_3	0.015	0.502	0.289	0.263
session3:HF_typeHF_3	-0.002	0.289	0.497	0.329
session4:HF_typeHF_3	-0.006	0.262	0.328	0.498
session2:HF_typeHF_5	0.015	0.499	0.288	0.261
session3:HF_typeHF_5	-0.002	0.288	0.494	0.327
session4:HF_typeHF_5	-0.006	0.261	0.327	0.496
session2:HF_typeHF_7	0.015	0.506	0.292	0.265
session3:HF_typeHF_7	-0.002	0.293	0.503	0.333
session4:HF_typeHF_7	-0.006	0.265	0.332	0.504
session2:HF_typeHF_8	0.032	0.504	0.291	0.264
session3:HF_typeHF_8	-0.004	0.291	0.499	0.330
session4:HF_typeHF_8	-0.009	0.264	0.331	0.502
activitystanding:HF_typeHF_2	0.009	0.087	0.081	0.128
activitywalking:HF_typeHF_2	0.003	0.392	0.158	0.447
activitystanding:HF_typeHF_3	0.009	0.044	0.038	0.063
activitywalking:HF_typeHF_3	0.003	0.197	0.079	0.224
activitystanding:HF_typeHF_5	0.009	0.044	0.037	0.063
activitywalking:HF_typeHF_5	0.003	0.196	0.079	0.223
activitystanding:HF_typeHF_7	0.009	0.045	0.038	0.064
activitywalking:HF_typeHF_7	0.003	0.199	0.080	0.227
activitystanding:HF_typeHF_8	0.018	0.044	0.038	0.063
activitywalking:HF_typeHF_8	0.008	0.198	0.080	0.225
sexM:HF_typeHF_2	-0.012	0.001	0.001	0.001
sexM:HF_typeHF_3	-0.012	0.001	0.001	0.001
sexM:HF_typeHF_5	-0.012	0.001	0.001	0.001
sexM:HF_typeHF_7	-0.011	0.001	0.001	0.001
sexM:HF_typeHF_8	-0.020	0.001	0.001	0.001
s2:HF_HF_3 s3:HF_HF_3 s4:HF_HF_3 s2:HF_HF_5				
day2				
session2				
session3				
session4				
activitystanding				
activitywalking				
sexM				
HF_typeHF_2				
HF_typeHF_3				
HF_typeHF_5				
HF_typeHF_7				
HF_typeHF_8				
day2:HF_typeHF_2				
day2:HF_typeHF_3				
day2:HF_typeHF_5				
day2:HF_typeHF_7				
day2:HF_typeHF_8				
session2:HF_typeHF_2				
session3:HF_typeHF_2				
session4:HF_typeHF_2				
session2:HF_typeHF_3				
session3:HF_typeHF_3	0.581			
session4:HF_typeHF_3	0.525	0.659		
session2:HF_typeHF_5	0.496	0.286	0.259	
session3:HF_typeHF_5	0.286	0.491	0.325	0.581

session4:HF_typeHF_5	0.260	0.325	0.492	0.526
session2:HF_typeHF_7	0.503	0.290	0.263	0.500
session3:HF_typeHF_7	0.291	0.500	0.330	0.290
session4:HF_typeHF_7	0.264	0.331	0.500	0.262
session2:HF_typeHF_8	0.501	0.289	0.262	0.498
session3:HF_typeHF_8	0.289	0.496	0.328	0.287
session4:HF_typeHF_8	0.263	0.329	0.498	0.261
activitystanding:HF_typeHF_2	0.044	0.038	0.063	0.044
activitywalking:HF_typeHF_2	0.197	0.079	0.224	0.196
activitystanding:HF_typeHF_3	0.089	0.081	0.128	0.044
activitywalking:HF_typeHF_3	0.393	0.158	0.446	0.195
activitystanding:HF_typeHF_5	0.044	0.037	0.062	0.089
activitywalking:HF_typeHF_5	0.195	0.078	0.221	0.393
activitystanding:HF_typeHF_7	0.044	0.038	0.063	0.044
activitywalking:HF_typeHF_7	0.198	0.080	0.225	0.197
activitystanding:HF_typeHF_8	0.044	0.037	0.063	0.044
activitywalking:HF_typeHF_8	0.197	0.079	0.223	0.196
sexM:HF_typeHF_2	0.001	0.001	0.001	0.001
sexM:HF_typeHF_3	0.002	0.001	0.003	0.001
sexM:HF_typeHF_5	0.001	0.001	0.001	0.001
sexM:HF_typeHF_7	0.001	0.001	0.001	0.001
sexM:HF_typeHF_8	0.001	0.001	0.001	0.001
s3:HF_HF_5 s4:HF_HF_5 s2:HF_HF_7 s3:HF_HF_7				
day2				
session2				
session3				
session4				
activitystanding				
activitywalking				
sexM				
HF_typeHF_2				
HF_typeHF_3				
HF_typeHF_5				
HF_typeHF_7				
HF_typeHF_8				
day2:HF_typeHF_2				
day2:HF_typeHF_3				
day2:HF_typeHF_5				
day2:HF_typeHF_7				
day2:HF_typeHF_8				
session2:HF_typeHF_2				
session3:HF_typeHF_2				
session4:HF_typeHF_2				
session2:HF_typeHF_3				
session3:HF_typeHF_3				
session4:HF_typeHF_3				
session2:HF_typeHF_5				
session3:HF_typeHF_5				
session4:HF_typeHF_5	0.661			
session2:HF_typeHF_7	0.289	0.262		
session3:HF_typeHF_7	0.498	0.329	0.579	
session4:HF_typeHF_7	0.329	0.499	0.524	0.661
session2:HF_typeHF_8	0.288	0.261	0.505	0.293
session3:HF_typeHF_8	0.494	0.327	0.291	0.503

session4:HF_typeHF_8	0.327	0.496	0.265	0.333
activitystanding:HF_typeHF_2	0.037	0.063	0.045	0.038
activitywalking:HF_typeHF_2	0.079	0.223	0.199	0.080
activitystanding:HF_typeHF_3	0.037	0.062	0.044	0.038
activitywalking:HF_typeHF_3	0.078	0.222	0.198	0.080
activitystanding:HF_typeHF_5	0.079	0.128	0.044	0.038
activitywalking:HF_typeHF_5	0.157	0.447	0.196	0.079
activitystanding:HF_typeHF_7	0.038	0.063	0.088	0.082
activitywalking:HF_typeHF_7	0.079	0.224	0.392	0.160
activitystanding:HF_typeHF_8	0.037	0.063	0.044	0.038
activitywalking:HF_typeHF_8	0.079	0.223	0.198	0.080
sexM:HF_typeHF_2	0.001	0.001	0.001	0.001
sexM:HF_typeHF_3	0.001	0.001	0.001	0.001
sexM:HF_typeHF_5	0.001	0.001	0.001	0.001
sexM:HF_typeHF_7	0.001	0.001	0.001	0.001
sexM:HF_typeHF_8	0.001	0.001	0.001	0.001
s4:HF_HF_7 s2:HF_HF_8 s3:HF_HF_8 s4:HF_HF_8				
day2				
session2				
session3				
session4				
activitystanding				
activitywalking				
sexM				
HF_typeHF_2				
HF_typeHF_3				
HF_typeHF_5				
HF_typeHF_7				
HF_typeHF_8				
day2:HF_typeHF_2				
day2:HF_typeHF_3				
day2:HF_typeHF_5				
day2:HF_typeHF_7				
day2:HF_typeHF_8				
session2:HF_typeHF_2				
session3:HF_typeHF_2				
session4:HF_typeHF_2				
session2:HF_typeHF_3				
session3:HF_typeHF_3				
session4:HF_typeHF_3				
session2:HF_typeHF_5				
session3:HF_typeHF_5				
session4:HF_typeHF_5				
session2:HF_typeHF_7				
session3:HF_typeHF_7				
session4:HF_typeHF_7				
session2:HF_typeHF_8	0.265			
session3:HF_typeHF_8	0.332	0.580		
session4:HF_typeHF_8	0.504	0.526	0.659	
activitystanding:HF_typeHF_2	0.064	0.044	0.038	0.064
activitywalking:HF_typeHF_2	0.227	0.198	0.080	0.226
activitystanding:HF_typeHF_3	0.063	0.044	0.038	0.063
activitywalking:HF_typeHF_3	0.225	0.197	0.079	0.224
activitystanding:HF_typeHF_5	0.063	0.044	0.037	0.063

activitywalking:HF_typeHF_5	0.224	0.196	0.079	0.223
activitystanding:HF_typeHF_7	0.127	0.045	0.038	0.064
activitywalking:HF_typeHF_7	0.448	0.199	0.080	0.227
activitystanding:HF_typeHF_8	0.064	0.090	0.076	0.126
activitywalking:HF_typeHF_8	0.226	0.395	0.152	0.445
sexM:HF_typeHF_2	0.001	0.001	0.001	0.001
sexM:HF_typeHF_3	0.001	0.001	0.001	0.001
sexM:HF_typeHF_5	0.001	0.001	0.001	0.001
sexM:HF_typeHF_7	0.001	0.001	0.001	0.001
sexM:HF_typeHF_8	0.001	0.001	0.000	0.001

actvtys:HF_HF_2 actvtyw:HF_HF_2 actvtys:HF_HF_3

day2
session2
session3
session4
activitystanding
activitywalking
sexM
HF_typeHF_2
HF_typeHF_3
HF_typeHF_5
HF_typeHF_7
HF_typeHF_8
day2:HF_typeHF_2
day2:HF_typeHF_3
day2:HF_typeHF_5
day2:HF_typeHF_7
day2:HF_typeHF_8
session2:HF_typeHF_2
session3:HF_typeHF_2
session4:HF_typeHF_2
session2:HF_typeHF_3
session3:HF_typeHF_3
session4:HF_typeHF_3
session2:HF_typeHF_5
session3:HF_typeHF_5
session4:HF_typeHF_5
session2:HF_typeHF_7
session3:HF_typeHF_7
session4:HF_typeHF_7
session2:HF_typeHF_8
session3:HF_typeHF_8
session4:HF_typeHF_8
activitystanding:HF_typeHF_2
activitywalking:HF_typeHF_2 0.400
activitystanding:HF_typeHF_3 0.505 0.202
activitywalking:HF_typeHF_3 0.202 0.500 0.401
activitystanding:HF_typeHF_5 0.503 0.201 0.499
activitywalking:HF_typeHF_5 0.201 0.497 0.200
activitystanding:HF_typeHF_7 0.511 0.204 0.507
activitywalking:HF_typeHF_7 0.205 0.506 0.203
activitystanding:HF_typeHF_8 0.507 0.203 0.503
activitywalking:HF_typeHF_8 0.203 0.502 0.202
sexM:HF_typeHF_2 0.000 0.001 0.000

sexM:HF_typeHF_3	0.000	0.001	0.000
sexM:HF_typeHF_5	0.000	0.001	0.000
sexM:HF_typeHF_7	0.000	0.001	0.000
sexM:HF_typeHF_8	0.000	0.001	0.000
	actvttyw:HF_HF_3	actvtys:HF_HF_5	actvttyw:HF_HF_5
day2			
session2			
session3			
session4			
activitystanding			
activitywalking			
sexM			
HF_typeHF_2			
HF_typeHF_3			
HF_typeHF_5			
HF_typeHF_7			
HF_typeHF_8			
day2:HF_typeHF_2			
day2:HF_typeHF_3			
day2:HF_typeHF_5			
day2:HF_typeHF_7			
day2:HF_typeHF_8			
session2:HF_typeHF_2			
session3:HF_typeHF_2			
session4:HF_typeHF_2			
session2:HF_typeHF_3			
session3:HF_typeHF_3			
session4:HF_typeHF_3			
session2:HF_typeHF_5			
session3:HF_typeHF_5			
session4:HF_typeHF_5			
session2:HF_typeHF_7			
session3:HF_typeHF_7			
session4:HF_typeHF_7			
session2:HF_typeHF_8			
session3:HF_typeHF_8			
session4:HF_typeHF_8			
activitystanding:HF_typeHF_2			
activitywalking:HF_typeHF_2			
activitystanding:HF_typeHF_3			
activitywalking:HF_typeHF_3			
activitystanding:HF_typeHF_5	0.200		
activitywalking:HF_typeHF_5	0.494	0.401	
activitystanding:HF_typeHF_7	0.203	0.504	0.202
activitywalking:HF_typeHF_7	0.503	0.202	0.500
activitystanding:HF_typeHF_8	0.201	0.500	0.200
activitywalking:HF_typeHF_8	0.499	0.201	0.496
sexM:HF_typeHF_2	0.001	0.000	0.001
sexM:HF_typeHF_3	0.002	0.000	0.001
sexM:HF_typeHF_5	0.001	0.000	0.001
sexM:HF_typeHF_7	0.001	0.000	0.001
sexM:HF_typeHF_8	0.001	0.000	0.001
	actvtys:HF_HF_7	actvttyw:HF_HF_7	actvtys:HF_HF_8
day2			

```

session2
session3
session4
activitystanding
activitywalking
sexM
HF_typeHF_2
HF_typeHF_3
HF_typeHF_5
HF_typeHF_7
HF_typeHF_8
day2:HF_typeHF_2
day2:HF_typeHF_3
day2:HF_typeHF_5
day2:HF_typeHF_7
day2:HF_typeHF_8
session2:HF_typeHF_2
session3:HF_typeHF_2
session4:HF_typeHF_2
session2:HF_typeHF_3
session3:HF_typeHF_3
session4:HF_typeHF_3
session2:HF_typeHF_5
session3:HF_typeHF_5
session4:HF_typeHF_5
session2:HF_typeHF_7
session3:HF_typeHF_7
session4:HF_typeHF_7
session2:HF_typeHF_8
session3:HF_typeHF_8
session4:HF_typeHF_8
activitystanding:HF_typeHF_2
activitywalking:HF_typeHF_2
activitystanding:HF_typeHF_3
activitywalking:HF_typeHF_3
activitystanding:HF_typeHF_5
activitywalking:HF_typeHF_5
activitystanding:HF_typeHF_7
activitywalking:HF_typeHF_7    0.401
activitystanding:HF_typeHF_8    0.508            0.204
activitywalking:HF_typeHF_8    0.204            0.505            0.401
sexM:HF_typeHF_2                0.000            0.001            0.000
sexM:HF_typeHF_3                0.000            0.001            0.000
sexM:HF_typeHF_5                0.000            0.001            0.000
sexM:HF_typeHF_7                0.000            0.001            0.000
sexM:HF_typeHF_8                0.000            0.001            0.000
                                actvtyw:HF_HF_8 sM:HF_HF_2 sM:HF_HF_3 sM:HF_HF_5
day2
session2
session3
session4
activitystanding
activitywalking
sexM

```

```

HF_typeHF_2
HF_typeHF_3
HF_typeHF_5
HF_typeHF_7
HF_typeHF_8
day2:HF_typeHF_2
day2:HF_typeHF_3
day2:HF_typeHF_5
day2:HF_typeHF_7
day2:HF_typeHF_8
session2:HF_typeHF_2
session3:HF_typeHF_2
session4:HF_typeHF_2
session2:HF_typeHF_3
session3:HF_typeHF_3
session4:HF_typeHF_3
session2:HF_typeHF_5
session3:HF_typeHF_5
session4:HF_typeHF_5
session2:HF_typeHF_7
session3:HF_typeHF_7
session4:HF_typeHF_7
session2:HF_typeHF_8
session3:HF_typeHF_8
session4:HF_typeHF_8
activitystanding:HF_typeHF_2
activitywalking:HF_typeHF_2
activitystanding:HF_typeHF_3
activitywalking:HF_typeHF_3
activitystanding:HF_typeHF_5
activitywalking:HF_typeHF_5
activitystanding:HF_typeHF_7
activitywalking:HF_typeHF_7
activitystanding:HF_typeHF_8
activitywalking:HF_typeHF_8
sexM:HF_typeHF_2      0.001
sexM:HF_typeHF_3      0.001      0.500
sexM:HF_typeHF_5      0.001      0.499      0.498
sexM:HF_typeHF_7      0.001      0.502      0.501      0.499
sexM:HF_typeHF_8      0.002      0.501      0.500      0.499
                                sM:HF_HF_7
day2
session2
session3
session4
activitystanding
activitywalking
sexM
HF_typeHF_2
HF_typeHF_3
HF_typeHF_5
HF_typeHF_7
HF_typeHF_8
day2:HF_typeHF_2

```

```

day2:HF_typeHF_3
day2:HF_typeHF_5
day2:HF_typeHF_7
day2:HF_typeHF_8
session2:HF_typeHF_2
session3:HF_typeHF_2
session4:HF_typeHF_2
session2:HF_typeHF_3
session3:HF_typeHF_3
session4:HF_typeHF_3
session2:HF_typeHF_5
session3:HF_typeHF_5
session4:HF_typeHF_5
session2:HF_typeHF_7
session3:HF_typeHF_7
session4:HF_typeHF_7
session2:HF_typeHF_8
session3:HF_typeHF_8
session4:HF_typeHF_8
activitystanding:HF_typeHF_2
activitywalking:HF_typeHF_2
activitystanding:HF_typeHF_3
activitywalking:HF_typeHF_3
activitystanding:HF_typeHF_5
activitywalking:HF_typeHF_5
activitystanding:HF_typeHF_7
activitywalking:HF_typeHF_7
activitystanding:HF_typeHF_8
activitywalking:HF_typeHF_8
sexM:HF_typeHF_2
sexM:HF_typeHF_3
sexM:HF_typeHF_5
sexM:HF_typeHF_7
sexM:HF_typeHF_8          0.502

```

Standardized Within-Group Residuals:

Min	Q1	Med	Q3	Max
-5.84148874	-0.38896040	0.09017881	0.53978284	3.56910479

Number of Observations: 2638

Number of Groups:

person	HF_type	%in%	person
24			144

```

anova_md17 <- anova mdl6_reml, mdl7_reml)
anova_md17_clean <- data.frame(
  Model = c("Model 6: Assumes Compound Symmetry", "Model 7: Assumes Auto-regressive AR(1)"),
  Df = as.numeric(anova_md17$df),
  AIC = round(as.numeric(anova_md17$AIC)),
  BIC = round(as.numeric(anova_md17$BIC)),
  LogLik = round(as.numeric(anova_md17$logLik)),
  LRT_Stat = c(NA, round(anova_md17$L.Ratio[2], 2)),
  p_value = c(NA, anova_md17$p-value`[2]))

```

```
print(anova_md17)
```

```

      Model df      AIC      BIC    logLik   Test  L.Ratio p-value
mdl6_reml    1 53  311.9480 622.4968 -102.9740
mdl7_reml    2 54 -401.9296 -85.5213  254.9648 1 vs 2 715.8776  <.0001

```

```

anova_md17_clean <- anova_md17_clean %>%
  mutate(
    LRT_Stat = ifelse(is.na(LRT_Stat), "--", as.character(LRT_Stat)),
    p_value = ifelse(is.na(p_value), "--",
      # eps defines the threshold for "<"
      # digits defines the rounding for values above that threshold
      format.pval(p_value, eps = 0.0001, digits = 4))
  )

```

```

col_labels <- c("Model Version",
  "\\shortstack{Degrees of\\\\Freedom}",
  "AIC",
  "BIC",
  "Log-Likelihood",
  "\\shortstack{Likelihood Ratio\\\\Test Stat}",
  "P-Value")

```

```

kable(anova_md17_clean,
  format="latex",
  col.names = col_labels,
  align = "lrrrrrr",
  booktabs = TRUE, #, #right aligns numbers
  escape = FALSE,
  caption = "\\footnotesize{Likelihood test between Model 6 and Model 7, testing an AR1 structure w",
  kable_styling(latex_options = c("scale_down", "hold_position"),
    full_width = FALSE,
    table.envir = "table") %>%
  add_header_above(c("Likelihood Ratio Test for Covariance Structures" = 7),
    bold = TRUE, align = "l") %>%
  row_spec(0, bold = TRUE)

```

Table 7: Likelihood test between Model 6 and Model 7, testing an AR1 structure where magnitude of correlation depends on amount of time between observations (instead of flat correlation). Model 7 has significantly lower AIC and BIC and a very small p-value, so it is statistically significant. Note that models are run using REML, as they have the same fixed effects.

Likelihood Ratio Test for Covariance Structures						
Model Version	Degrees of Freedom	AIC	BIC	Log-Likelihood	Likelihood Ratio Test Stat	P-Value
Model 6: Assumes Compound Symmetry	53	312	622	-103	—	—
Model 7: Assumes Auto-regressive AR(1)	54	-402	-86	255	715.88	< 1e-04

```

mdl7b_reml <- lme(
  fixed = log(HF_value) ~ (day + session + activity + sex) * HF_type,
  random = list(
    person = ~ 1 + day,
    HF_type = ~ 1
  ),

```

```

correlation = corAR1(form = ~ 1 | person/HF_type/day/activity),
data = data_clean,
method = "REML"
)
print(summary mdl7b_reml))

```

Linear mixed-effects model fit by REML

```

Data: data_clean
      AIC      BIC   logLik
-551.9934 -235.5851 329.9967

```

Random effects:

```

Formula: ~1 + day | person
Structure: General positive-definite, Log-Cholesky parametrization
      StdDev      Corr
(Intercept) 0.06587512 (Intr)
day2         0.13096188 -0.843

```

```

Formula: ~1 | HF_type %in% person
      (Intercept) Residual
StdDev:   0.1776895 0.2365776

```

Correlation Structure: AR(1)

```

Formula: ~1 | person/HF_type/day/activity
Parameter estimate(s):
      Phi
0.6594993

```

Fixed effects: log(HF_value) ~ (day + session + activity + sex) * HF_type

	Value	Std. Error	DF	t-value	p-value
(Intercept)	4.483784	0.06561790	2458	68.33172	0.0000
day2	-0.167377	0.04299213	2458	-3.89319	0.0001
session2	-0.067562	0.02047854	2458	-3.29917	0.0010
session3	-0.061275	0.01838532	2458	-3.33280	0.0009
session4	-0.185926	0.02489924	2458	-7.46715	0.0000
activitystanding	-0.123030	0.03931825	2458	-3.12909	0.0018
activitywalking	0.025512	0.04241395	2458	0.60151	0.5476
sexM	0.024406	0.08129255	22	0.30023	0.7668
HF_typeHF_2	-0.715902	0.09016155	110	-7.94021	0.0000
HF_typeHF_3	-0.619149	0.09085283	110	-6.81486	0.0000
HF_typeHF_5	-0.419077	0.09098278	110	-4.60612	0.0000
HF_typeHF_7	-0.157179	0.09004388	110	-1.74559	0.0837
HF_typeHF_8	-0.915731	0.09023994	110	-10.14774	0.0000
day2:HF_typeHF_2	0.506103	0.04755338	2458	10.64285	0.0000
day2:HF_typeHF_3	0.479271	0.04808463	2458	9.96724	0.0000
day2:HF_typeHF_5	0.430988	0.04863429	2458	8.86182	0.0000
day2:HF_typeHF_7	0.338016	0.04706917	2458	7.18127	0.0000
day2:HF_typeHF_8	0.499496	0.04761329	2458	10.49069	0.0000
session2:HF_typeHF_2	0.125302	0.02866554	2458	4.37116	0.0000
session3:HF_typeHF_2	0.093015	0.02597652	2458	3.58075	0.0003
session4:HF_typeHF_2	0.261777	0.03520897	2458	7.43496	0.0000
session2:HF_typeHF_3	0.056036	0.02885723	2458	1.94184	0.0523
session3:HF_typeHF_3	0.032059	0.02610501	2458	1.22806	0.2195
session4:HF_typeHF_3	0.153005	0.03550657	2458	4.30921	0.0000
session2:HF_typeHF_5	0.088742	0.02905853	2458	3.05390	0.0023

session3:HF_typeHF_5	0.051634	0.02623803	2458	1.96790	0.0492		
session4:HF_typeHF_5	0.179020	0.03558653	2458	5.03055	0.0000		
session2:HF_typeHF_7	0.052485	0.02862688	2458	1.83341	0.0669		
session3:HF_typeHF_7	0.010586	0.02580860	2458	0.41016	0.6817		
session4:HF_typeHF_7	0.060073	0.03506205	2458	1.71334	0.0868		
session2:HF_typeHF_8	0.103789	0.02865595	2458	3.62188	0.0003		
session3:HF_typeHF_8	0.108254	0.02602904	2458	4.15897	0.0000		
session4:HF_typeHF_8	0.216937	0.03508758	2458	6.18273	0.0000		
activitystanding:HF_typeHF_2	-0.060630	0.05544349	2458	-1.09355	0.2743		
activitywalking:HF_typeHF_2	-0.366914	0.05986955	2458	-6.12857	0.0000		
activitystanding:HF_typeHF_3	0.235171	0.05579413	2458	4.21497	0.0000		
activitywalking:HF_typeHF_3	0.251179	0.06020184	2458	4.17228	0.0000		
activitystanding:HF_typeHF_5	0.100296	0.05607680	2458	1.78854	0.0738		
activitywalking:HF_typeHF_5	-0.024548	0.06052943	2458	-0.40556	0.6851		
activitystanding:HF_typeHF_7	-0.009614	0.05520019	2458	-0.17416	0.8618		
activitywalking:HF_typeHF_7	-0.108670	0.05958032	2458	-1.82392	0.0683		
activitystanding:HF_typeHF_8	0.225002	0.05553931	2458	4.05122	0.0001		
activitywalking:HF_typeHF_8	0.190642	0.05994288	2458	3.18039	0.0015		
sexM:HF_typeHF_2	-0.113291	0.11306928	110	-1.00196	0.3186		
sexM:HF_typeHF_3	0.042413	0.11327630	110	0.37442	0.7088		
sexM:HF_typeHF_5	0.001784	0.11347985	110	0.01572	0.9875		
sexM:HF_typeHF_7	0.102673	0.11285746	110	0.90976	0.3649		
sexM:HF_typeHF_8	0.064093	0.11309340	110	0.56672	0.5721		
Correlation:							
	(Intr)	day2	sessn2	sessn3	sessn4	actvtys	actvtyw
day2	-0.307						
session2	-0.105	0.012					
session3	-0.139	-0.004	0.450				
session4	-0.175	-0.010	0.308	0.606			
activitystanding	-0.296	0.025	0.026	-0.002	-0.012		
activitywalking	-0.291	0.000	0.057	-0.001	0.142	0.447	
sexM	-0.610	-0.014	0.002	0.002	0.002	-0.002	0.000
HF_typeHF_2	-0.688	0.145	0.077	0.101	0.127	0.215	0.212
HF_typeHF_3	-0.683	0.144	0.076	0.101	0.127	0.214	0.210
HF_typeHF_5	-0.682	0.144	0.076	0.100	0.126	0.213	0.210
HF_typeHF_7	-0.689	0.146	0.077	0.101	0.127	0.216	0.212
HF_typeHF_8	-0.688	0.145	0.077	0.101	0.127	0.215	0.212
day2:HF_typeHF_2	0.181	-0.555	-0.011	0.003	0.009	-0.023	0.000
day2:HF_typeHF_3	0.179	-0.548	-0.011	0.003	0.009	-0.022	0.000
day2:HF_typeHF_5	0.177	-0.543	-0.011	0.003	0.009	-0.022	0.000
day2:HF_typeHF_7	0.182	-0.560	-0.011	0.003	0.010	-0.023	0.000
day2:HF_typeHF_8	0.180	-0.554	-0.011	0.003	0.009	-0.023	0.000
session2:HF_typeHF_2	0.075	-0.009	-0.714	-0.321	-0.220	-0.018	-0.041
session3:HF_typeHF_2	0.099	0.003	-0.318	-0.708	-0.429	0.001	0.001
session4:HF_typeHF_2	0.124	0.007	-0.218	-0.429	-0.707	0.009	-0.101
session2:HF_typeHF_3	0.075	-0.009	-0.710	-0.319	-0.219	-0.018	-0.041
session3:HF_typeHF_3	0.098	0.003	-0.317	-0.704	-0.427	0.001	0.001
session4:HF_typeHF_3	0.122	0.007	-0.216	-0.425	-0.701	0.009	-0.100
session2:HF_typeHF_5	0.074	-0.009	-0.705	-0.317	-0.217	-0.018	-0.040
session3:HF_typeHF_5	0.098	0.003	-0.315	-0.701	-0.425	0.001	0.001
session4:HF_typeHF_5	0.122	0.007	-0.216	-0.424	-0.700	0.009	-0.100
session2:HF_typeHF_7	0.075	-0.009	-0.715	-0.322	-0.221	-0.018	-0.041
session3:HF_typeHF_7	0.099	0.003	-0.320	-0.712	-0.432	0.001	0.001
session4:HF_typeHF_7	0.124	0.007	-0.219	-0.430	-0.710	0.009	-0.101

session2:HF_typeHF_8	0.075	-0.009	-0.715	-0.321	-0.220	-0.018	-0.041
session3:HF_typeHF_8	0.098	0.003	-0.318	-0.706	-0.428	0.001	0.001
session4:HF_typeHF_8	0.124	0.007	-0.219	-0.430	-0.710	0.009	-0.101
activitystanding:HF_typeHF_2	0.210	-0.018	-0.018	0.001	0.009	-0.709	-0.317
activitywalking:HF_typeHF_2	0.206	0.000	-0.040	0.001	-0.101	-0.317	-0.708
activitystanding:HF_typeHF_3	0.209	-0.018	-0.018	0.001	0.009	-0.705	-0.315
activitywalking:HF_typeHF_3	0.205	0.000	-0.040	0.001	-0.100	-0.315	-0.705
activitystanding:HF_typeHF_5	0.208	-0.018	-0.018	0.001	0.009	-0.701	-0.314
activitywalking:HF_typeHF_5	0.204	0.000	-0.040	0.001	-0.100	-0.313	-0.701
activitystanding:HF_typeHF_7	0.211	-0.018	-0.018	0.001	0.009	-0.712	-0.319
activitywalking:HF_typeHF_7	0.207	0.000	-0.041	0.001	-0.101	-0.318	-0.712
activitystanding:HF_typeHF_8	0.210	-0.018	-0.018	0.001	0.009	-0.708	-0.317
activitywalking:HF_typeHF_8	0.206	0.000	-0.040	0.001	-0.101	-0.317	-0.708
sexM:HF_typeHF_2	0.425	0.010	-0.001	-0.001	-0.002	0.001	0.000
sexM:HF_typeHF_3	0.424	0.010	-0.002	-0.001	-0.002	0.001	0.000
sexM:HF_typeHF_5	0.423	0.010	-0.001	-0.001	-0.002	0.001	0.000
sexM:HF_typeHF_7	0.425	0.010	-0.001	-0.001	-0.002	0.001	0.000
sexM:HF_typeHF_8	0.425	0.010	-0.001	-0.001	-0.002	0.001	0.000
sexM	HF_HF_2	HF_HF_3	HF_HF_5	HF_HF_7	HF_HF_8		
day2							
session2							
session3							
session4							
activitystanding							
activitywalking							
sexM							
HF_typeHF_2	0.430						
HF_typeHF_3	0.426	0.497					
HF_typeHF_5	0.426	0.497	0.493				
HF_typeHF_7	0.430	0.502	0.498	0.497			
HF_typeHF_8	0.429	0.501	0.497	0.496	0.501		
day2:HF_typeHF_2	0.013	-0.263	-0.131	-0.131	-0.132	-0.132	
day2:HF_typeHF_3	0.013	-0.131	-0.275	-0.130	-0.130	-0.131	
day2:HF_typeHF_5	0.013	-0.130	-0.128	-0.277	-0.129	-0.129	
day2:HF_typeHF_7	0.013	-0.133	-0.132	-0.132	-0.260	-0.133	
day2:HF_typeHF_8	0.013	-0.132	-0.131	-0.131	-0.131	-0.263	
session2:HF_typeHF_2	-0.001	-0.107	-0.055	-0.054	-0.055	-0.055	
session3:HF_typeHF_2	-0.001	-0.144	-0.071	-0.071	-0.072	-0.072	
session4:HF_typeHF_2	-0.002	-0.179	-0.089	-0.089	-0.090	-0.090	
session2:HF_typeHF_3	-0.001	-0.054	-0.107	-0.054	-0.054	-0.054	
session3:HF_typeHF_3	-0.001	-0.071	-0.143	-0.071	-0.071	-0.071	
session4:HF_typeHF_3	-0.002	-0.089	-0.180	-0.088	-0.089	-0.089	
session2:HF_typeHF_5	-0.001	-0.054	-0.054	-0.108	-0.054	-0.054	
session3:HF_typeHF_5	-0.001	-0.071	-0.071	-0.144	-0.071	-0.071	
session4:HF_typeHF_5	-0.002	-0.089	-0.089	-0.179	-0.089	-0.089	
session2:HF_typeHF_7	-0.001	-0.055	-0.055	-0.054	-0.106	-0.055	
session3:HF_typeHF_7	-0.001	-0.072	-0.072	-0.072	-0.143	-0.072	
session4:HF_typeHF_7	-0.002	-0.090	-0.090	-0.089	-0.178	-0.090	
session2:HF_typeHF_8	-0.001	-0.055	-0.055	-0.054	-0.055	-0.108	
session3:HF_typeHF_8	-0.001	-0.072	-0.071	-0.071	-0.072	-0.141	
session4:HF_typeHF_8	-0.002	-0.090	-0.090	-0.089	-0.090	-0.179	
activitystanding:HF_typeHF_2	0.001	-0.304	-0.152	-0.151	-0.153	-0.153	
activitywalking:HF_typeHF_2	0.000	-0.298	-0.149	-0.149	-0.150	-0.150	
activitystanding:HF_typeHF_3	0.001	-0.152	-0.303	-0.150	-0.152	-0.152	

activitywalking:HF_typeHF_3	0.000	-0.149	-0.297	-0.148	-0.149	-0.149
activitystanding:HF_typeHF_5	0.001	-0.151	-0.150	-0.305	-0.151	-0.151
activitywalking:HF_typeHF_5	0.000	-0.148	-0.147	-0.298	-0.149	-0.148
activitystanding:HF_typeHF_7	0.001	-0.153	-0.152	-0.152	-0.303	-0.153
activitywalking:HF_typeHF_7	0.000	-0.151	-0.150	-0.149	-0.297	-0.151
activitystanding:HF_typeHF_8	0.001	-0.152	-0.151	-0.151	-0.153	-0.305
activitywalking:HF_typeHF_8	0.000	-0.150	-0.149	-0.149	-0.150	-0.299
sexM:HF_typeHF_2	-0.696	-0.618	-0.306	-0.306	-0.309	-0.308
sexM:HF_typeHF_3	-0.694	-0.308	-0.619	-0.305	-0.309	-0.308
sexM:HF_typeHF_5	-0.693	-0.308	-0.305	-0.614	-0.308	-0.307
sexM:HF_typeHF_7	-0.697	-0.310	-0.307	-0.307	-0.622	-0.309
sexM:HF_typeHF_8	-0.696	-0.309	-0.306	-0.306	-0.309	-0.618

d2:HF_HF_2 d2:HF_HF_3 d2:HF_HF_5 d2:HF_HF_7

day2

session2

session3

session4

activitystanding

activitywalking

sexM

HF_typeHF_2

HF_typeHF_3

HF_typeHF_5

HF_typeHF_7

HF_typeHF_8

day2:HF_typeHF_2

day2:HF_typeHF_3 0.500

day2:HF_typeHF_5 0.494 0.488

day2:HF_typeHF_7 0.507 0.501 0.496

day2:HF_typeHF_8 0.505 0.499 0.494 0.506

session2:HF_typeHF_2 0.018 0.008 0.008 0.008

session3:HF_typeHF_2 -0.002 -0.002 -0.002 -0.002

session4:HF_typeHF_2 -0.011 -0.006 -0.007 -0.007

session2:HF_typeHF_3 0.008 0.018 0.008 0.008

session3:HF_typeHF_3 -0.002 -0.002 -0.002 -0.002

session4:HF_typeHF_3 -0.007 -0.008 -0.007 -0.007

session2:HF_typeHF_5 0.008 0.008 0.018 0.008

session3:HF_typeHF_5 -0.002 -0.002 -0.002 -0.002

session4:HF_typeHF_5 -0.007 -0.006 -0.011 -0.007

session2:HF_typeHF_7 0.008 0.008 0.008 0.018

session3:HF_typeHF_7 -0.002 -0.002 -0.002 -0.002

session4:HF_typeHF_7 -0.007 -0.006 -0.007 -0.010

session2:HF_typeHF_8 0.008 0.008 0.008 0.008

session3:HF_typeHF_8 -0.002 -0.002 -0.002 -0.002

session4:HF_typeHF_8 -0.007 -0.006 -0.007 -0.007

activitystanding:HF_typeHF_2 0.031 0.016 0.016 0.016

activitywalking:HF_typeHF_2 0.000 0.000 0.000 0.000

activitystanding:HF_typeHF_3 0.016 0.030 0.016 0.016

activitywalking:HF_typeHF_3 0.000 0.000 0.000 0.000

activitystanding:HF_typeHF_5 0.016 0.016 0.030 0.016

activitywalking:HF_typeHF_5 0.000 0.000 0.000 0.000

activitystanding:HF_typeHF_7 0.016 0.016 0.016 0.029

activitywalking:HF_typeHF_7 0.000 0.000 0.000 -0.001

activitystanding:HF_typeHF_8 0.016 0.016 0.016 0.016

activitywalking:HF_typeHF_8	0.000	0.000	0.000	0.000
sexM:HF_typeHF_2	-0.018	-0.011	-0.011	-0.009
sexM:HF_typeHF_3	-0.011	-0.009	-0.011	-0.009
sexM:HF_typeHF_5	-0.011	-0.011	-0.019	-0.009
sexM:HF_typeHF_7	-0.009	-0.009	-0.009	-0.010
sexM:HF_typeHF_8	-0.011	-0.011	-0.011	-0.009
	d2:HF_HF_8	s2:HF_HF_2	s3:HF_HF_2	s4:HF_HF_2
day2				
session2				
session3				
session4				
activitystanding				
activitywalking				
sexM				
HF_typeHF_2				
HF_typeHF_3				
HF_typeHF_5				
HF_typeHF_7				
HF_typeHF_8				
day2:HF_typeHF_2				
day2:HF_typeHF_3				
day2:HF_typeHF_5				
day2:HF_typeHF_7				
day2:HF_typeHF_8				
session2:HF_typeHF_2	0.008			
session3:HF_typeHF_2	-0.002	0.454		
session4:HF_typeHF_2	-0.007	0.306	0.609	
session2:HF_typeHF_3	0.008	0.507	0.226	0.155
session3:HF_typeHF_3	-0.002	0.226	0.498	0.302
session4:HF_typeHF_3	-0.007	0.155	0.301	0.496
session2:HF_typeHF_5	0.008	0.503	0.224	0.154
session3:HF_typeHF_5	-0.002	0.225	0.496	0.300
session4:HF_typeHF_5	-0.007	0.154	0.300	0.495
session2:HF_typeHF_7	0.008	0.511	0.228	0.156
session3:HF_typeHF_7	-0.002	0.229	0.504	0.305
session4:HF_typeHF_7	-0.007	0.156	0.305	0.502
session2:HF_typeHF_8	0.018	0.511	0.228	0.156
session3:HF_typeHF_8	-0.003	0.227	0.500	0.303
session4:HF_typeHF_8	-0.010	0.156	0.304	0.502
activitystanding:HF_typeHF_2	0.016	0.025	-0.001	-0.011
activitywalking:HF_typeHF_2	0.000	0.055	0.000	0.141
activitystanding:HF_typeHF_3	0.016	0.013	-0.001	-0.006
activitywalking:HF_typeHF_3	0.000	0.029	0.000	0.071
activitystanding:HF_typeHF_5	0.016	0.013	-0.001	-0.006
activitywalking:HF_typeHF_5	0.000	0.029	0.000	0.070
activitystanding:HF_typeHF_7	0.016	0.013	-0.001	-0.006
activitywalking:HF_typeHF_7	0.000	0.029	0.000	0.072
activitystanding:HF_typeHF_8	0.030	0.013	-0.001	-0.006
activitywalking:HF_typeHF_8	0.000	0.029	0.000	0.071
sexM:HF_typeHF_2	-0.011	0.001	0.001	0.001
sexM:HF_typeHF_3	-0.011	0.001	0.001	0.001
sexM:HF_typeHF_5	-0.011	0.001	0.001	0.001
sexM:HF_typeHF_7	-0.009	0.001	0.001	0.001
sexM:HF_typeHF_8	-0.018	0.001	0.001	0.001

	s2:HF_HF_3	s3:HF_HF_3	s4:HF_HF_3	s2:HF_HF_5
day2				
session2				
session3				
session4				
activitystanding				
activitywalking				
sexM				
HF_typeHF_2				
HF_typeHF_3				
HF_typeHF_5				
HF_typeHF_7				
HF_typeHF_8				
day2:HF_typeHF_2				
day2:HF_typeHF_3				
day2:HF_typeHF_5				
day2:HF_typeHF_7				
day2:HF_typeHF_8				
session2:HF_typeHF_2				
session3:HF_typeHF_2				
session4:HF_typeHF_2				
session2:HF_typeHF_3				
session3:HF_typeHF_3	0.453			
session4:HF_typeHF_3	0.305	0.607		
session2:HF_typeHF_5	0.500	0.223	0.152	
session3:HF_typeHF_5	0.224	0.494	0.298	0.452
session4:HF_typeHF_5	0.153	0.299	0.491	0.305
session2:HF_typeHF_7	0.508	0.227	0.155	0.504
session3:HF_typeHF_7	0.227	0.502	0.303	0.226
session4:HF_typeHF_7	0.155	0.303	0.498	0.154
session2:HF_typeHF_8	0.507	0.226	0.155	0.504
session3:HF_typeHF_8	0.225	0.497	0.300	0.224
session4:HF_typeHF_8	0.155	0.303	0.498	0.154
activitystanding:HF_typeHF_2	0.013	-0.001	-0.006	0.013
activitywalking:HF_typeHF_2	0.029	0.000	0.071	0.029
activitystanding:HF_typeHF_3	0.026	-0.001	-0.011	0.013
activitywalking:HF_typeHF_3	0.055	0.000	0.140	0.028
activitystanding:HF_typeHF_5	0.013	-0.001	-0.006	0.026
activitywalking:HF_typeHF_5	0.028	0.000	0.070	0.055
activitystanding:HF_typeHF_7	0.013	-0.001	-0.006	0.013
activitywalking:HF_typeHF_7	0.029	0.000	0.071	0.029
activitystanding:HF_typeHF_8	0.013	-0.001	-0.006	0.013
activitywalking:HF_typeHF_8	0.029	0.000	0.071	0.029
sexM:HF_typeHF_2	0.001	0.001	0.001	0.001
sexM:HF_typeHF_3	0.001	0.001	0.002	0.001
sexM:HF_typeHF_5	0.001	0.001	0.001	0.001
sexM:HF_typeHF_7	0.001	0.001	0.001	0.001
sexM:HF_typeHF_8	0.001	0.001	0.001	0.001
	s3:HF_HF_5	s4:HF_HF_5	s2:HF_HF_7	s3:HF_HF_7
day2				
session2				
session3				
session4				
activitystanding				

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activitywalking
sexM
HF_typeHF_2
HF_typeHF_3
HF_typeHF_5
HF_typeHF_7
HF_typeHF_8
day2:HF_typeHF_2
day2:HF_typeHF_3
day2:HF_typeHF_5
day2:HF_typeHF_7
day2:HF_typeHF_8
session2:HF_typeHF_2
session3:HF_typeHF_2
session4:HF_typeHF_2
session2:HF_typeHF_3
session3:HF_typeHF_3
session4:HF_typeHF_3
session2:HF_typeHF_5
session3:HF_typeHF_5
session4:HF_typeHF_5      0.609
session2:HF_typeHF_7      0.225      0.154
session3:HF_typeHF_7      0.499      0.302      0.451
session4:HF_typeHF_7      0.302      0.497      0.304      0.607
session2:HF_typeHF_8      0.225      0.154      0.511      0.229
session3:HF_typeHF_8      0.495      0.300      0.227      0.503
session4:HF_typeHF_8      0.301      0.497      0.157      0.306
activitystanding:HF_typeHF_2 -0.001      -0.006      0.013      -0.001
activitywalking:HF_typeHF_2  0.000      0.071      0.029      0.000
activitystanding:HF_typeHF_3 -0.001      -0.006      0.013      -0.001
activitywalking:HF_typeHF_3  0.000      0.070      0.029      0.000
activitystanding:HF_typeHF_5 -0.001      -0.011      0.013      -0.001
activitywalking:HF_typeHF_5  0.000      0.141      0.029      0.000
activitystanding:HF_typeHF_7 -0.001      -0.006      0.025      -0.001
activitywalking:HF_typeHF_7  0.000      0.071      0.055      0.000
activitystanding:HF_typeHF_8 -0.001      -0.006      0.013      -0.001
activitywalking:HF_typeHF_8  0.000      0.070      0.029      0.000
sexM:HF_typeHF_2      0.001      0.001      0.001      0.001
sexM:HF_typeHF_3      0.001      0.001      0.001      0.001
sexM:HF_typeHF_5      0.001      0.001      0.001      0.001
sexM:HF_typeHF_7      0.001      0.001      0.001      0.001
sexM:HF_typeHF_8      0.001      0.001      0.001      0.001
s4:HF_HF_7 s2:HF_HF_8 s3:HF_HF_8 s4:HF_HF_8
day2
session2
session3
session4
activitystanding
activitywalking
sexM
HF_typeHF_2
HF_typeHF_3
HF_typeHF_5
HF_typeHF_7

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HF_typeHF_8				
day2:HF_typeHF_2				
day2:HF_typeHF_3				
day2:HF_typeHF_5				
day2:HF_typeHF_7				
day2:HF_typeHF_8				
session2:HF_typeHF_2				
session3:HF_typeHF_2				
session4:HF_typeHF_2				
session2:HF_typeHF_3				
session3:HF_typeHF_3				
session4:HF_typeHF_3				
session2:HF_typeHF_5				
session3:HF_typeHF_5				
session4:HF_typeHF_5				
session2:HF_typeHF_7				
session3:HF_typeHF_7				
session4:HF_typeHF_7				
session2:HF_typeHF_8	0.157			
session3:HF_typeHF_8	0.304	0.451		
session4:HF_typeHF_8	0.504	0.316	0.604	
activitystanding:HF_typeHF_2	-0.006	0.013	-0.001	-0.006
activitywalking:HF_typeHF_2	0.072	0.029	0.000	0.072
activitystanding:HF_typeHF_3	-0.006	0.013	-0.001	-0.006
activitywalking:HF_typeHF_3	0.071	0.029	0.000	0.071
activitystanding:HF_typeHF_5	-0.006	0.013	-0.001	-0.006
activitywalking:HF_typeHF_5	0.071	0.029	0.000	0.071
activitystanding:HF_typeHF_7	-0.012	0.013	-0.001	-0.006
activitywalking:HF_typeHF_7	0.140	0.029	0.000	0.072
activitystanding:HF_typeHF_8	-0.006	0.025	-0.001	-0.011
activitywalking:HF_typeHF_8	0.071	0.057	-0.003	0.141
sexM:HF_typeHF_2	0.001	0.001	0.001	0.001
sexM:HF_typeHF_3	0.001	0.001	0.001	0.001
sexM:HF_typeHF_5	0.001	0.001	0.001	0.001
sexM:HF_typeHF_7	0.001	0.001	0.001	0.001
sexM:HF_typeHF_8	0.001	0.001	0.000	0.001
actvtys:HF_HF_2 actvttyw:HF_HF_2 actvtys:HF_HF_3				
day2				
session2				
session3				
session4				
activitystanding				
activitywalking				
sexM				
HF_typeHF_2				
HF_typeHF_3				
HF_typeHF_5				
HF_typeHF_7				
HF_typeHF_8				
day2:HF_typeHF_2				
day2:HF_typeHF_3				
day2:HF_typeHF_5				
day2:HF_typeHF_7				
day2:HF_typeHF_8				

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session2:HF_typeHF_2
session3:HF_typeHF_2
session4:HF_typeHF_2
session2:HF_typeHF_3
session3:HF_typeHF_3
session4:HF_typeHF_3
session2:HF_typeHF_5
session3:HF_typeHF_5
session4:HF_typeHF_5
session2:HF_typeHF_7
session3:HF_typeHF_7
session4:HF_typeHF_7
session2:HF_typeHF_8
session3:HF_typeHF_8
session4:HF_typeHF_8
activitystanding:HF_typeHF_2
activitywalking:HF_typeHF_2 0.446
activitystanding:HF_typeHF_3 0.500 0.223
activitywalking:HF_typeHF_3 0.223 0.499 0.447
activitystanding:HF_typeHF_5 0.497 0.222 0.494
activitywalking:HF_typeHF_5 0.222 0.496 0.221
activitystanding:HF_typeHF_7 0.505 0.226 0.502
activitywalking:HF_typeHF_7 0.226 0.504 0.224
activitystanding:HF_typeHF_8 0.502 0.224 0.499
activitywalking:HF_typeHF_8 0.224 0.501 0.223
sexM:HF_typeHF_2 -0.001 0.000 -0.001
sexM:HF_typeHF_3 -0.001 0.000 -0.001
sexM:HF_typeHF_5 -0.001 0.000 -0.001
sexM:HF_typeHF_7 -0.001 0.000 -0.001
sexM:HF_typeHF_8 -0.001 0.000 -0.001
actvttyw:HF_HF_3 actvttys:HF_HF_5 actvttyw:HF_HF_5

day2
session2
session3
session4
activitystanding
activitywalking
sexM
HF_typeHF_2
HF_typeHF_3
HF_typeHF_5
HF_typeHF_7
HF_typeHF_8
day2:HF_typeHF_2
day2:HF_typeHF_3
day2:HF_typeHF_5
day2:HF_typeHF_7
day2:HF_typeHF_8
session2:HF_typeHF_2
session3:HF_typeHF_2
session4:HF_typeHF_2
session2:HF_typeHF_3
session3:HF_typeHF_3
session4:HF_typeHF_3

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session2:HF_typeHF_5
session3:HF_typeHF_5
session4:HF_typeHF_5
session2:HF_typeHF_7
session3:HF_typeHF_7
session4:HF_typeHF_7
session2:HF_typeHF_8
session3:HF_typeHF_8
session4:HF_typeHF_8
activitystanding:HF_typeHF_2
activitywalking:HF_typeHF_2
activitystanding:HF_typeHF_3
activitywalking:HF_typeHF_3
activitystanding:HF_typeHF_5 0.221
activitywalking:HF_typeHF_5 0.494          0.446
activitystanding:HF_typeHF_7 0.224          0.499          0.223
activitywalking:HF_typeHF_7 0.502          0.223          0.499
activitystanding:HF_typeHF_8 0.223          0.496          0.222
activitywalking:HF_typeHF_8 0.499          0.222          0.496
sexM:HF_typeHF_2          0.000          -0.001          0.000
sexM:HF_typeHF_3          0.000          -0.001          0.000
sexM:HF_typeHF_5          0.000          -0.001          0.000
sexM:HF_typeHF_7          0.000          -0.001          0.000
sexM:HF_typeHF_8          0.000          -0.001          0.000
                                actvtys:HF_HF_7 actvtys:HF_HF_7 actvtys:HF_HF_8
day2
session2
session3
session4
activitystanding
activitywalking
sexM
HF_typeHF_2
HF_typeHF_3
HF_typeHF_5
HF_typeHF_7
HF_typeHF_8
day2:HF_typeHF_2
day2:HF_typeHF_3
day2:HF_typeHF_5
day2:HF_typeHF_7
day2:HF_typeHF_8
session2:HF_typeHF_2
session3:HF_typeHF_2
session4:HF_typeHF_2
session2:HF_typeHF_3
session3:HF_typeHF_3
session4:HF_typeHF_3
session2:HF_typeHF_5
session3:HF_typeHF_5
session4:HF_typeHF_5
session2:HF_typeHF_7
session3:HF_typeHF_7
session4:HF_typeHF_7

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session2:HF_typeHF_8
session3:HF_typeHF_8
session4:HF_typeHF_8
activitystanding:HF_typeHF_2
activitywalking:HF_typeHF_2
activitystanding:HF_typeHF_3
activitywalking:HF_typeHF_3
activitystanding:HF_typeHF_5
activitywalking:HF_typeHF_5
activitystanding:HF_typeHF_7
activitywalking:HF_typeHF_7 0.447
activitystanding:HF_typeHF_8 0.504 0.225
activitywalking:HF_typeHF_8 0.225 0.504 0.447
sexM:HF_typeHF_2 -0.001 0.000 -0.001
sexM:HF_typeHF_3 -0.001 0.000 -0.001
sexM:HF_typeHF_5 -0.001 0.000 -0.001
sexM:HF_typeHF_7 -0.001 0.000 -0.001
sexM:HF_typeHF_8 -0.001 0.000 0.000
actvttyw:HF_HF_8 sM:HF_HF_2 sM:HF_HF_3 sM:HF_HF_5

day2
session2
session3
session4
activitystanding
activitywalking
sexM
HF_typeHF_2
HF_typeHF_3
HF_typeHF_5
HF_typeHF_7
HF_typeHF_8
day2:HF_typeHF_2
day2:HF_typeHF_3
day2:HF_typeHF_5
day2:HF_typeHF_7
day2:HF_typeHF_8
session2:HF_typeHF_2
session3:HF_typeHF_2
session4:HF_typeHF_2
session2:HF_typeHF_3
session3:HF_typeHF_3
session4:HF_typeHF_3
session2:HF_typeHF_5
session3:HF_typeHF_5
session4:HF_typeHF_5
session2:HF_typeHF_7
session3:HF_typeHF_7
session4:HF_typeHF_7
session2:HF_typeHF_8
session3:HF_typeHF_8
session4:HF_typeHF_8
activitystanding:HF_typeHF_2
activitywalking:HF_typeHF_2
activitystanding:HF_typeHF_3

```


activitywalking:HF_typeHF_3				
activitystanding:HF_typeHF_5				
activitywalking:HF_typeHF_5				
activitystanding:HF_typeHF_7				
activitywalking:HF_typeHF_7				
activitystanding:HF_typeHF_8				
activitywalking:HF_typeHF_8				
sexM:HF_typeHF_2	0.000			
sexM:HF_typeHF_3	0.000	0.500		
sexM:HF_typeHF_5	0.000	0.499	0.498	
sexM:HF_typeHF_7	0.000	0.501	0.500	0.499
sexM:HF_typeHF_8	0.001	0.501	0.500	0.499
	sM:HF_HF_7			
day2				
session2				
session3				
session4				
activitystanding				
activitywalking				
sexM				
HF_typeHF_2				
HF_typeHF_3				
HF_typeHF_5				
HF_typeHF_7				
HF_typeHF_8				
day2:HF_typeHF_2				
day2:HF_typeHF_3				
day2:HF_typeHF_5				
day2:HF_typeHF_7				
day2:HF_typeHF_8				
session2:HF_typeHF_2				
session3:HF_typeHF_2				
session4:HF_typeHF_2				
session2:HF_typeHF_3				
session3:HF_typeHF_3				
session4:HF_typeHF_3				
session2:HF_typeHF_5				
session3:HF_typeHF_5				
session4:HF_typeHF_5				
session2:HF_typeHF_7				
session3:HF_typeHF_7				
session4:HF_typeHF_7				
session2:HF_typeHF_8				
session3:HF_typeHF_8				
session4:HF_typeHF_8				
activitystanding:HF_typeHF_2				
activitywalking:HF_typeHF_2				
activitystanding:HF_typeHF_3				
activitywalking:HF_typeHF_3				
activitystanding:HF_typeHF_5				
activitywalking:HF_typeHF_5				
activitystanding:HF_typeHF_7				
activitywalking:HF_typeHF_7				
activitystanding:HF_typeHF_8				

```

activitywalking:HF_typeHF_8
sexM:HF_typeHF_2
sexM:HF_typeHF_3
sexM:HF_typeHF_5
sexM:HF_typeHF_7
sexM:HF_typeHF_8          0.501

```

Standardized Within-Group Residuals:

Min	Q1	Med	Q3	Max
-5.64369441	-0.41892442	0.06251821	0.54174900	3.41944555

Number of Observations: 2638

Number of Groups:

person	HF_type	%in%	person
24			144

```

anova_md17b <- anova(md17_reml, md17b_reml)
# anova_md17b_clean <- data.frame(
#   Model = c("Model 6: Assumes Compound Symmetry", "Model 7: Assumes Auto-regressive AR(1)"),
#   Df = as.numeric(anova_md17b$df),
#   AIC = round(as.numeric(anova_md17b$AIC)),
#   BIC = round(as.numeric(anova_md17b$BIC)),
#   LogLik = round(as.numeric(anova_md17b$logLik)),
#   LRT_Stat = c("--", round(anova_md17b$L.Ratio[2], 2)),
#   p_value = anova_md17b$p-value[2])

print(anova_md17b)

```

	Model	df	AIC	BIC	logLik
md17_reml	1	54	-401.9296	-85.52129	254.9648
md17b_reml	2	54	-551.9934	-235.58506	329.9967

The selected hierarchical structure (Model 6) effectively partitions the data's variance into three distinct components: global participant-level baselines, individual thermal adaptation patterns (represented by the random slope for day), and nested sensor-level deviations. This framework accounts for the inherent dependencies in the data while acknowledging that heat flow sensitivity is not uniform across the study population.

However, this still imposes a strict assumption of conditional independence for the remaining residual errors. It has a compound symmetry structure, assuming a constant correlation structure where the correlation between session 1 and session 2 is identical to the correlation between session 1 and session 4, ignoring the fact that observations closer in time are typically more highly correlated. As such, we examined implementing an autoregressive (AR1) correlation structure for the residuals in Model 7. We assumed this was nested within person, sensor, and day, ensuring independence between people, sensor location, and days. We did not assume independence between activity because the activities were done in order, with session 1 consisting of sitting, then standing, then walking, then a break, and then session two beginning with the same cycle (although sessions 2 and 4 did not have walking). This assumes that these measurements were equally spaced in time, which is likely not true - timestamp data could improve this assumption.

The correlation structure has a ϕ of 0.5745, meaning that the correlation between session 1 and 2 is ~ 0.57 , and session 1 & 3 is $\sim 0.57^2$. This is fairly high, suggesting the previous data point strongly influences the next one. Since we have the data segmented by session and activity, it suggests that sitting in session 1 impacts standing in session 1, and walking in session 1 impacts sitting in session 2. Additionally, Table 7 shows that Model 7 has significantly lower AIC and BIC, and is statistically significant in the likelihood ratio test against Model 6, suggesting an AR1 covariance structure is a significant improvement on the Compound Symmetry structure, and that there is a strong temporal auto-correlation.

```
mdl8_reml <- lme(
  fixed = log(HF_value) ~ (day + session + activity + sex) * HF_type,
  random = list(
    person = ~ 1 + day,
    HF_type = ~ 1
  ),
  correlation = corAR1(form = ~ 1 | person/HF_type/day),
  weights = varIdent(form = ~ 1 | HF_type),
  data = data_clean,
  method = "REML",
  control = lmeControl(msMaxIter = 200) #basically, increase iterations
)
summary(mdl8_reml)
```

Linear mixed-effects model fit by REML

Data: data_clean
 AIC BIC logLik
 -943.6603 -597.9549 530.8301

Random effects:

Formula: ~1 + day | person
 Structure: General positive-definite, Log-Cholesky parametrization
 StdDev Corr
 (Intercept) 0.08206226 (Intr)
 day2 0.13858581 -0.919

Formula: ~1 | HF_type %in% person
 (Intercept) Residual
 StdDev: 0.1599517 0.2770674

Correlation Structure: AR(1)

Formula: ~1 | person/HF_type/day
 Parameter estimate(s):
 Phi
 0.6268998

Variance function:

Structure: Different standard deviations per stratum
 Formula: ~1 | HF_type
 Parameter estimates:

	HF_16	HF_2	HF_3	HF_5	HF_7	HF_8
	1.0000000	1.2659210	0.7236136	0.3495410	0.9474615	0.9663790

Fixed effects: log(HF_value) ~ (day + session + activity + sex) * HF_type

	Value	Std.Error	DF	t-value	p-value
(Intercept)	4.489829	0.07016096	2458	63.99327	0.0000
day2	-0.143983	0.05552491	2458	-2.59313	0.0096
session2	-0.061995	0.03640623	2458	-1.70287	0.0887
session3	-0.093244	0.03972797	2458	-2.34705	0.0190
session4	-0.227054	0.05004767	2458	-4.53675	0.0000
activitystanding	-0.090242	0.01737844	2458	-5.19276	0.0000
activitywalking	0.037818	0.02554203	2458	1.48061	0.1388
sexM	-0.003900	0.08197708	22	-0.04758	0.9625
HF_typeHF_2	-0.717918	0.10311020	110	-6.96263	0.0000
HF_typeHF_3	-0.632310	0.09018661	110	-7.01113	0.0000
HF_typeHF_5	-0.425352	0.08421237	110	-5.05094	0.0000

HF_typeHF_7	-0.164855	0.09429294	110	-1.74832	0.0832		
HF_typeHF_8	-0.915086	0.09498788	110	-9.63372	0.0000		
day2:HF_typeHF_2	0.467856	0.07670804	2458	6.09918	0.0000		
day2:HF_typeHF_3	0.463269	0.05939741	2458	7.79947	0.0000		
day2:HF_typeHF_5	0.410477	0.05122773	2458	8.01278	0.0000		
day2:HF_typeHF_7	0.308130	0.06516484	2458	4.72847	0.0000		
day2:HF_typeHF_8	0.459607	0.06641339	2458	6.92040	0.0000		
session2:HF_typeHF_2	0.101553	0.05843919	2458	1.73775	0.0824		
session3:HF_typeHF_2	0.112863	0.06417811	2458	1.75859	0.0788		
session4:HF_typeHF_2	0.311092	0.08063282	2458	3.85814	0.0001		
session2:HF_typeHF_3	0.035569	0.04498925	2458	0.79062	0.4292		
session3:HF_typeHF_3	0.076137	0.04926337	2458	1.54552	0.1223		
session4:HF_typeHF_3	0.210838	0.06205554	2458	3.39757	0.0007		
session2:HF_typeHF_5	0.070522	0.03862771	2458	1.82568	0.0680		
session3:HF_typeHF_5	0.086942	0.04219522	2458	2.06046	0.0395		
session4:HF_typeHF_5	0.228198	0.05313121	2458	4.29500	0.0000		
session2:HF_typeHF_7	0.034914	0.04983452	2458	0.70061	0.4836		
session3:HF_typeHF_7	0.029736	0.05441924	2458	0.54642	0.5848		
session4:HF_typeHF_7	0.103268	0.06852505	2458	1.50701	0.1319		
session2:HF_typeHF_8	0.070063	0.05048580	2458	1.38777	0.1653		
session3:HF_typeHF_8	0.156155	0.05531619	2458	2.82296	0.0048		
session4:HF_typeHF_8	0.269100	0.06946587	2458	3.87384	0.0001		
activitystanding:HF_typeHF_2	-0.088725	0.02772177	2458	-3.20057	0.0014		
activitywalking:HF_typeHF_2	-0.377094	0.04103853	2458	-9.18878	0.0000		
activitystanding:HF_typeHF_3	0.189652	0.02143230	2458	8.84889	0.0000		
activitywalking:HF_typeHF_3	0.220204	0.03159297	2458	6.97004	0.0000		
activitystanding:HF_typeHF_5	0.064069	0.01842463	2458	3.47738	0.0005		
activitywalking:HF_typeHF_5	-0.049993	0.02710705	2458	-1.84427	0.0653		
activitystanding:HF_typeHF_7	-0.031394	0.02365825	2458	-1.32698	0.1846		
activitywalking:HF_typeHF_7	-0.114153	0.03491448	2458	-3.26952	0.0011		
activitystanding:HF_typeHF_8	0.176273	0.02406689	2458	7.32429	0.0000		
activitywalking:HF_typeHF_8	0.144897	0.03547447	2458	4.08455	0.0000		
sexM:HF_typeHF_2	-0.072418	0.12002677	110	-0.60335	0.5475		
sexM:HF_typeHF_3	0.070122	0.10978193	110	0.63874	0.5243		
sexM:HF_typeHF_5	0.020344	0.10556862	110	0.19270	0.8475		
sexM:HF_typeHF_7	0.131129	0.11298234	110	1.16061	0.2483		
sexM:HF_typeHF_8	0.093172	0.11372910	110	0.81925	0.4144		
Correlation:							
	(Intr)	day2	sessn2	sessn3	sessn4	actvtys	actvtyw
day2	-0.392						
session2	-0.296	0.021					
session3	-0.330	-0.004	0.606				
session4	-0.371	-0.009	0.550	0.689			
activitystanding	-0.179	0.012	0.102	0.099	0.155		
activitywalking	-0.251	0.003	0.414	0.197	0.477	0.412	
sexM	-0.572	-0.019	0.003	0.002	0.003	0.000	0.003
HF_typeHF_2	-0.635	0.191	0.201	0.225	0.252	0.122	0.171
HF_typeHF_3	-0.727	0.218	0.230	0.257	0.289	0.140	0.195
HF_typeHF_5	-0.778	0.234	0.246	0.275	0.309	0.149	0.209
HF_typeHF_7	-0.695	0.209	0.220	0.246	0.276	0.133	0.187
HF_typeHF_8	-0.690	0.207	0.218	0.244	0.274	0.132	0.185
day2:HF_typeHF_2	0.203	-0.536	-0.015	0.003	0.007	-0.009	-0.002
day2:HF_typeHF_3	0.262	-0.692	-0.019	0.003	0.008	-0.011	-0.003
day2:HF_typeHF_5	0.304	-0.803	-0.022	0.004	0.010	-0.013	-0.003

day2:HF_typeHF_7	0.239	-0.631	-0.018	0.003	0.008	-0.010	-0.002
day2:HF_typeHF_8	0.234	-0.619	-0.017	0.003	0.008	-0.010	-0.002
session2:HF_typeHF_2	0.184	-0.013	-0.623	-0.378	-0.342	-0.064	-0.258
session3:HF_typeHF_2	0.204	0.002	-0.375	-0.619	-0.427	-0.061	-0.122
session4:HF_typeHF_2	0.230	0.006	-0.341	-0.428	-0.621	-0.096	-0.296
session2:HF_typeHF_3	0.239	-0.017	-0.809	-0.491	-0.445	-0.083	-0.335
session3:HF_typeHF_3	0.266	0.003	-0.489	-0.806	-0.556	-0.079	-0.159
session4:HF_typeHF_3	0.299	0.007	-0.443	-0.556	-0.806	-0.125	-0.384
session2:HF_typeHF_5	0.279	-0.019	-0.942	-0.571	-0.518	-0.097	-0.390
session3:HF_typeHF_5	0.311	0.003	-0.571	-0.942	-0.649	-0.093	-0.185
session4:HF_typeHF_5	0.349	0.009	-0.518	-0.649	-0.942	-0.146	-0.449
session2:HF_typeHF_7	0.216	-0.015	-0.731	-0.443	-0.401	-0.075	-0.302
session3:HF_typeHF_7	0.241	0.003	-0.443	-0.730	-0.503	-0.072	-0.144
session4:HF_typeHF_7	0.271	0.007	-0.401	-0.504	-0.730	-0.113	-0.348
session2:HF_typeHF_8	0.213	-0.015	-0.721	-0.437	-0.396	-0.074	-0.298
session3:HF_typeHF_8	0.237	0.003	-0.435	-0.718	-0.495	-0.071	-0.141
session4:HF_typeHF_8	0.267	0.007	-0.396	-0.497	-0.720	-0.112	-0.343
activitystanding:HF_typeHF_2	0.112	-0.008	-0.064	-0.062	-0.097	-0.627	-0.258
activitywalking:HF_typeHF_2	0.156	-0.002	-0.257	-0.123	-0.297	-0.257	-0.622
activitystanding:HF_typeHF_3	0.145	-0.010	-0.083	-0.080	-0.126	-0.811	-0.334
activitywalking:HF_typeHF_3	0.203	-0.002	-0.334	-0.159	-0.385	-0.333	-0.808
activitystanding:HF_typeHF_5	0.169	-0.011	-0.097	-0.093	-0.146	-0.943	-0.389
activitywalking:HF_typeHF_5	0.237	-0.003	-0.390	-0.186	-0.449	-0.388	-0.942
activitystanding:HF_typeHF_7	0.132	-0.009	-0.075	-0.072	-0.114	-0.735	-0.303
activitywalking:HF_typeHF_7	0.184	-0.002	-0.303	-0.144	-0.349	-0.302	-0.732
activitystanding:HF_typeHF_8	0.129	-0.009	-0.074	-0.071	-0.112	-0.722	-0.298
activitywalking:HF_typeHF_8	0.181	-0.002	-0.298	-0.142	-0.343	-0.297	-0.720
sexM:HF_typeHF_2	0.380	0.013	-0.002	-0.001	-0.002	0.000	-0.002
sexM:HF_typeHF_3	0.416	0.014	-0.002	-0.001	-0.002	0.000	-0.002
sexM:HF_typeHF_5	0.432	0.015	-0.002	-0.001	-0.002	0.000	-0.002
sexM:HF_typeHF_7	0.404	0.014	-0.002	-0.001	-0.002	0.000	-0.002
sexM:HF_typeHF_8	0.401	0.014	-0.002	-0.001	-0.002	0.000	-0.002

sexM	HF_HF_2	HF_HF_3	HF_HF_5	HF_HF_7	HF_HF_8
------	---------	---------	---------	---------	---------

day2

session2

session3

session4

activitystanding

activitywalking

sexM

HF_typeHF_2	0.379					
HF_typeHF_3	0.433	0.495				
HF_typeHF_5	0.463	0.530	0.606			
HF_typeHF_7	0.414	0.473	0.541	0.579		
HF_typeHF_8	0.411	0.470	0.537	0.575	0.513	
day2:HF_typeHF_2	0.014	-0.359	-0.159	-0.170	-0.151	-0.151
day2:HF_typeHF_3	0.018	-0.179	-0.326	-0.220	-0.195	-0.195
day2:HF_typeHF_5	0.021	-0.208	-0.237	-0.297	-0.226	-0.226
day2:HF_typeHF_7	0.016	-0.163	-0.186	-0.199	-0.332	-0.177
day2:HF_typeHF_8	0.016	-0.160	-0.183	-0.197	-0.174	-0.337
session2:HF_typeHF_2	-0.002	-0.325	-0.143	-0.153	-0.137	-0.136
session3:HF_typeHF_2	-0.001	-0.363	-0.159	-0.170	-0.152	-0.151
session4:HF_typeHF_2	-0.002	-0.406	-0.179	-0.192	-0.171	-0.170
session2:HF_typeHF_3	-0.002	-0.163	-0.285	-0.199	-0.178	-0.177

session3:HF_typeHF_3	-0.001	-0.181	-0.319	-0.222	-0.198	-0.197
session4:HF_typeHF_3	-0.002	-0.204	-0.358	-0.249	-0.223	-0.221
session2:HF_typeHF_5	-0.003	-0.190	-0.217	-0.262	-0.207	-0.206
session3:HF_typeHF_5	-0.001	-0.211	-0.242	-0.292	-0.231	-0.230
session4:HF_typeHF_5	-0.002	-0.238	-0.272	-0.328	-0.260	-0.258
session2:HF_typeHF_7	-0.002	-0.147	-0.168	-0.180	-0.300	-0.160
session3:HF_typeHF_7	-0.001	-0.164	-0.188	-0.201	-0.337	-0.178
session4:HF_typeHF_7	-0.002	-0.184	-0.211	-0.226	-0.377	-0.200
session2:HF_typeHF_8	-0.002	-0.145	-0.166	-0.178	-0.159	-0.304
session3:HF_typeHF_8	-0.001	-0.161	-0.185	-0.198	-0.176	-0.336
session4:HF_typeHF_8	-0.002	-0.182	-0.208	-0.223	-0.199	-0.379
activitystanding:HF_typeHF_2	0.000	-0.197	-0.087	-0.094	-0.084	-0.083
activitywalking:HF_typeHF_2	-0.002	-0.272	-0.122	-0.130	-0.116	-0.115
activitystanding:HF_typeHF_3	0.000	-0.099	-0.173	-0.121	-0.108	-0.107
activitywalking:HF_typeHF_3	-0.002	-0.138	-0.241	-0.169	-0.151	-0.150
activitystanding:HF_typeHF_5	0.000	-0.115	-0.132	-0.159	-0.126	-0.125
activitywalking:HF_typeHF_5	-0.003	-0.161	-0.184	-0.221	-0.176	-0.175
activitystanding:HF_typeHF_7	0.000	-0.090	-0.102	-0.110	-0.183	-0.097
activitywalking:HF_typeHF_7	-0.002	-0.125	-0.143	-0.153	-0.254	-0.136
activitystanding:HF_typeHF_8	0.000	-0.088	-0.101	-0.108	-0.096	-0.184
activitywalking:HF_typeHF_8	-0.002	-0.123	-0.141	-0.151	-0.134	-0.255
sexM:HF_typeHF_2	-0.665	-0.567	-0.295	-0.316	-0.283	-0.280
sexM:HF_typeHF_3	-0.727	-0.282	-0.601	-0.345	-0.309	-0.306
sexM:HF_typeHF_5	-0.757	-0.293	-0.335	-0.615	-0.321	-0.318
sexM:HF_typeHF_7	-0.706	-0.275	-0.314	-0.336	-0.592	-0.298
sexM:HF_typeHF_8	-0.702	-0.272	-0.311	-0.333	-0.298	-0.585

d2:HF_HF_2 d2:HF_HF_3 d2:HF_HF_5 d2:HF_HF_7

day2

session2

session3

session4

activitystanding

activitywalking

sexM

HF_typeHF_2

HF_typeHF_3

HF_typeHF_5

HF_typeHF_7

HF_typeHF_8

day2:HF_typeHF_2

day2:HF_typeHF_3 0.503

day2:HF_typeHF_5 0.584 0.755

day2:HF_typeHF_7 0.457 0.590 0.684

day2:HF_typeHF_8 0.450 0.582 0.675 0.527

session2:HF_typeHF_2 0.027 0.012 0.014 0.011

session3:HF_typeHF_2 -0.001 -0.002 -0.002 -0.002

session4:HF_typeHF_2 -0.008 -0.005 -0.006 -0.005

session2:HF_typeHF_3 0.012 0.026 0.018 0.014

session3:HF_typeHF_3 -0.002 -0.002 -0.003 -0.002

session4:HF_typeHF_3 -0.005 -0.007 -0.008 -0.006

session2:HF_typeHF_5 0.014 0.018 0.024 0.017

session3:HF_typeHF_5 -0.002 -0.003 -0.003 -0.003

session4:HF_typeHF_5 -0.006 -0.008 -0.010 -0.007

session2:HF_typeHF_7 0.011 0.014 0.016 0.027

session3:HF_typeHF_7	-0.002	-0.002	-0.003	-0.001
session4:HF_typeHF_7	-0.005	-0.006	-0.007	-0.008
session2:HF_typeHF_8	0.011	0.014	0.016	0.013
session3:HF_typeHF_8	-0.002	-0.002	-0.003	-0.002
session4:HF_typeHF_8	-0.005	-0.006	-0.007	-0.006
activitystanding:HF_typeHF_2	0.014	0.007	0.008	0.006
activitywalking:HF_typeHF_2	0.006	0.002	0.002	0.002
activitystanding:HF_typeHF_3	0.007	0.015	0.010	0.008
activitywalking:HF_typeHF_3	0.002	0.005	0.002	0.002
activitystanding:HF_typeHF_5	0.008	0.011	0.014	0.010
activitywalking:HF_typeHF_5	0.002	0.003	0.004	0.002
activitystanding:HF_typeHF_7	0.006	0.008	0.010	0.013
activitywalking:HF_typeHF_7	0.001	0.002	0.002	0.005
activitystanding:HF_typeHF_8	0.006	0.008	0.009	0.007
activitywalking:HF_typeHF_8	0.001	0.002	0.002	0.002
sexM:HF_typeHF_2	-0.022	-0.014	-0.016	-0.011
sexM:HF_typeHF_3	-0.012	-0.014	-0.018	-0.012
sexM:HF_typeHF_5	-0.012	-0.016	-0.022	-0.013
sexM:HF_typeHF_7	-0.010	-0.013	-0.015	-0.012
sexM:HF_typeHF_8	-0.011	-0.015	-0.017	-0.012
	d2:HF_HF_8	s2:HF_HF_2	s3:HF_HF_2	s4:HF_HF_2
day2				
session2				
session3				
session4				
activitystanding				
activitywalking				
sexM				
HF_typeHF_2				
HF_typeHF_3				
HF_typeHF_5				
HF_typeHF_7				
HF_typeHF_8				
day2:HF_typeHF_2				
day2:HF_typeHF_3				
day2:HF_typeHF_5				
day2:HF_typeHF_7				
day2:HF_typeHF_8				
session2:HF_typeHF_2	0.011			
session3:HF_typeHF_2	-0.002	0.608		
session4:HF_typeHF_2	-0.005	0.551	0.690	
session2:HF_typeHF_3	0.014	0.504	0.304	0.276
session3:HF_typeHF_3	-0.002	0.305	0.499	0.345
session4:HF_typeHF_3	-0.006	0.276	0.344	0.501
session2:HF_typeHF_5	0.016	0.587	0.354	0.321
session3:HF_typeHF_5	-0.003	0.356	0.583	0.403
session4:HF_typeHF_5	-0.007	0.322	0.402	0.585
session2:HF_typeHF_7	0.013	0.455	0.274	0.249
session3:HF_typeHF_7	-0.002	0.276	0.452	0.312
session4:HF_typeHF_7	-0.006	0.250	0.312	0.453
session2:HF_typeHF_8	0.026	0.449	0.271	0.246
session3:HF_typeHF_8	-0.003	0.271	0.445	0.307
session4:HF_typeHF_8	-0.008	0.247	0.307	0.447
activitystanding:HF_typeHF_2	0.006	0.102	0.106	0.159

activitywalking:HF_typeHF_2	0.001	0.412	0.195	0.474
activitystanding:HF_typeHF_3	0.008	0.052	0.049	0.078
activitywalking:HF_typeHF_3	0.002	0.208	0.099	0.239
activitystanding:HF_typeHF_5	0.009	0.060	0.058	0.091
activitywalking:HF_typeHF_5	0.002	0.243	0.115	0.279
activitystanding:HF_typeHF_7	0.007	0.047	0.045	0.071
activitywalking:HF_typeHF_7	0.002	0.188	0.089	0.216
activitystanding:HF_typeHF_8	0.014	0.046	0.044	0.070
activitywalking:HF_typeHF_8	0.006	0.185	0.088	0.213
sexM:HF_typeHF_2	-0.012	0.001	0.001	0.001
sexM:HF_typeHF_3	-0.014	0.001	0.001	0.001
sexM:HF_typeHF_5	-0.014	0.001	0.001	0.001
sexM:HF_typeHF_7	-0.012	0.001	0.001	0.001
sexM:HF_typeHF_8	-0.022	0.001	0.001	0.001
		s2:HF_HF_3	s3:HF_HF_3	s4:HF_HF_3 s2:HF_HF_5
day2				
session2				
session3				
session4				
activitystanding				
activitywalking				
sexM				
HF_typeHF_2				
HF_typeHF_3				
HF_typeHF_5				
HF_typeHF_7				
HF_typeHF_8				
day2:HF_typeHF_2				
day2:HF_typeHF_3				
day2:HF_typeHF_5				
day2:HF_typeHF_7				
day2:HF_typeHF_8				
session2:HF_typeHF_2				
session3:HF_typeHF_2				
session4:HF_typeHF_2				
session2:HF_typeHF_3				
session3:HF_typeHF_3	0.608			
session4:HF_typeHF_3	0.550	0.689		
session2:HF_typeHF_5	0.763	0.461	0.418	
session3:HF_typeHF_5	0.462	0.759	0.523	0.607
session4:HF_typeHF_5	0.419	0.524	0.760	0.550
session2:HF_typeHF_7	0.591	0.357	0.324	0.689
session3:HF_typeHF_7	0.358	0.589	0.406	0.417
session4:HF_typeHF_7	0.325	0.406	0.589	0.378
session2:HF_typeHF_8	0.584	0.353	0.320	0.680
session3:HF_typeHF_8	0.352	0.579	0.399	0.410
session4:HF_typeHF_8	0.320	0.401	0.581	0.373
activitystanding:HF_typeHF_2	0.052	0.050	0.078	0.061
activitywalking:HF_typeHF_2	0.208	0.099	0.239	0.243
activitystanding:HF_typeHF_3	0.104	0.103	0.157	0.078
activitywalking:HF_typeHF_3	0.413	0.196	0.475	0.315
activitystanding:HF_typeHF_5	0.078	0.075	0.118	0.103
activitywalking:HF_typeHF_5	0.315	0.150	0.362	0.413
activitystanding:HF_typeHF_7	0.061	0.058	0.092	0.071

activitywalking:HF_typeHF_7	0.245	0.116	0.281	0.285
activitystanding:HF_typeHF_8	0.060	0.057	0.090	0.070
activitywalking:HF_typeHF_8	0.241	0.114	0.277	0.281
sexM:HF_typeHF_2	0.002	0.001	0.001	0.002
sexM:HF_typeHF_3	0.002	0.001	0.003	0.002
sexM:HF_typeHF_5	0.002	0.001	0.002	0.002
sexM:HF_typeHF_7	0.002	0.001	0.002	0.002
sexM:HF_typeHF_8	0.002	0.001	0.002	0.002
	s3:HF_HF_5	s4:HF_HF_5	s2:HF_HF_7	s3:HF_HF_7
day2				
session2				
session3				
session4				
activitystanding				
activitywalking				
sexM				
HF_typeHF_2				
HF_typeHF_3				
HF_typeHF_5				
HF_typeHF_7				
HF_typeHF_8				
day2:HF_typeHF_2				
day2:HF_typeHF_3				
day2:HF_typeHF_5				
day2:HF_typeHF_7				
day2:HF_typeHF_8				
session2:HF_typeHF_2				
session3:HF_typeHF_2				
session4:HF_typeHF_2				
session2:HF_typeHF_3				
session3:HF_typeHF_3				
session4:HF_typeHF_3				
session2:HF_typeHF_5				
session3:HF_typeHF_5				
session4:HF_typeHF_5	0.689			
session2:HF_typeHF_7	0.417	0.378		
session3:HF_typeHF_7	0.687	0.474	0.606	
session4:HF_typeHF_7	0.474	0.688	0.549	0.690
session2:HF_typeHF_8	0.412	0.373	0.527	0.319
session3:HF_typeHF_8	0.676	0.466	0.318	0.524
session4:HF_typeHF_8	0.468	0.679	0.289	0.363
activitystanding:HF_typeHF_2	0.058	0.092	0.047	0.045
activitywalking:HF_typeHF_2	0.115	0.279	0.188	0.089
activitystanding:HF_typeHF_3	0.075	0.119	0.061	0.058
activitywalking:HF_typeHF_3	0.150	0.363	0.244	0.116
activitystanding:HF_typeHF_5	0.100	0.156	0.071	0.068
activitywalking:HF_typeHF_5	0.197	0.476	0.285	0.135
activitystanding:HF_typeHF_7	0.068	0.107	0.103	0.105
activitywalking:HF_typeHF_7	0.136	0.328	0.413	0.198
activitystanding:HF_typeHF_8	0.067	0.106	0.054	0.052
activitywalking:HF_typeHF_8	0.134	0.323	0.218	0.104
sexM:HF_typeHF_2	0.001	0.002	0.001	0.001
sexM:HF_typeHF_3	0.001	0.002	0.002	0.001
sexM:HF_typeHF_5	0.001	0.002	0.002	0.001

sexM:HF_typeHF_7	0.001	0.002	0.001	0.001
sexM:HF_typeHF_8	0.001	0.002	0.002	0.001
	s4:HF_HF_7	s2:HF_HF_8	s3:HF_HF_8	s4:HF_HF_8
day2				
session2				
session3				
session4				
activitystanding				
activitywalking				
sexM				
HF_typeHF_2				
HF_typeHF_3				
HF_typeHF_5				
HF_typeHF_7				
HF_typeHF_8				
day2:HF_typeHF_2				
day2:HF_typeHF_3				
day2:HF_typeHF_5				
day2:HF_typeHF_7				
day2:HF_typeHF_8				
session2:HF_typeHF_2				
session3:HF_typeHF_2				
session4:HF_typeHF_2				
session2:HF_typeHF_3				
session3:HF_typeHF_3				
session4:HF_typeHF_3				
session2:HF_typeHF_5				
session3:HF_typeHF_5				
session4:HF_typeHF_5				
session2:HF_typeHF_7				
session3:HF_typeHF_7				
session4:HF_typeHF_7				
session2:HF_typeHF_8	0.289			
session3:HF_typeHF_8	0.362	0.607		
session4:HF_typeHF_8	0.526	0.551	0.688	
activitystanding:HF_typeHF_2	0.071	0.046	0.044	0.070
activitywalking:HF_typeHF_2	0.217	0.186	0.088	0.214
activitystanding:HF_typeHF_3	0.092	0.060	0.057	0.091
activitywalking:HF_typeHF_3	0.281	0.241	0.114	0.278
activitystanding:HF_typeHF_5	0.107	0.070	0.067	0.105
activitywalking:HF_typeHF_5	0.328	0.281	0.133	0.324
activitystanding:HF_typeHF_7	0.157	0.054	0.052	0.082
activitywalking:HF_typeHF_7	0.475	0.218	0.103	0.251
activitystanding:HF_typeHF_8	0.082	0.105	0.099	0.155
activitywalking:HF_typeHF_8	0.251	0.414	0.190	0.473
sexM:HF_typeHF_2	0.001	0.001	0.001	0.001
sexM:HF_typeHF_3	0.002	0.002	0.001	0.001
sexM:HF_typeHF_5	0.002	0.002	0.001	0.001
sexM:HF_typeHF_7	0.001	0.001	0.001	0.001
sexM:HF_typeHF_8	0.001	0.001	0.000	0.001
	actvtys:HF_HF_2	actvtyw:HF_HF_2	actvtys:HF_HF_3	
day2				
session2				
session3				

```

session4
activitystanding
activitywalking
sexM
HF_typeHF_2
HF_typeHF_3
HF_typeHF_5
HF_typeHF_7
HF_typeHF_8
day2:HF_typeHF_2
day2:HF_typeHF_3
day2:HF_typeHF_5
day2:HF_typeHF_7
day2:HF_typeHF_8
session2:HF_typeHF_2
session3:HF_typeHF_2
session4:HF_typeHF_2
session2:HF_typeHF_3
session3:HF_typeHF_3
session4:HF_typeHF_3
session2:HF_typeHF_5
session3:HF_typeHF_5
session4:HF_typeHF_5
session2:HF_typeHF_7
session3:HF_typeHF_7
session4:HF_typeHF_7
session2:HF_typeHF_8
session3:HF_typeHF_8
session4:HF_typeHF_8
activitystanding:HF_typeHF_2
activitywalking:HF_typeHF_2 0.409
activitystanding:HF_typeHF_3 0.508 0.208
activitywalking:HF_typeHF_3 0.209 0.503 0.411
activitystanding:HF_typeHF_5 0.591 0.242 0.765
activitywalking:HF_typeHF_5 0.243 0.586 0.315
activitystanding:HF_typeHF_7 0.460 0.188 0.596
activitywalking:HF_typeHF_7 0.189 0.455 0.245
activitystanding:HF_typeHF_8 0.453 0.185 0.586
activitywalking:HF_typeHF_8 0.186 0.448 0.241
sexM:HF_typeHF_2 0.000 0.001 0.000
sexM:HF_typeHF_3 0.000 0.001 0.000
sexM:HF_typeHF_5 0.000 0.001 0.000
sexM:HF_typeHF_7 0.000 0.001 0.000
sexM:HF_typeHF_8 0.000 0.001 0.000
actvtyw:HF_HF_3 actvtys:HF_HF_5 actvtyw:HF_HF_5

day2
session2
session3
session4
activitystanding
activitywalking
sexM
HF_typeHF_2
HF_typeHF_3

```

```

HF_typeHF_5
HF_typeHF_7
HF_typeHF_8
day2:HF_typeHF_2
day2:HF_typeHF_3
day2:HF_typeHF_5
day2:HF_typeHF_7
day2:HF_typeHF_8
session2:HF_typeHF_2
session3:HF_typeHF_2
session4:HF_typeHF_2
session2:HF_typeHF_3
session3:HF_typeHF_3
session4:HF_typeHF_3
session2:HF_typeHF_5
session3:HF_typeHF_5
session4:HF_typeHF_5
session2:HF_typeHF_7
session3:HF_typeHF_7
session4:HF_typeHF_7
session2:HF_typeHF_8
session3:HF_typeHF_8
session4:HF_typeHF_8
activitystanding:HF_typeHF_2
activitywalking:HF_typeHF_2
activitystanding:HF_typeHF_3
activitywalking:HF_typeHF_3
activitystanding:HF_typeHF_5 0.314
activitywalking:HF_typeHF_5 0.762 0.412
activitystanding:HF_typeHF_7 0.245 0.693 0.285
activitywalking:HF_typeHF_7 0.591 0.284 0.689
activitystanding:HF_typeHF_8 0.241 0.681 0.280
activitywalking:HF_typeHF_8 0.582 0.280 0.678
sexM:HF_typeHF_2 0.002 0.000 0.002
sexM:HF_typeHF_3 0.002 0.000 0.002
sexM:HF_typeHF_5 0.002 0.000 0.002
sexM:HF_typeHF_7 0.002 0.000 0.002
sexM:HF_typeHF_8 0.002 0.000 0.002
actvtys:HF_HF_7 actvtyw:HF_HF_7 actvtys:HF_HF_8

day2
session2
session3
session4
activitystanding
activitywalking
sexM
HF_typeHF_2
HF_typeHF_3
HF_typeHF_5
HF_typeHF_7
HF_typeHF_8
day2:HF_typeHF_2
day2:HF_typeHF_3
day2:HF_typeHF_5

```

```

day2:HF_typeHF_7
day2:HF_typeHF_8
session2:HF_typeHF_2
session3:HF_typeHF_2
session4:HF_typeHF_2
session2:HF_typeHF_3
session3:HF_typeHF_3
session4:HF_typeHF_3
session2:HF_typeHF_5
session3:HF_typeHF_5
session4:HF_typeHF_5
session2:HF_typeHF_7
session3:HF_typeHF_7
session4:HF_typeHF_7
session2:HF_typeHF_8
session3:HF_typeHF_8
session4:HF_typeHF_8
activitystanding:HF_typeHF_2
activitywalking:HF_typeHF_2
activitystanding:HF_typeHF_3
activitywalking:HF_typeHF_3
activitystanding:HF_typeHF_5
activitywalking:HF_typeHF_5
activitystanding:HF_typeHF_7
activitywalking:HF_typeHF_7 0.411
activitystanding:HF_typeHF_8 0.530 0.218
activitywalking:HF_typeHF_8 0.218 0.527 0.411
sexM:HF_typeHF_2 0.000 0.001 0.000
sexM:HF_typeHF_3 0.000 0.002 0.000
sexM:HF_typeHF_5 0.000 0.002 0.000
sexM:HF_typeHF_7 0.000 0.001 0.000
sexM:HF_typeHF_8 0.000 0.002 0.000
actvtyw:HF_HF_8 sM:HF_HF_2 sM:HF_HF_3 sM:HF_HF_5

day2
session2
session3
session4
activitystanding
activitywalking
sexM
HF_typeHF_2
HF_typeHF_3
HF_typeHF_5
HF_typeHF_7
HF_typeHF_8
day2:HF_typeHF_2
day2:HF_typeHF_3
day2:HF_typeHF_5
day2:HF_typeHF_7
day2:HF_typeHF_8
session2:HF_typeHF_2
session3:HF_typeHF_2
session4:HF_typeHF_2
session2:HF_typeHF_3

```

```

session3:HF_typeHF_3
session4:HF_typeHF_3
session2:HF_typeHF_5
session3:HF_typeHF_5
session4:HF_typeHF_5
session2:HF_typeHF_7
session3:HF_typeHF_7
session4:HF_typeHF_7
session2:HF_typeHF_8
session3:HF_typeHF_8
session4:HF_typeHF_8
activitystanding:HF_typeHF_2
activitywalking:HF_typeHF_2
activitystanding:HF_typeHF_3
activitywalking:HF_typeHF_3
activitystanding:HF_typeHF_5
activitywalking:HF_typeHF_5
activitystanding:HF_typeHF_7
activitywalking:HF_typeHF_7
activitystanding:HF_typeHF_8
activitywalking:HF_typeHF_8
sexM:HF_typeHF_2      0.001
sexM:HF_typeHF_3      0.002      0.498
sexM:HF_typeHF_5      0.002      0.517      0.566
sexM:HF_typeHF_7      0.001      0.483      0.528      0.549
sexM:HF_typeHF_8      0.002      0.480      0.525      0.546
                        sM:HF_HF_7

day2
session2
session3
session4
activitystanding
activitywalking
sexM
HF_typeHF_2
HF_typeHF_3
HF_typeHF_5
HF_typeHF_7
HF_typeHF_8
day2:HF_typeHF_2
day2:HF_typeHF_3
day2:HF_typeHF_5
day2:HF_typeHF_7
day2:HF_typeHF_8
session2:HF_typeHF_2
session3:HF_typeHF_2
session4:HF_typeHF_2
session2:HF_typeHF_3
session3:HF_typeHF_3
session4:HF_typeHF_3
session2:HF_typeHF_5
session3:HF_typeHF_5
session4:HF_typeHF_5
session2:HF_typeHF_7

```

```

session3:HF_typeHF_7
session4:HF_typeHF_7
session2:HF_typeHF_8
session3:HF_typeHF_8
session4:HF_typeHF_8
activitystanding:HF_typeHF_2
activitywalking:HF_typeHF_2
activitystanding:HF_typeHF_3
activitywalking:HF_typeHF_3
activitystanding:HF_typeHF_5
activitywalking:HF_typeHF_5
activitystanding:HF_typeHF_7
activitywalking:HF_typeHF_7
activitystanding:HF_typeHF_8
activitywalking:HF_typeHF_8
sexM:HF_typeHF_2
sexM:HF_typeHF_3
sexM:HF_typeHF_5
sexM:HF_typeHF_7
sexM:HF_typeHF_8          0.509

```

Standardized Within-Group Residuals:

	Min	Q1	Med	Q3	Max
	-5.7487484	-0.4549235	0.1115377	0.5816359	3.3710954

Number of Observations: 2638

Number of Groups:

person	HF_type	%in%	person
24			144

```

anova_md18 <- anova(md17_reml, md18_reml)
print(anova_md18)

```

	Model	df	AIC	BIC	logLik	Test	L.Ratio	p-value
md17_reml	1	54	-401.9296	-85.5213	254.9648			
md18_reml	2	59	-943.6603	-597.9549	530.8301	1 vs 2	551.7307	<.0001

```

anova_md18_clean <- data.frame(
  Model = c("Model 7: Assumes Auto-regressive Structure (AR(1))", "Model 8: Assumes Heterogeneous AR St.
  Df = as.numeric(anova_md18$df),
  AIC = round(as.numeric(anova_md18$AIC)),
  BIC = round(as.numeric(anova_md18$BIC)),
  LogLik = round(as.numeric(anova_md18$logLik)),
  LRT_Stat = c(NA, round(anova_md18$L.Ratio[2], 2)),
  p_value = c(NA, anova_md18$p-value[2]))

anova_md18_clean <- anova_md18_clean %>%
  mutate(
    LRT_Stat = ifelse(is.na(LRT_Stat), "--", as.character(LRT_Stat)),
    p_value = ifelse(is.na(p_value), "--",
      # eps defines the threshold for "<"
      # digits defines the rounding for values above that threshold
      format.pval(p_value, eps = 0.0001, digits = 4))
  )

```

```
col_labels <- c("Model Version",
  "\\shortstack{Degrees of\\\\Freedom}",
  "AIC",
  "BIC",
  "Log-Likelihood",
  "\\shortstack{Likelihood Ratio\\\\Test Stat}",
  "P-Value")

kable(anova_md18_clean,
  format="latex",
  col.names = col_labels,
  align = "lrrrrrr",
  booktabs = TRUE, #, #right aligns numbers
  escape = FALSE,
  caption = "\\footnotesize{Likelihood test between Model 7 and Model 8, testing an ARH1 structure",
  kable_styling(latex_options = c("scale_down", "hold_position"),
    full_width = FALSE,
    table.envir = "table") %>%
  add_header_above(c("Likelihood Ratio Test for Covariance Structures" = 7),
    bold = TRUE, align = "l") %>%
  row_spec(0, bold = TRUE)
```

Table 8: Likelihood test between Model 7 and Model 8, testing an ARH1 structure that allows each sensor to have its own unique residual variance instead of assuming uniform residual noise. Model 8 has significantly lower AIC and BIC and a very small p-value, so it is statistically significant. Note that models are run using REML, as they have the same fixed effects.

Likelihood Ratio Test for Covariance Structures						
Model Version	Degrees of Freedom	AIC	BIC	Log-Likelihood	Likelihood Ratio Test Stat	P-Value
Model 7: Assumes Auto-regressive Structure (AR(1))	54	-402	-86	255	–	–
Model 8: Assumes Heterogeneous AR Structure (ARH(1))	59	-944	-598	531	551.73	< 1e-04

Model 7 (AR(1)) showed that there is a temporal aspect to the data, where consecutive sessions are more closely linked than further ones. However, AR(1) assumes that the amount of variance is identical for all sensors. However, we would expect sensors on different body parts to have different variance. As such, we tested a heterogeneous AR structure ARH(1) by combining Model 7's AR(1) structure with heterogeneous residual variances across sensors.

Model 8's ϕ is 0.6269, higher than Model 7's (0.5745), suggesting the temporal aspect of the data was higher than previously suggested and was masked by the noise levels of the sensors. The estimates for the heterogeneous variances relative to sensor 16 (baseline, dominant hand) are: 1.27 (sensor 2, chest (side)), 0.72 (sensor 3, upper chest), 0.35 (sensor 5, upper arm), 0.95 (sensor 7, lower arm), and 0.97 (sensor 8, abdomen). This implies that sensors on the abdomen, lower arm, and dominant hand have similar noise. The sensor on the chest (side) is significantly noisier than those sensors, and the sensors on the upper chest and arm are significantly less noisy. There does not seem to be an obvious trend or reason for this; it may be helpful to consult experts in the field. Additionally, Table 8 shows that the ARH(1) structure (Model 8) has significantly lower AIC and BIC than Model 7's AR(1) structure, and is statistically significant in the likelihood ratio test.

```
mdl8b_reml <- lme(
  fixed = log(HF_value) ~ (day + session + activity + sex) * HF_type,
  random = list(
    person = ~ 1 + day,
    HF_type = ~ 1
```



```

),
correlation = corAR1(form = ~ 1 | person/HF_type/day/activity),
weights = varIdent(form = ~ 1 | HF_type),
data = data_clean,
method = "REML",
control = lmeControl(msMaxIter = 200) #basically, increase iterations
)

anova.mdl18_reml, mdl18b_reml)

```

	Model	df	AIC	BIC	logLik
mdl18_reml	1	59	-943.6603	-597.9549	530.8301
mdl18b_reml	2	59	-1014.1562	-668.4508	566.0781

6.7 Pruning Fixed Effects

The final model utilizes an ARH(1) covariance structure to account for the temporal aspect of the data between sequential activities and sessions, while allowing for unique levels of noise at each sensor. Additionally, it has random intercepts for individuals and nested sensors to account for sensor-specific baselines per individual, and a person-level random slope for day to capture individual level responses to temperature changes. The model has become quite complex, so we re-examine fixed effects to determine if there are any that can be removed.

```

mdl18_ml <- lme(
  fixed = log(HF_value) ~ (day + session + activity + sex) * HF_type,
  random = list(
    person = ~ 1 + day,
    HF_type = ~ 1
  ),
  correlation = corAR1(form = ~ 1 | person/HF_type/day),
  weights = varIdent(form = ~ 1 | HF_type),
  data = data_clean,
  method = "ML",
  control = lmeControl(msMaxIter = 200) #basically, increase iterations
)

```

```

mdl19a_ml_no_interact_sensor_sex <- lme(
  fixed = log(HF_value) ~ (day + session + activity) * HF_type + sex,
  random = list(
    person = ~ 1 + day,
    HF_type = ~ 1
  ),
  correlation = corAR1(form = ~ 1 | person/HF_type/day),
  weights = varIdent(form = ~ 1 | HF_type),
  data = data_clean,
  method = "ML",
  control = lmeControl(msMaxIter = 200) #basically, increase iterations
)

```

```

mdl19a2_ml_no_sex <- lme(
  fixed = log(HF_value) ~ (day + session + activity) * HF_type ,
  random = list(
    person = ~ 1 + day,
    HF_type = ~ 1
  ),
)

```

```

correlation = corAR1(form = ~ 1 | person/HF_type/day),
weights = varIdent(form = ~ 1 | HF_type),
data = data_clean,
method = "ML",
control = lmeControl(msMaxIter = 200) #basically, increase iterations
)

mdl9b_ml_no_interact_sensor_day <- lme(
  fixed = log(HF_value) ~ (sex + session + activity) * HF_type + day,
  random = list(
    person = ~ 1 + day,
    HF_type = ~ 1
  ),
  correlation = corAR1(form = ~ 1 | person/HF_type/day),
  weights = varIdent(form = ~ 1 | HF_type),
  data = data_clean,
  method = "ML",
  control = lmeControl(msMaxIter = 200) #basically, increase iterations
)

mdl9b2_ml_no_day <- lme(
  fixed = log(HF_value) ~ (sex + session + activity) * HF_type ,
  random = list(
    person = ~ 1 + day,
    HF_type = ~ 1
  ),
  correlation = corAR1(form = ~ 1 | person/HF_type/day),
  weights = varIdent(form = ~ 1 | HF_type),
  data = data_clean,
  method = "ML",
  control = lmeControl(msMaxIter = 200) #basically, increase iterations
)

mdl9c_ml_no_interact_sensor_activity <- lme(
  fixed = log(HF_value) ~ (day + session + sex) * HF_type + activity,
  random = list(
    person = ~ 1 + day,
    HF_type = ~ 1
  ),
  correlation = corAR1(form = ~ 1 | person/HF_type/day),
  weights = varIdent(form = ~ 1 | HF_type),
  data = data_clean,
  method = "ML",
  control = lmeControl(msMaxIter = 200) #basically, increase iterations
)

mdl9c2_ml_no_activity <- lme(
  fixed = log(HF_value) ~ (day + session + sex) * HF_type ,
  random = list(
    person = ~ 1 + day,
    HF_type = ~ 1
  ),
  correlation = corAR1(form = ~ 1 | person/HF_type/day),
  weights = varIdent(form = ~ 1 | HF_type),

```

```
data = data_clean,
method = "ML",
control = lmeControl(msMaxIter = 200) #basically, increase iterations
)
```

```
mdl9d_ml_no_interact_sensor_session <- lme(
  fixed = log(HF_value) ~ (day + activity + sex) * HF_type + session,
  random = list(
    person = ~ 1 + day,
    HF_type = ~ 1
  ),
  correlation = corAR1(form = ~ 1 | person/HF_type/day),
  weights = varIdent(form = ~ 1 | HF_type),
  data = data_clean,
  method = "ML",
  control = lmeControl(msMaxIter = 200) #basically, increase iterations
)
```

```
mdl9d2_ml_no_session <- lme(
  fixed = log(HF_value) ~ (day + activity + sex) * HF_type ,
  random = list(
    person = ~ 1 + day,
    HF_type = ~ 1
  ),
  correlation = corAR1(form = ~ 1 | person/HF_type/day),
  weights = varIdent(form = ~ 1 | HF_type),
  data = data_clean,
  method = "ML",
  control = lmeControl(msMaxIter = 200) #basically, increase iterations
)
```

```
anova(mdl8_ml, mdl9a_ml_no_interact_sensor_sex)
```

	Model	df	AIC	BIC	logLik	Test
mdl8_ml	1	59	-1192.542	-845.7532	655.271	
mdl9a_ml_no_interact_sensor_sex	2	54	-1198.410	-881.0101	653.205	1 vs 2
			L.Ratio p-value			
mdl8_ml						
mdl9a_ml_no_interact_sensor_sex	4.132014		0.5306			

#not statistically significant, higher BIC and AIC if include --> remove

```
anova(mdl8_ml, mdl9a2_ml_no_sex)
```

	Model	df	AIC	BIC	logLik	Test	L.Ratio	p-value
mdl8_ml	1	59	-1192.542	-845.7532	655.271			
mdl9a2_ml_no_sex	2	53	-1198.978	-887.4559	652.489	1 vs 2	5.564012	0.4738

#not statistically significant, higher BIC and AIC if include --> remove

```
anova(mdl8_ml, mdl9b_ml_no_interact_sensor_day)
```

	Model	df	AIC	BIC	logLik	Test
mdl8_ml	1	59	-1192.542	-845.7532	655.2710	
mdl9b_ml_no_interact_sensor_day	2	54	-1135.017	-817.6170	621.5085	1 vs 2
			L.Ratio p-value			
mdl8_ml						

```
mdl9b_ml_no_interact_sensor_day 67.52511 <.0001
```

```
#lower AIC and BIC if include, and p-val <0.0001 --> include
```

```
anova(mdl8_ml, mdl9b2_ml_no_day)
```

	Model	df	AIC	BIC	logLik	Test	L.Ratio	p-value
mdl8_ml	1	59	-1192.542	-845.7532	655.2710			
mdl9b2_ml_no_day	2	53	-1106.610	-795.0880	606.3051	1 vs 2	97.93187	<.0001

```
#lower AIC and BIC if include, and p-val <0.0001 --> include
```

```
anova(mdl8_ml, mdl9c_ml_no_interact_sensor_activity)
```

	Model	df	AIC	BIC	logLik	Test	L.Ratio	p-value
mdl8_ml	1	59	-1192.542	-845.7532	655.2710			
mdl9c_ml_no_interact_sensor_activity	2	49	-808.695	-520.6840	453.3475			

Test L.Ratio p-value

mdl8_ml
mdl9c_ml_no_interact_sensor_activity 1 vs 2 403.847 <.0001

```
#lower AIC and BIC if include, and p-val <0.0001 --> include
```

```
anova(mdl8_ml, mdl9c2_ml_no_activity)
```

	Model	df	AIC	BIC	logLik	Test	L.Ratio	p-value
mdl8_ml	1	59	-1192.5420	-845.7532	655.2710			
mdl9c2_ml_no_activity	2	47	-769.4748	-493.2193	431.7374	1 vs 2	447.0672	

p-value

mdl8_ml
mdl9c2_ml_no_activity <.0001

```
#lower AIC and BIC if include, and p-val <0.0001 --> include
```

```
anova(mdl8_ml, mdl9d_ml_no_interact_sensor_session)
```

	Model	df	AIC	BIC	logLik	Test	L.Ratio	p-value
mdl8_ml	1	59	-1192.542	-845.7532	655.2710			
mdl9d_ml_no_interact_sensor_session	2	44	-1187.052	-928.4297	637.5259			

Test L.Ratio p-value

mdl8_ml
mdl9d_ml_no_interact_sensor_session 1 vs 2 35.49017 0.0021

```
#BIC worsens (increases) significant if include, but AIC decreases slightly. p-value is 0.0021, stat si
```

```
anova(mdl8_ml, mdl9d2_ml_no_session)
```

	Model	df	AIC	BIC	logLik	Test	L.Ratio	p-value
mdl8_ml	1	59	-1192.542	-845.7532	655.2710			
mdl9d2_ml_no_session	2	41	-1190.153	-949.1638	636.0763	1 vs 2	38.38944	

p-value

mdl8_ml
mdl9d2_ml_no_session 0.0034

```
#BIC worsens (increases) significant if include, but AIC decreases slightly (2 points). p-value is 0.00
```

```
anova_summary_manual_md18 <- bind_rows(  
  get_baseline_metrics(mdl8_ml, "Model 8 (Full Model)"),  
  get_anova_outputs(mdl8_ml, mdl9a_ml_no_interact_sensor_sex, "Drop Sensor and Sex Interaction"),  
  get_anova_outputs(mdl8_ml, mdl9a2_ml_no_sex, "Drop Sex Fully"),
```

```

get_anova_outputs mdl8_ml, mdl9b_ml_no_interact_sensor_day, "Drop Sensor and Day Interaction"),
get_anova_outputs mdl8_ml, mdl9b2_ml_no_day, "Drop Day Fully"),
get_anova_outputs mdl8_ml, mdl9c_ml_no_interact_sensor_activity, "Drop Sensor and Activity Interaction"),
get_anova_outputs mdl8_ml, mdl9c2_ml_no_activity, "Drop Activity Fully"),
get_anova_outputs mdl8_ml, mdl9d_ml_no_interact_sensor_session, "Drop Sensor and Session Interaction"),
get_anova_outputs mdl8_ml, mdl9d2_ml_no_session, "Drop Session Fully"),
)

print(anova_summary_manual_md18)

```

	Model	Df	AIC	BIC	LogLik	LRT_Stat
1	Model 8 (Full Model)	59	-1193	-846	655	NA
2	Drop Sensor and Sex Interaction	54	-1198	-881	653	4.13
3	Drop Sex Fully	53	-1199	-887	652	5.56
4	Drop Sensor and Day Interaction	54	-1135	-818	622	67.53
5	Drop Day Fully	53	-1107	-795	606	97.93
6	Drop Sensor and Activity Interaction	49	-809	-521	453	403.85
7	Drop Activity Fully	47	-769	-493	432	447.07
8	Drop Sensor and Session Interaction	44	-1187	-928	638	35.49
9	Drop Session Fully	41	-1190	-949	636	38.39

	p_value
1	NA
2	5.305701e-01
3	4.737540e-01
4	3.351761e-13
5	6.774187e-19
6	1.428621e-80
7	3.961802e-88
8	2.092879e-03
9	3.438823e-03

```

anova_summary_manual_md18_clean <- anova_summary_manual_md18 %>%
  mutate(
    LRT_Stat = ifelse(is.na(LRT_Stat), "--", as.character(LRT_Stat)),
    p_value = ifelse(is.na(p_value), "--",
      # eps defines the threshold for "<"
      # digits defines the rounding for values above that threshold
      format.pval(p_value, eps = 0.0001, digits = 4))
  )

```

```

col_labels <- c("Model Version",
  "\\shortstack{Degrees of\\\\Freedom}",
  "AIC",
  "BIC",
  "Log-Likelihood",
  "\\shortstack{Likelihood Ratio\\\\Test Stat}",
  "P-Value")

```

```

kable(anova_summary_manual_md18_clean,
  format="latex",
  col.names = col_labels,
  align = "lrrrrrr",
  booktabs = TRUE, #, #right aligns numbers
  escape = FALSE,
  caption = "\\footnotesize{Model selection of variables via backwards elimination. Model 8 includes

```

```

kable_styling(latex_options = c("scale_down", "hold_position"),
              full_width = FALSE,
              table.envir = "table") %>%
add_header_above(c("Backwards Selection of Interaction Terms and Variables: Likelihood Ratio Test S
                    bold = TRUE, align = "l") %>%
row_spec(0, bold = TRUE) %>% #highlight main row
row_spec(1, italic = TRUE)

```

Table 9: Model selection of variables via backwards elimination. Model 8 includes the four interaction terms previously noted. Each subsequent row represents a model where either only one interaction term was dropped from Model 8 (Drop X and Y Interaction), or the whole variable and interaction (Drop Sex Fully means dropped the variable sex, and the interaction term sex and sensor). The LRT Statistic and P-Value are derived from Likelihood Ratio Tests comparing the reduced models to Model 8. Note all models were fit using Maximum Likelihood (ML), to compare across models.

Backwards Selection of Interaction Terms and Variables: Likelihood Ratio Test Summary						
Model Version	Degrees of Freedom	AIC	BIC	Log-Likelihood	Likelihood Ratio Test Stat	P-Value
<i>Model 8 (Full Model)</i>	59	-1193	-846	655	–	–
Drop Sensor and Sex Interaction	54	-1198	-881	653	4.13	0.530570
Drop Sex Fully	53	-1199	-887	652	5.56	0.473754
Drop Sensor and Day Interaction	54	-1135	-818	622	67.53	< 1e-04
Drop Day Fully	53	-1107	-795	606	97.93	< 1e-04
Drop Sensor and Activity Interaction	49	-809	-521	453	403.85	< 1e-04
Drop Activity Fully	47	-769	-493	432	447.07	< 1e-04
Drop Sensor and Session Interaction	44	-1187	-928	638	35.49	0.002093
Drop Session Fully	41	-1190	-949	636	38.39	0.003439

Table 9 suggests that we should remove the variables Sex and the interaction term of Sensor and Sex, as the model without those terms has better AIC and BIC, and the p-value from the likelihood ratio test is not statistically significant (>0.4). We also removed the variable Session and the interaction term of Sensor and Session. Although they minimally improve the AIC (by <10 points), they worsen the BIC significantly and introduce ~ 18 degrees of freedom.

The final model is Model 9: ($\log(\text{HF_value}) \sim \text{activity} + \text{day} + \text{sensor} + \text{sensor} * \text{activity} + \text{sensor} * \text{day}$), which evaluates the fixed effects of activity, day, sensor, and the interactions of sensor with activity and day. It addresses the hierarchical nature of the data by incorporating a random intercept for person, a nested random intercept for sensor, and a person-level random slope for day. Finally, the error structure is defined by an ARH(1) covariance structure, allowing temporal auto-correlation segmented by person, sensor, and day, and heterogeneous residual variances to account for different levels of volatility across sensors.

```

mdl_9_fin <- lme(
  fixed = log(HF_value) ~ (day + activity + sex) * HF_type ,
  random = list(
    person = ~ 1 + day,
    HF_type = ~ 1
  ),
  correlation = corAR1(form = ~ 1 | person/HF_type/day),
  weights = varIdent(form = ~ 1 | HF_type),
  data = data_clean,
  method = "REML",
  control = lmeControl(msMaxIter = 200) #basically, increase iterations
)

summary(mdl_9_fin)

```

Linear mixed-effects model fit by REML

Data: data_clean

AIC BIC logLik
-1037.279 -796.7589 559.6394

Random effects:

Formula: ~1 + day | person

Structure: General positive-definite, Log-Cholesky parametrization

StdDev Corr
(Intercept) 0.08198097 (Intr)
day2 0.13882909 -0.922

Formula: ~1 | HF_type %in% person

(Intercept) Residual
StdDev: 0.1591161 0.2849547

Correlation Structure: AR(1)

Formula: ~1 | person/HF_type/day

Parameter estimate(s):

Phi
0.6324753

Variance function:

Structure: Different standard deviations per stratum

Formula: ~1 | HF_type

Parameter estimates:

HF_16 HF_2 HF_3 HF_5 HF_7 HF_8
1.0000000 1.2388910 0.7058033 0.3412306 0.9310399 0.9463243

Fixed effects: log(HF_value) ~ (day + activity + sex) * HF_type

	Value	Std.Error	DF	t-value	p-value
(Intercept)	4.382541	0.06525206	2476	67.16326	0.0000
day2	-0.146868	0.05693764	2476	-2.57945	0.0100
activitystanding	-0.077875	0.01749847	2476	-4.45040	0.0000
activitywalking	0.094058	0.02127396	2476	4.42128	0.0000
sexM	-0.003530	0.08263706	22	-0.04272	0.9663
HF_typeHF_2	-0.570742	0.09382998	110	-6.08273	0.0000
HF_typeHF_3	-0.538773	0.08400422	110	-6.41365	0.0000
HF_typeHF_5	-0.317337	0.07947313	110	-3.99301	0.0001
HF_typeHF_7	-0.116802	0.08715457	110	-1.34018	0.1830
HF_typeHF_8	-0.781547	0.08765932	110	-8.91573	0.0000
day2:HF_typeHF_2	0.470108	0.07826288	2476	6.00679	0.0000
day2:HF_typeHF_3	0.467161	0.06086916	2476	7.67484	0.0000
day2:HF_typeHF_5	0.412815	0.05278869	2476	7.82013	0.0000
day2:HF_typeHF_7	0.308993	0.06682584	2476	4.62386	0.0000
day2:HF_typeHF_8	0.464246	0.06796492	2476	6.83067	0.0000
activitystanding:HF_typeHF_2	-0.105925	0.02754265	2476	-3.84585	0.0001
activitywalking:HF_typeHF_2	-0.461892	0.03378722	2476	-13.67060	0.0000
activitystanding:HF_typeHF_3	0.178397	0.02139428	2476	8.33853	0.0000
activitywalking:HF_typeHF_3	0.170678	0.02612520	2476	6.53307	0.0000
activitystanding:HF_typeHF_5	0.051458	0.01850259	2476	2.78110	0.0055
activitywalking:HF_typeHF_5	-0.110354	0.02252214	2476	-4.89980	0.0000
activitystanding:HF_typeHF_7	-0.037174	0.02362961	2476	-1.57320	0.1158
activitywalking:HF_typeHF_7	-0.144755	0.02890232	2476	-5.00842	0.0000
activitystanding:HF_typeHF_8	0.162279	0.02399148	2476	6.76404	0.0000
activitywalking:HF_typeHF_8	0.091377	0.02923513	2476	3.12559	0.0018

sexM:HF_typeHF_2	-0.072653	0.12069664	110	-0.60195	0.5484	
sexM:HF_typeHF_3	0.069682	0.11020538	110	0.63229	0.5285	
sexM:HF_typeHF_5	0.019993	0.10593391	110	0.18873	0.8507	
sexM:HF_typeHF_7	0.130835	0.11358630	110	1.15186	0.2519	
sexM:HF_typeHF_8	0.093280	0.11428393	110	0.81621	0.4161	
Correlation:						
	(Intr)	day2	actvtys	actvtyw	sexM	HF_HF_2
day2	-0.433					
activitystanding	-0.134	0.013				
activitywalking	-0.097	-0.001	0.408			
sexM	-0.618	-0.019	-0.001	0.001		
HF_typeHF_2	-0.643	0.219	0.093	0.067	0.419	
HF_typeHF_3	-0.718	0.245	0.104	0.075	0.468	0.500
HF_typeHF_5	-0.759	0.259	0.110	0.079	0.494	0.528
HF_typeHF_7	-0.692	0.236	0.101	0.072	0.451	0.481
HF_typeHF_8	-0.688	0.235	0.100	0.072	0.448	0.479
day2:HF_typeHF_2	0.229	-0.547	-0.009	0.001	0.014	-0.402
day2:HF_typeHF_3	0.295	-0.704	-0.012	0.001	0.018	-0.206
day2:HF_typeHF_5	0.340	-0.812	-0.014	0.001	0.022	-0.238
day2:HF_typeHF_7	0.269	-0.641	-0.011	0.001	0.017	-0.187
day2:HF_typeHF_8	0.264	-0.630	-0.011	0.001	0.017	-0.185
activitystanding:HF_typeHF_2	0.085	-0.008	-0.635	-0.259	0.001	-0.147
activitywalking:HF_typeHF_2	0.061	0.000	-0.257	-0.630	-0.001	-0.105
activitystanding:HF_typeHF_3	0.110	-0.010	-0.818	-0.334	0.001	-0.076
activitywalking:HF_typeHF_3	0.079	0.001	-0.332	-0.814	-0.001	-0.055
activitystanding:HF_typeHF_5	0.127	-0.012	-0.946	-0.386	0.001	-0.088
activitywalking:HF_typeHF_5	0.091	0.001	-0.385	-0.945	-0.001	-0.063
activitystanding:HF_typeHF_7	0.099	-0.009	-0.741	-0.302	0.001	-0.069
activitywalking:HF_typeHF_7	0.071	0.001	-0.300	-0.736	-0.001	-0.049
activitystanding:HF_typeHF_8	0.098	-0.009	-0.729	-0.297	0.001	-0.068
activitywalking:HF_typeHF_8	0.070	0.001	-0.297	-0.728	-0.001	-0.049
sexM:HF_typeHF_2	0.412	0.013	0.001	-0.001	-0.668	-0.625
sexM:HF_typeHF_3	0.452	0.015	0.001	-0.001	-0.731	-0.314
sexM:HF_typeHF_5	0.470	0.015	0.001	-0.001	-0.761	-0.326
sexM:HF_typeHF_7	0.438	0.014	0.001	-0.001	-0.709	-0.305
sexM:HF_typeHF_8	0.435	0.014	0.001	-0.001	-0.705	-0.302
	HF_HF_3	HF_HF_5	HF_HF_7	HF_HF_8	d2:HF_HF_2	
day2						
activitystanding						
activitywalking						
sexM						
HF_typeHF_2						
HF_typeHF_3						
HF_typeHF_5	0.590					
HF_typeHF_7	0.538	0.568				
HF_typeHF_8	0.535	0.565	0.515			
day2:HF_typeHF_2	-0.179	-0.189	-0.172	-0.172		
day2:HF_typeHF_3	-0.358	-0.244	-0.221	-0.221	0.514	
day2:HF_typeHF_5	-0.265	-0.325	-0.255	-0.255	0.594	
day2:HF_typeHF_7	-0.209	-0.221	-0.368	-0.200	0.466	
day2:HF_typeHF_8	-0.206	-0.218	-0.198	-0.374	0.461	
activitystanding:HF_typeHF_2	-0.066	-0.070	-0.064	-0.063	0.014	
activitywalking:HF_typeHF_2	-0.047	-0.050	-0.046	-0.045	0.000	
activitystanding:HF_typeHF_3	-0.128	-0.090	-0.082	-0.082	0.007	

activitywalking:HF_typeHF_3	-0.091	-0.065	-0.059	-0.059	0.000
activitystanding:HF_typeHF_5	-0.099	-0.116	-0.095	-0.094	0.009
activitywalking:HF_typeHF_5	-0.071	-0.084	-0.068	-0.068	-0.001
activitystanding:HF_typeHF_7	-0.077	-0.082	-0.136	-0.074	0.007
activitywalking:HF_typeHF_7	-0.055	-0.058	-0.097	-0.053	0.000
activitystanding:HF_typeHF_8	-0.076	-0.080	-0.073	-0.137	0.007
activitywalking:HF_typeHF_8	-0.055	-0.058	-0.053	-0.098	0.000
sexM:HF_typeHF_2	-0.320	-0.338	-0.309	-0.306	-0.022
sexM:HF_typeHF_3	-0.646	-0.370	-0.338	-0.336	-0.012
sexM:HF_typeHF_5	-0.364	-0.653	-0.352	-0.349	-0.013
sexM:HF_typeHF_7	-0.340	-0.360	-0.644	-0.326	-0.010
sexM:HF_typeHF_8	-0.338	-0.357	-0.326	-0.636	-0.011
	d2:HF_HF_3	d2:HF_HF_5	d2:HF_HF_7	d2:HF_HF_8	
day2					
activitystanding					
activitywalking					
sexM					
HF_typeHF_2					
HF_typeHF_3					
HF_typeHF_5					
HF_typeHF_7					
HF_typeHF_8					
day2:HF_typeHF_2					
day2:HF_typeHF_3					
day2:HF_typeHF_5	0.763				
day2:HF_typeHF_7	0.600	0.692			
day2:HF_typeHF_8	0.592	0.684	0.537		
activitystanding:HF_typeHF_2	0.007	0.009	0.007	0.007	
activitywalking:HF_typeHF_2	0.000	-0.001	0.000	0.000	
activitystanding:HF_typeHF_3	0.015	0.011	0.009	0.009	
activitywalking:HF_typeHF_3	-0.001	-0.001	-0.001	-0.001	
activitystanding:HF_typeHF_5	0.011	0.014	0.010	0.010	
activitywalking:HF_typeHF_5	-0.001	-0.001	-0.001	-0.001	
activitystanding:HF_typeHF_7	0.009	0.010	0.014	0.008	
activitywalking:HF_typeHF_7	0.000	-0.001	0.000	0.000	
activitystanding:HF_typeHF_8	0.009	0.010	0.008	0.014	
activitywalking:HF_typeHF_8	0.000	-0.001	0.000	0.000	
sexM:HF_typeHF_2	-0.014	-0.017	-0.011	-0.013	
sexM:HF_typeHF_3	-0.014	-0.019	-0.013	-0.014	
sexM:HF_typeHF_5	-0.016	-0.022	-0.013	-0.015	
sexM:HF_typeHF_7	-0.013	-0.015	-0.013	-0.012	
sexM:HF_typeHF_8	-0.015	-0.018	-0.012	-0.022	
	actvtys:HF_HF_2	actvtyw:HF_HF_2	actvtys:HF_HF_3		
day2					
activitystanding					
activitywalking					
sexM					
HF_typeHF_2					
HF_typeHF_3					
HF_typeHF_5					
HF_typeHF_7					
HF_typeHF_8					
day2:HF_typeHF_2					
day2:HF_typeHF_3					

day2:HF_typeHF_5			
day2:HF_typeHF_7			
day2:HF_typeHF_8			
activitystanding:HF_typeHF_2			
activitywalking:HF_typeHF_2	0.405		
activitystanding:HF_typeHF_3	0.520	0.210	
activitywalking:HF_typeHF_3	0.211	0.513	0.407
activitystanding:HF_typeHF_5	0.601	0.243	0.774
activitywalking:HF_typeHF_5	0.245	0.595	0.315
activitystanding:HF_typeHF_7	0.470	0.190	0.606
activitywalking:HF_typeHF_7	0.191	0.463	0.245
activitystanding:HF_typeHF_8	0.463	0.187	0.597
activitywalking:HF_typeHF_8	0.189	0.458	0.243
sexM:HF_typeHF_2	0.000	0.001	0.000
sexM:HF_typeHF_3	0.000	0.001	0.000
sexM:HF_typeHF_5	0.000	0.001	-0.001
sexM:HF_typeHF_7	0.000	0.001	-0.001
sexM:HF_typeHF_8	0.000	0.001	0.000
	actvttyw:HF_HF_3	actvttys:HF_HF_5	actvttyw:HF_HF_5
day2			
activitystanding			
activitywalking			
sexM			
HF_typeHF_2			
HF_typeHF_3			
HF_typeHF_5			
HF_typeHF_7			
HF_typeHF_8			
day2:HF_typeHF_2			
day2:HF_typeHF_3			
day2:HF_typeHF_5			
day2:HF_typeHF_7			
day2:HF_typeHF_8			
activitystanding:HF_typeHF_2			
activitywalking:HF_typeHF_2			
activitystanding:HF_typeHF_3			
activitywalking:HF_typeHF_3			
activitystanding:HF_typeHF_5	0.314		
activitywalking:HF_typeHF_5	0.769	0.407	
activitystanding:HF_typeHF_7	0.246	0.700	0.285
activitywalking:HF_typeHF_7	0.599	0.284	0.695
activitystanding:HF_typeHF_8	0.242	0.690	0.281
activitywalking:HF_typeHF_8	0.593	0.281	0.687
sexM:HF_typeHF_2	0.001	-0.001	0.001
sexM:HF_typeHF_3	0.000	-0.001	0.001
sexM:HF_typeHF_5	0.001	-0.001	0.001
sexM:HF_typeHF_7	0.001	-0.001	0.001
sexM:HF_typeHF_8	0.001	-0.001	0.001
	actvttys:HF_HF_7	actvttyw:HF_HF_7	actvttys:HF_HF_8
day2			
activitystanding			
activitywalking			
sexM			
HF_typeHF_2			

HF_typeHF_3				
HF_typeHF_5				
HF_typeHF_7				
HF_typeHF_8				
day2:HF_typeHF_2				
day2:HF_typeHF_3				
day2:HF_typeHF_5				
day2:HF_typeHF_7				
day2:HF_typeHF_8				
activitystanding:HF_typeHF_2				
activitywalking:HF_typeHF_2				
activitystanding:HF_typeHF_3				
activitywalking:HF_typeHF_3				
activitystanding:HF_typeHF_5				
activitywalking:HF_typeHF_5				
activitystanding:HF_typeHF_7				
activitywalking:HF_typeHF_7	0.408			
activitystanding:HF_typeHF_8	0.540	0.219		
activitywalking:HF_typeHF_8	0.220	0.536	0.406	
sexM:HF_typeHF_2	0.000	0.001	0.000	
sexM:HF_typeHF_3	0.000	0.001	0.000	
sexM:HF_typeHF_5	0.000	0.001	0.000	
sexM:HF_typeHF_7	0.000	0.001	0.000	
sexM:HF_typeHF_8	0.000	0.001	0.000	
actvtyw:HF_HF_8 sM:HF_HF_2 sM:HF_HF_3 sM:HF_HF_5				
day2				
activitystanding				
activitywalking				
sexM				
HF_typeHF_2				
HF_typeHF_3				
HF_typeHF_5				
HF_typeHF_7				
HF_typeHF_8				
day2:HF_typeHF_2				
day2:HF_typeHF_3				
day2:HF_typeHF_5				
day2:HF_typeHF_7				
day2:HF_typeHF_8				
activitystanding:HF_typeHF_2				
activitywalking:HF_typeHF_2				
activitystanding:HF_typeHF_3				
activitywalking:HF_typeHF_3				
activitystanding:HF_typeHF_5				
activitywalking:HF_typeHF_5				
activitystanding:HF_typeHF_7				
activitywalking:HF_typeHF_7				
activitystanding:HF_typeHF_8				
activitywalking:HF_typeHF_8				
sexM:HF_typeHF_2	0.001			
sexM:HF_typeHF_3	0.001	0.502		
sexM:HF_typeHF_5	0.001	0.522	0.572	
sexM:HF_typeHF_7	0.001	0.486	0.532	0.553
sexM:HF_typeHF_8	0.001	0.484	0.530	0.551

```

                                sM:HF_HF_7
day2
activitystanding
activitywalking
sexM
HF_typeHF_2
HF_typeHF_3
HF_typeHF_5
HF_typeHF_7
HF_typeHF_8
day2:HF_typeHF_2
day2:HF_typeHF_3
day2:HF_typeHF_5
day2:HF_typeHF_7
day2:HF_typeHF_8
activitystanding:HF_typeHF_2
activitywalking:HF_typeHF_2
activitystanding:HF_typeHF_3
activitywalking:HF_typeHF_3
activitystanding:HF_typeHF_5
activitywalking:HF_typeHF_5
activitystanding:HF_typeHF_7
activitywalking:HF_typeHF_7
activitystanding:HF_typeHF_8
activitywalking:HF_typeHF_8
sexM:HF_typeHF_2
sexM:HF_typeHF_3
sexM:HF_typeHF_5
sexM:HF_typeHF_7
sexM:HF_typeHF_8                                0.513

Standardized Within-Group Residuals:
      Min      Q1      Med      Q3      Max
-5.8180449 -0.4262270  0.1072506  0.5857570  3.2774555

Number of Observations: 2638
Number of Groups:
      person HF_type %in% person
      24          144

```

7 Discussion: Answer the questions

- Motivate the use of mixed linear models in this project and build one to assess
 - the inter-variability in heat flow,
 - * ??
 - the intra-variability in heat flow, and
 - * ??
 - the covariance structure between the sensors
 - * ??
- Allow for different levels of grouping: allow observations for the same participant to be correlated, and observations within the same participant and the same activity to be correlated too. Check whether this is needed by looking at the proportion of variance explained by your different levels

- Person yes (prev discussed in ICC)
 - Activity no (prev discussed in ICC)
 - Sensor yes (prev discussed)
- The context of the experiment is multivariate by nature. How will you use the univariate mixed model to model marginal and joint measurements from the different sensors?
- Provide interpretation of the most important parameters of your final model
- Aim of the study: Is there significant inter- and intra-individual variability in heat flow measurements in each of the sensors? How are measurements from different sensors related?
- To examine the intra-individual variability, it is necessary to analyze for the same person
 - Effect of the time: The difference between the same activities on the same day, but at different times (sessions)
 - Temperature effect: The difference between the same activities but on different days
- To examine the inter-individual variability, it is necessary to examine the difference between the same activities on the same day, but across participants